

THE ROLE OF CELLULAR SENESENCE AND THE SENESENCE-ASSOCIATED
SECRETORY PHENOTYPE IN VASCULAR AGING: AMELIORATION BY THE
NATURAL SENOLYTIC Fisetin

by

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ABSTRACT

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The Role of Cellular Senescence and the Senescence-Associated Secretory Phenotype in Vascular Aging: Amelioration by the Natural Senolytic Fisetin

Thesis directed by Distinguished Professor Douglas R. Seals

Cardiovascular diseases (CVDs) remain the leading cause of death globally, with advancing age being the primary risk factor for CVD development. Age-related vascular dysfunction, characterized by the stiffening of the large elastic arteries and vascular endothelial dysfunction, represents the primary antecedent to the development of CVDs. As such, understanding the mechanisms that drive age-related vascular dysfunction and establishing effective interventions to improve arterial health are biomedical research priorities. Cellular senescence, a multifactorial stress response, accumulates with advancing age and promotes vascular dysfunction through the production of the senescence-associated secretory phenotype (SASP) which drives chronic inflammation and excess oxidative stress. Senolytics, compounds that selectively eliminate senescent cells, hold promise in reducing excess cellular senescence and the SASP and subsequently improving or preventing age-related vascular dysfunction.

The purpose of this dissertation research was to identify promising senolytic interventions to suppress excess cellular senescence and the SASP and enhance vascular function with aging. In a review of the literature, I compiled the putative and established mechanisms by which vascular senescent cells and the SASP may contribute to vascular aging. I collected evidence of the effects of senolytic therapies on vascular health and identified groups that had high cellular senescence and SASP burden and impaired vascular function that might stand to

benefit the most from senolytic treatment. In a preclinical intervention study, I found that late-life supplementation with the natural senolytic fisetin enhanced endothelial function and lowered arterial stiffness in old mice. These improvements were mediated by a suppression of excess vascular cell senescence and associated with reduced oxidative stress and favorable remodeling of the arterial wall. In a mechanistic study, I identified aortic endothelial cells to be highly susceptible to becoming senescent in the context of aging and are readily eliminated by fisetin supplementation. Further, I demonstrated that the SASP was an underlying mechanism of age-related endothelial dysfunction and identified chemokine 12 (CXCL12) as a SASP factor that mediated endothelial dysfunction with aging that could be modulated with senolytic intervention.

Collectively, these studies establish cellular senescence and the SASP as causal contributors to the age-related decline in vascular function and demonstrate that senolytic interventions have efficacy in improving or preventing vascular function with aging. Thus, these findings open new avenues for the development of preventative and therapeutic interventions targeting cellular senescence and the SASP to improve cardiovascular health in aging populations.

DEDICATION

My dissertation is dedicated to beloved companion, Sunshine.

As far as dogs go, you have some room for improvements.

But, as a girl's best friend, you are perfect.

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During my time as a research assistant at the Buck Institute for Research on Aging, I was privileged to be graced with the presence of brilliant scientists who influenced my own research and career trajectories. Both expert scientists who worked at the Buck Institute and those who were invited to present at the Buck Institute had significant influence on my biomedical perspectives and priorities, which was incredibly important to me as a fresh out of college, early-stage investigator. However, it was one invited speaker, in particular, who had the most transformative effects on my research mindset. With just a brief seminar presentation, I was compelled to apply to the University of Colorado Boulder and pursue an advanced degree in Integrative Physiology. And after four years in Boulder, I find myself in awe of the impact of that monumental presentation on my life. As such, I'd like to thank my primary mentor, Dr. Doug Seals, for (unintentionally) convincing me to join the Integrative Physiology of Aging Laboratory. Dr. Seals has provided me with the resources, personnel, and environment that have allowed me to thrive during my graduate school experience. Specifically, Dr. Seals placed not one, but two resident Research Assistant Professors in charge of my scientific and professional training.

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in-person!) and grown to include many other trainees in the laboratory. This “senescence” meeting remains one of my favorite and most stimulating aspects of my graduate training.

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CHAPTER I

INTRODUCTION

Cardiovascular diseases (CVD) are the leading cause of morbidity and mortality in developed countries¹. The primary, non-modifiable risk factor for CVD development is advancing age, as evidenced by over 90% of cases of CVD occurring in middle-aged and older adults². Demographic projections predict continuous increases in the number and percentage of older adults (*i.e.*, people over 65 years of age) over the next several decades, such that by the year 2100 there will be 2 billion older adults accounting for nearly 1 in 4 of the population globally². The combination of age as a risk factor for CVD, and the projected increase in the number of older adults, will lead to greater CV events and CVD prevalence³. Thus, effective prevention and treatment strategies are a biomedical priority to mitigate the projected increase in the burden of CVD. As we work to develop effective prevention strategies, it is crucial to understand the role of aging in the onset and progression of CVD. Age-related vascular dysfunction is the primary antecedent to CVD development and is characterized by the manifestation of large elastic artery stiffening and impaired vascular endothelial function⁴.

Large elastic arteries expand and recoil elastically with each cardiac pulsation to buffer the energy of the pulse wave and propel blood distally to peripheral tissues⁵. The primary target for clinically relevant arterial stiffness assessments is the aorta given that it is the largest and most distensible artery in the body and also the most prone to abnormal stiffening in response to adverse stimuli^{5,6}. In young, healthy adults, a compliant aorta effectively buffers excess pulsatility caused by left ventricular ejection and exhibits a slow pulse wave velocity (PWV)⁵, the gold-standard measure of *in vivo* arterial stiffness⁶⁻⁸. A slow PWV allows pulse wave reflections to arrive at the heart during diastole, increasing diastolic coronary perfusion pressure

but not systolic ventricular load⁵. However, with increasing chronological age, greater arterial stiffening occurs resulting in a faster PWV⁵⁻⁸. This causes the heart to pump harder leading to higher blood pressure and greater pulsatile flow⁵. Increases in pulsatile flow may damage other tissues, especially high-perfusion organs with low-resistance beds such as the kidneys⁹, brain¹⁰, and eyes¹¹. As such, arterial stiffness is a strong independent predictor of CVD risk and a risk factor for several other chronic disorders^{6,7}.

Mechanistically, age-related arterial stiffness results primarily from structural changes that occur in the arterial wall with advancing age^{5,12,13}. Under healthy, physiological conditions, low arterial stiffness is reflected by favorable ratios of elastin and collagen, two vital structural components of the arterial wall^{12,13}. Elastin is the primary matrix protein responsible for the reversible elasticity of arteries, whereas collagen is the main load-bearing element in arterial walls and serves as a stiff reinforcing structural element^{12,13}. With advancing age, there is an increase in pathophysiological stimuli including the development of excess reactive oxygen species (ROS) and chronic, low-grade inflammation^{12,13}. Because of this, 3 unfavorable structural changes occur in the arterial wall: 1) degradation and fragmentation of elastin; 2) compensatory deposition of collagen to replace the fractured elastin; and 3) stimulation of advanced glycation end products (AGEs) that induce crosslinking of these structural components^{12,13}. Together, these processes reduce elasticity and confer stiffness of the arteries.

Vascular endothelial function is another key measure of vascular health¹⁴. The endothelium is the inner lining of blood vessels and is composed of a monolayer of endothelial cells¹⁵. The endothelium plays a critical role in regulating blood flow by dilating and constricting to control blood flow to peripheral tissues¹⁵. The vasoprotective molecule nitric oxide (NO) is produced by endothelial cells and mediates vasodilation by inducing relaxation of the vascular

smooth muscle¹⁶. Importantly, NO-mediated vasodilation can be induced by mechanical stimuli such as blood flow or chemical stimuli such as acetylcholine, bradykinin, prostaglandins and hyperpolarizing factors¹⁵. Both mechanical and chemical stimuli elicit NO production in endothelial cells by activation of endothelial NO synthase (eNOS) which catalyzes the conversion of L-arginine and oxygen with cofactor tetrahydrobiopterin (BH₄) to produce NO¹⁵. Endothelium-derived NO diffuses to vascular smooth muscle cells where it activates intracellular signaling cascades, leading to vasodilation¹⁵. With advancing age, endothelial function declines, as measured *in vivo* by flow-mediated dilation in people, and is mediated by decreased NO bioavailability¹⁶. Reduced NO bioavailability is due to an increase in ROS from various sources including dysfunctional mitochondria and increased pro-oxidant enzymes^{12,13,17}. Specifically, ROS scavenge NO by forming peroxynitrite which then oxidizes BH₄ and turns it into its inactive form, resulting in ‘uncoupling’ of eNOS and leading to the synthesis of ROS rather than NO¹⁸.

Two macro-mechanistic processes that drive age-related arterial stiffness and endothelial dysfunction are oxidative stress and inflammation, which diminish the availability of the NO and promote arterial structural remodeling^{12,13,17}. Although these vascular aging phenotypes and macro-mechanisms are well-characterized, the age-related biological changes that contribute to the accumulation of oxidative stress and inflammation are not fully understood. As such, elucidating the upstream mechanisms responsible is essential for identifying preventative and therapeutic strategies for mitigating CVD risk with aging¹⁹.

As such, we identified the hallmark of aging cellular senescence, as a putative integrative mechanism with inflammation and oxidative stress that may drive age-related vascular dysfunction²⁰⁻²³. Cellular senescence is a multifactorial stress response characterized by a state of essentially irreversible cell cycle arrest and a pro-inflammatory secretome, known as the

senescence-associated secretory phenotype (SASP)^{20,21,24,25}. Senescent cells exist at low levels during physiological conditions and some of their basal functions include tissue regeneration and tumor suppression^{20,21}. Transient levels of the SASP contribute to these basal responses by secreting a pro-inflammatory milieu that recruits immune responses to sites of injury or to clear potentially cancerous cells²⁴⁻²⁶. However, with increased chronological age, senescent cells accumulate due to increased exposure to adverse stimuli and a decline in immune capacities. As senescent cells accumulate, SASP levels also become excessive, creating an adverse environment which can lead to the development of chronic diseases and disorders²⁴⁻²⁶. Further, the SASP can also independently induce cellular senescence in neighboring cells and in circulation to promote systemic cellular senescence induction, if not readily eliminated²⁴⁻²⁷.

States of cellular senescence may be induced by many triggers including telomere shortening and dysfunction, oncogene activation, DNA damage and stress, oxidative stress and extrinsic factors like cytotoxic drugs and radiation exposure^{20,21}. Senescent cells induced by these triggers lead to pleiotropic biological effects depending on the biological context^{20,21}. Some physiological, short-term effects of cellular senescence include tumor suppression, tissue and wound healing, and embryonic development^{20,21,24,26}. Conversely, chronic and long-term effects of excess cellular senescence include promoting cancer progression, inflammation, physiological decline and tissue aging^{20,21,24,26}. Senescent cells have various phenotypic changes compared to healthy proliferating cells^{28,29}. In addition to the SASP, key vascular senescent cells, like endothelial and vascular smooth muscle cells, also exhibit higher oxidative stress, increased activation of central inflammatory regulators such as NF- κ B, reduced NO bioavailability, dysfunctional organelles like the mitochondria and lysosomes, increased expression of cell

regulators and an up-regulation on senescent cell anti-apoptotic pathways (SCAPs) to confer cell cycle arrest²⁹⁻³¹.

Importantly, cellular senescence burden can be quantified by biomarkers such as the *Cdkn2a* gene and its encoded protein, the cell cycle inhibitor p16^{Ink4a}^{32,33}. While most mouse tissues exhibit moderate to substantial increase in senescent cell burden with advancing age, the aorta has been reported as tissue with the greatest change in *Cdkn2a* expression with age (~30 times greater expression in old mice versus young mice)³⁴. This is consistent with prior reports that endothelial cells readily undergo cellular senescence and also suggests that the vasculature is especially vulnerable to cellular senescence stimuli³⁵.

The upregulation of p16^{Ink4a} in senescent cells has also been leveraged to develop a transgenic mouse model that serves as a valuable tool to study cellular senescence *in vivo* and *ex vivo*²⁶. The p16-3MR mouse model, developed by the laboratory of the late Dr. Judith Campisi, contains a trimodality reporter fusion protein (3MR), with functional domains of luciferase, red fluorescent protein, and herpes simplex virus²⁶. The reporter is located at the promoter region of the *Cdkn2a* gene such that when p16^{Ink4a} is highly expressed, the fusion protein is as well²⁶. These functional domains allow for the detection of p16^{Ink4a}-high expressing cells by luminescence and fluorescence in addition to allowing for genetic elimination of p16^{Ink4a}-high cells using the anti-viral drug ganciclovir (GCV)²⁶.

In addition to genetic clearance of senescent cells, the development of pharmaceutical compounds known as senotherapeutics has emerged as an innovative class of new drugs (or lifestyle interventions)^{36,37}. There are two main classes of pharmacological senotherapeutics: senolytics and senomorphics³⁶. Senolytic therapies are interventions that selectively suppress

excess senescent cells by targeting SCAP networks³⁸, whereas senomorphics are compounds that suppress SASP signaling by inhibiting central inflammatory activation³⁶.

To establish initial evidence that age-related accumulation of senescent cells may be an underlying mechanism of vascular dysfunction, our laboratory leveraged the p16-3MR model to compare GCV-induced genetic cellular senescence clearance with the established pharmaceutical senolytic therapy known as ABT-263²². In this proof-of-concept study, we found that both genetic and pharmaceutical clearance of senescent cells reduced arterial stiffness and improved endothelial function in old mice by improving NO bioavailability, reducing oxidative stress, and favorably remodeling the arterial wall²². This important proof-of-concept study was the first to establish causality of senescent cells in age-related vascular dysfunction. Despite these major advancements, the senolytic interventions used in this study lack translational potential given that genetic cellular senescence clearance is not viable in people and ABT-263 is a repurposed anti-cancer drug and therefore has limited broad-level translatability to healthy older adults^{39,40}.

The recent development of naturally occurring compounds with senolytic properties has offered a wide variety of putative therapeutics that may target diverse types of senescent cells and SCAPs whilst being considered safer, more accessible, and more affordable than synthetic, pharmaceutical senolytics^{36,38,41}. Currently, several natural, food-derived compounds (*i.e.*, fisetin^{42,43}, quercetin⁴⁴⁻⁴⁶, piperlongumine⁴⁷, procyanidin C1⁴⁸, resveratrol^{49,50}, gingerenone A⁵¹, curcumin⁵²) have been identified to have putative senolytic properties. Yet, determining whether these compounds have direct senolytic properties or benefit senescent cells through alternative capacities (*i.e.*, senomorphic, anti-inflammatory, antioxidant, *etc.*) remains to be investigated.

Recent advancements in omics-based approaches and tools to isolate the role of cellular senescence and the circulating SASP milieu have been crucial to unravel the mechanisms-of-

action of senolytics on senescent cells. Remarkable strides in omics-based technologies, specifically transcriptomics^{53,54} and proteomics^{55,56}, offer powerful tools to uncover and define the molecular signatures of senescent cells and the molecular effects of senolytics in vascular tissue. By analyzing the comprehensive molecular profiles of senescent cells, omics-based approaches allow identification of specific genetic alterations, gene expression patterns, protein abundances, and metabolite levels associated with cellular senescence and senolytic intervention in the vasculature⁵⁷. These omics-based discoveries provide insights into the mechanisms underlying cellular senescence-induced vascular impairments, facilitating the development of novel diagnostic biomarkers and therapeutic targets⁵⁷. Furthermore, the integration of multiple omics datasets enables a systems-level understanding of cellular senescence in the vasculature, paving the way for precision medicine approaches to prevent or treat vascular aging and its associated complications⁵⁷.

However, even with these cutting-edge technologies, the majority of published research has been aimed at describing associations between senescent cells, the SASP and health outcomes providing only descriptive inferences. In the context of senolytic interventions, clinical and preclinical studies often characterize senescent cell burden, circulating SASP levels, and functional outcomes independently⁴⁴⁻⁴⁶. As such, it is challenging to ascertain whether improvements in physiological function are due to a direct suppression in cellular senescence and/or the SASP. Further, these studies often identify specific cellular senescence or SASP factors that are differentially changed with senolytic treatment but often fail to implicate the causal role of the individual factor either as a prognostic biomarker or therapeutic target of cellular senescence or the SASP. As a result, the causal implications of senolytic therapies on

senescent cell burden, the circulating SASP milieu, and vascular dysfunction with aging have not been thoroughly investigated.

The development of novel experimental approaches that allow for the isolation of cellular senescence and the SASP is essential for addressing this research gap. The p16-3MR mouse model has emerged as the gold-standard approach to establish pharmacological senolytics compared to genetic cellular senescence clearance with GCV using ‘pharmaco-dissection’ techniques²⁶. Leveraging this model, *in vivo* treatment of putative senolytic intervention can be tested for senolytic capacities in the vasculature by performing *ex vivo* GCV incubations prior to assessing measures of vascular function^{22,58}. For example, in isolated mouse arteries, *ex vivo* incubations with GCV prior to measuring vascular function results in improved outcomes in old vehicle (control) mice, given the clearance of the present excess senescent cells⁵⁸. However, in old mice treated with putative senolytic interventions, *ex vivo* incubations with GCV may not further improve vascular responses if the senolytic effectively cleared senescent cells, thus establishing a causal of cellular senescence in vascular function and providing evidence that the putative senolytic elicited its effects through senescent cell clearance^{22,58}.

In addition to cellular senescence-isolating models, there has been a recent development in experimental models that aim to investigate the role of the circulating milieu, including the SASP, on vascular function^{27,59-63}. Studies leveraging heterochronic parabiosis (*i.e.*, an experimental technique in which two animals of different ages are surgically joined, allowing them to share a circulatory system)^{60,61,64} and heterochronic blood exchange (*i.e.*, in which blood is transferred from one animal to another animal of different age)^{27,63} have reported that through the circulating milieu, aged animals may transfer age-related phenotypes to young animals and young animals may promote rejuvenation in aged animals, including in relations to vascular

responses. Further, these studies have established that cellular senescence phenotypes are regulated by the circulating milieu in multiple tissues²⁷. Overall, these studies have provided the primary evidence of the role of the circulating SASP milieu on age-related physiological declines. However, some major limitations of these models include their invasive nature of the procedures, the inability to isolate the circulating milieu from each donor, and the inability to isolate the direct effects on specific tissues⁶⁵.

As such, our laboratory sought to develop a novel bioassay to isolate the role of the circulating milieu on vascular function⁵⁹. In this approach, aortas and carotid arteries are isolated from young (3-5 months) and old (25 months), intervention-naïve mice⁵⁹. Isolated arteries are then exposed to exogenous components of the circulating milieu (*i.e.*, serum, plasma, extracellular vesicles, *etc.*) whilst being maintained under physiological conditions (heated, with nutrients, and pressurized)⁵⁹. Following exposure to the circulating milieu, vascular function is assessed to measure the influence of the circulating milieu on vascular responses⁵⁹. In complement to the model, similar approaches may be taken in cultured endothelial cells exposed to the circulating milieu to determine mechanisms by which the circulating milieu affects endothelial cells, including NO bioavailability, general and mitochondrial-specific oxidative stress, and other biochemical outcomes⁶⁶. Leveraging these approaches, our laboratory has shown that the circulating milieu from aged animals and mid-life and older adults induced vascular impairments in young arteries and proliferating cells, whereas the circulating milieu from young animals and adults improves function in old mouse arteries and cells^{59,66}. Importantly, this approach may also be used to assess the effects of the circulating SASP milieu on vascular function and investigate the direct effects of the SASP and senolytic interventions on circulating milieu-mediated vascular responses and investigate underlying mechanisms.

The purpose of this dissertation was to elucidate the role of cellular senescence and the SASP on vascular aging using senolytic interventions. In Chapter II, I thoroughly examined scientific literature to evaluate the mechanisms of cellular senescence and the SASP on vascular dysfunction and evidence of senotherapeutic strategies. I also sought to identify clinical groups with high senescent cell burden and vascular dysfunction that may stand to benefit the most from senotherapeutic intervention. In Chapter III, I investigated the efficacy of late-life fisetin supplementation on cellular senescence and vascular function with aging in mice and explored potential underlying mechanisms. Finally, in Chapter IV, I sought to characterize senescent cells and the SASP in the aged vasculature and determine the mechanisms-of-action of fisetin supplementation on vascular senescent cells. Further, I identified candidate SASP factors inducing vascular damage with aging and tested the effects of the circulating SASP milieu and specific SASP factors on endothelial function.

SPECIFIC AIMS

Specific Aim 1 (Chapter II):

1) Summarize the role of cellular senescence and the SASP in vascular dysfunction. 2) Review established and putative mechanisms through which excess senescent cells and the SASP drive vascular dysfunction. 3) Identify clinical groups with high senescence burden and early vascular aging that may benefit most from senotherapeutics. 4) Determine translational challenges in senotherapeutic dosing and clinical implementation.

Specific Aim 2 (Chapter III):

1) Test the senolytic properties of the natural senolytic compound fisetin on cultured senescent endothelial cells. 2) Investigate the senolytic properties of late-life fisetin supplementation on vascular cell senescence in old (27 months) C57BL/6 and p16-3MR mice. 3) Determine if late-life supplementation with fisetin improves vascular function in old mice through: a) suppressed senescent cell burden, b) improved NO bioavailability, c) reduced whole-cell and mitochondrial oxidative stress, and d) favorable arterial wall remodeling.

Specific Aim 3 (Chapter IV):

1) Characterize vascular senescent cell transcriptomic profiles in aortas from old (27 months) relative to young (6 months) C57BL/6 mice. 2) Assess the effects of fisetin supplementation on vascular cell senescence burden and identify target cell types and molecular mechanisms. 3) Investigate the role of the circulating SASP milieu and specific SASP factors on endothelial dysfunction with fisetin supplementation.

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CHAPTER II

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Mechanisms of cellular senescence-induced vascular aging: Evidence of senotherapeutic strategies

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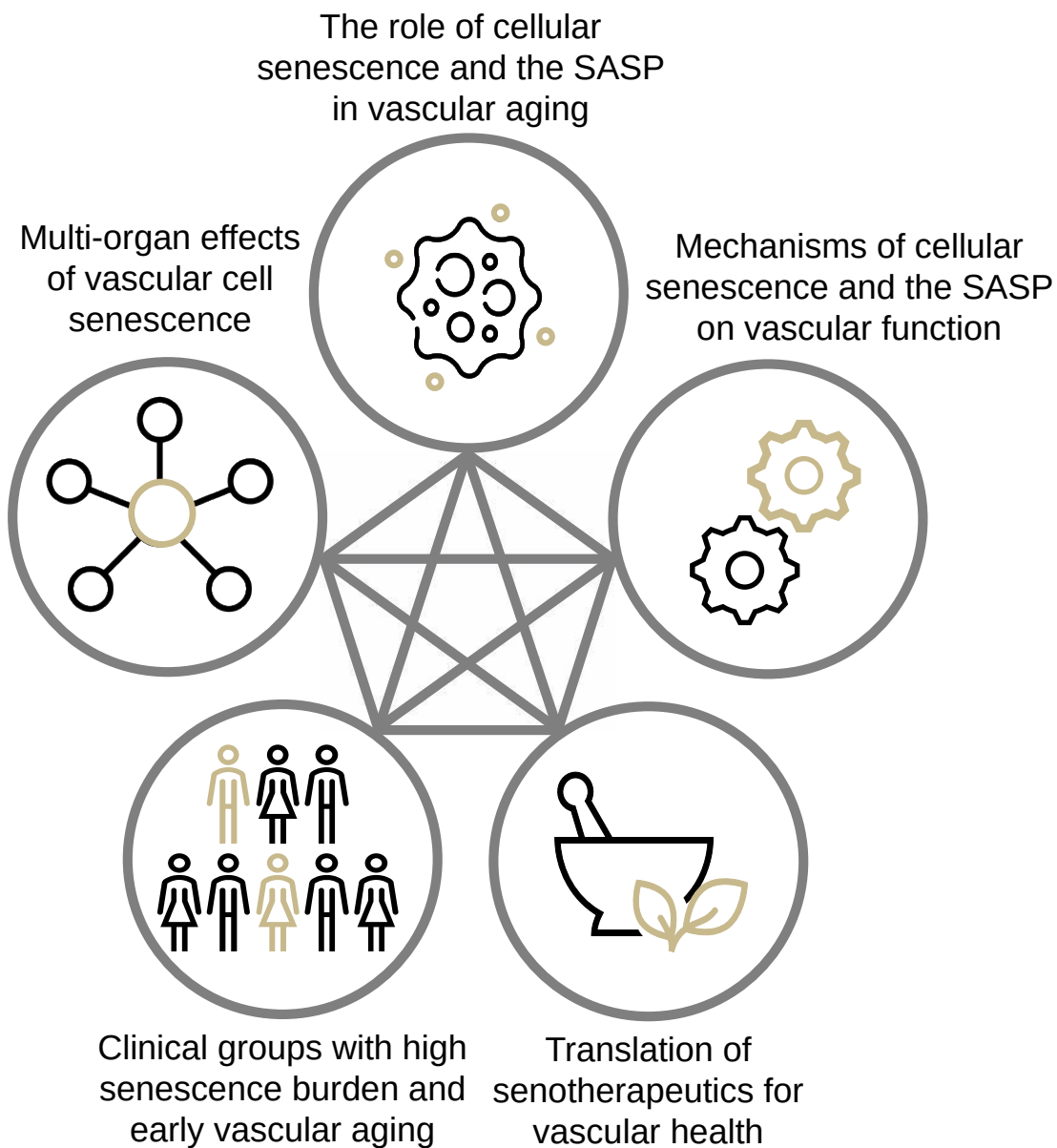
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GRAPHICAL ABSTRACT

ABSTRACT

Cardiovascular diseases (CVD) remain the leading cause of death worldwide with advancing age being the primary, nonmodifiable risk factor. Vascular dysfunction, namely arterial stiffening and endothelial dysfunction, is the key antecedent to the development of clinical CVD with aging. Fundamental aging macro-mechanistic processes that drive vascular aging include excess oxidative stress, chronic inflammation, and declines in the vasodilatory molecule nitric oxide. An important hallmark of aging that contributes to the vascular aging processes is cellular senescence – a stress response characterized by cell cycle arrest and accompanied by the production and secretion of proinflammatory molecules (i.e., the senescence-associated secretory phenotype [SASP]). Excess senescent cell and the SASP have deleterious effects on vascular function and in states of CVD, making it a putative therapeutic target for improving vascular function and preventing or reversing CVD. This review will focus of the role of cellular senescence in age-related vascular dysfunction and CVD. We will examine established and emerging mechanisms underlying cellular senescence-induced vascular dysfunction. We will then discuss groups with impaired vascular function and high cellular senescence burden and examine strategies to reduce or remove excess senescent cells and the SASP in the groups who are likely to benefit from most these therapies. Finally, we will highlight the systemic effects of vascular senescent cell suppression on other tissues and organs given the integrative role of the vasculature in physiology. Together, this review will underscore the imperative role of cellular senescence in vascular dysfunction and the need for deeper understanding of the translational use of cellular senescence and SASP targeting therapies in groups with high senescent cell burden.

KEYWORDS: Vascular dysfunction, cellular senescence, senotherapeutics, vascular aging

INTRODUCTION

Cardiovascular diseases (CVD) are the leading cause of death worldwide and despite major advances in CV medicine, advancing age remains the major CVD risk factor²⁻⁴.

Demographic projections predict continuous increases in the number and percentage of older adults (i.e., people over 65 years of age) over the next several decades¹. The combination of age as a risk factor for CVD, and the projected increase in number of older adults, will equvalate to greater CV events and CVD prevalence². As such, it is crucial to identify novel mechanisms of CVD and therapeutics to prevent and treat CVD.

Advanced age results in vascular dysfunction characterized by the stiffening of the large elastic arteries (i.e., aorta) and impaired vascular endothelial function, both of which are key antecedents of CVD⁵. Age-related vascular dysfunction is a consequence of several fundamental macro-mechanistic processes including chronic inflammation, excess oxidative stress, and declines in the bioavailability of the vasodilatory molecule nitric oxide (NO)⁶⁻⁸. Whilst the etiology of these fundamental macro-mechanistic processes has not been fully elucidated, the processes have been well characterized, and modulating these pathways improves vascular function in older mammals^{9,10}. Cellular senescence, a hallmark of aging, has emerged as a plausible contributor of inflammation, oxidative stress, and suppressed NO⁸. Cellular senescence is a stress response characterized by stable cell cycle arrest and the production of a pro-inflammatory milieu referred to as the senescence-associated secretory phenotype (SASP)¹¹. Advancing age increases the abundance of senescent cells in the vasculature¹² which contributes to age-related vascular dysfunction¹³, and several CVD¹⁴. Therefore, targeting cellular senescence and the SASP using senotherapeutics – therapies that suppress excess senescent cells

and/or the SASP – is an appealing strategy to treat and prevent vascular dysfunction and mitigate CVD risk⁸.

In this review, we will discuss: 1) a summary of the role of cellular senescence and the SASP in vascular dysfunction 2) established and putative mechanisms through which excess senescent cell and the SASP drive vascular dysfunction; 3) clinical groups with high senescence burden that may benefit most from senotherapeutics; and 4) translational challenges in senotherapeutic dosing and clinical implementation.

DISCUSSION

Age-related vascular dysfunction and mechanisms of vascular aging.

Large elastic artery stiffening and vascular endothelial dysfunction are two independent predictors of future CV events and mortality^{15,16}. Stiffening of the large elastic arteries (i.e., the aorta) occurs mostly through changes in mechanical properties of the vessel wall¹⁷. During aging, vascular smooth muscle cells (VSMC) undergo phenotypic switching from synthetic to contractile states which promote unfavorable vascular remodeling such as changes in the stress/strain components of the arteries¹⁸. Aortic stiffness is assessed non-invasively in mice and humans by the reference standard *in vivo* measure of pulse wave velocity (PWV)¹⁹. This assessment quantifies the stiffness of the aorta by measuring the time elapsed for the pressure wave produced by the contraction of the heart during systole to propagate bidirectionally to the two pulse pressure sites¹⁹. Importantly, PWV increases with advancing age and increases pressure pulsatility thereby affecting systolic blood pressure in a feed-forward cycle¹⁷. As such, increased arterial stiffness contributes to hypertension and is independently predictive of future CVD¹⁶.

Endothelial dysfunction is another primary manifestation of vascular aging characterized, in part, by impaired endothelium-dependent dilation (EDD)²⁰. Endothelial cells make up a monolayer that lines the inner-most wall of blood vessels²¹. In response to mechanical shear stress induced by antegrade blood flow or exposure to pharmacological agents (*i.e.*, acetylcholine), endothelial cells produce NO, which diffuses into the VSMC to cause vasodilation and regulate vascular tone²². In humans, EDD is commonly measured using the non-invasive reference-standard approach of brachial artery flow-mediated dilation (FMD), which is primarily dependent on the endothelial response to produce NO, and is a widely used measure to characterize the health of the macrovascular endothelium¹⁵. Another key characteristic of age-related endothelial dysfunction includes dysregulation of delivery of essential nutrients and oxygen to peripheral tissues²¹. Given that the endothelium lines the circulation, endothelial cells serve as an important barrier and regulator of biological factors that cross into peripheral tissues²¹.

Fundamental macro-mechanistic processes underlying arterial stiffening and endothelial dysfunction include excess oxidative stress, chronic, low-grade inflammation, and reduced NO bioavailability²³. With aging, oxidative stress results from a combination of excess reactive oxygen species (ROS) production and decreased antioxidant defenses due primarily to reductions in antioxidant enzyme activity/abundance^{9,24}. Excess ROS readily reacts with NO to form peroxynitrite and lower NO bioavailability²⁴. Peroxynitrite may also oxidize tetrahydrobiopterin (BH₄), an essential cofactor for NO production by endothelial NO synthase (eNOS)²⁵. Importantly, this results in eNOS uncoupling whereby eNOS generates ROS instead of NO²⁵. Together, these processes promote ROS bioactivity and reduce NO bioavailability to impair endothelial function²⁰. Excess ROS may also activate proinflammatory and prooxidant signaling

pathways²⁶. This intrinsic inflammatory activation in the microenvironment or in the circulation can activate other ROS producing systems, creating a feed-forward cycle of inflammation and oxidative stress that exacerbates vascular dysfunction. States of high oxidative stress and inflammation contribute to arterial stiffening by promoting adverse structural remodeling of the vascular walls⁸. Notably, four unfavorable structural changes occur in the vascular wall with aging: 1) there is a degradation and fragmentation of elastin; 2) there is a compensatory deposition of collagen to replace the fractured elastin; 3), the stimulation of advanced glycation end products (AGEs) promotes crosslinking of these structural components; and 4) AGE-induced calcium (Ca^{2+}) deposition leading to arterial calcification^{8,27}. There are also functional contributions to arterial stiffening driven by reduced vasodilatory factors (i.e., NO and hyperpolarizing factors) and increased vasoconstrictive factors (i.e., angiotensin II) that confer VSMC stiffness^{8,28}. Together, these processes reduce elasticity and contribute to the stiffness of the arteries. Although these major macro-mechanistic events are well-accepted as underlying mechanism of vascular aging²³, the integrative cellular/molecular processes regulating excessive oxidative stress, chronic inflammation, and limited NO bioavailability with aging are incompletely understood.

The role of cellular senescence in vascular dysfunction.

Cellular senescence is a complex biological process characterized by the irreversible growth arrest of cells in response to various stimuli, including DNA damage, telomere shortening, oxidative stress, and oncogene activation^{11,29}. Senescent cells exhibit a distinct phenotype compared to healthy, proliferating cells, including an enlarged and flattened morphology, increased metabolic activity, and an amplified secretion of a wide range of proinflammatory cytokines, chemokines, and growth factors collectively referred to as the

senescence-associated secretory phenotype (SASP)^{30,31}. Senescent cells and the SASP contribute to many physiological processes, including embryonic development, wound healing, and tumor suppression^{29,32}. However, the accumulation of senescent cells and the SASP has been implicated in various age-related pathologies, including CVD³³⁻³⁵. Increased cellular senescence is evident in clinical vascular pathologies including atherosclerosis^{36,37}, pulmonary hypertension³⁸, and aortic aneurysms³⁹. In the absence of disease, the vasculature remains the tissue with the greatest senescent cell burden in old mice and in models of premature aging in mice¹², suggesting that the vasculature is particularly vulnerable to cellular senescence. Key vascular cells, such as endothelial cells⁴⁰⁻⁴² and VSMC^{43,44} become senescent with chronological aging and with other stressors *in vivo*. Importantly, vascular senescent cells are identified by key characteristics and biomarkers including cell cycle regulators (i.e., *Cdkn1a*, *Cdkn2a*, p16, p19 and p21), altered lysosomal activity (senescence-associated β -galactosidase [SA- β -Gal]), DNA damage markers (53BP1 and γ -H2AX), telomere length and telomerase activity, senescent cell anti-apoptotic pathways (SCAPs), and SASP profile (IL-6, IL-1 β , tumor necrosis factor α [TNF- α], among many other)^{31,45} (**Figure 1**). Together, this evidence suggests that the burden of cellular senescence is greatest in the vasculature compared to other tissues and contributes to impairments in function and the development of CVD. Therefore, targeting senescence in the vasculature may improve function and reduce age-related vascular pathologies. Below we discuss mechanisms of vascular cell senescence, groups with high senescent cell burden, and therapeutic treatments that target senescent cells to promote healthy vascular aging.

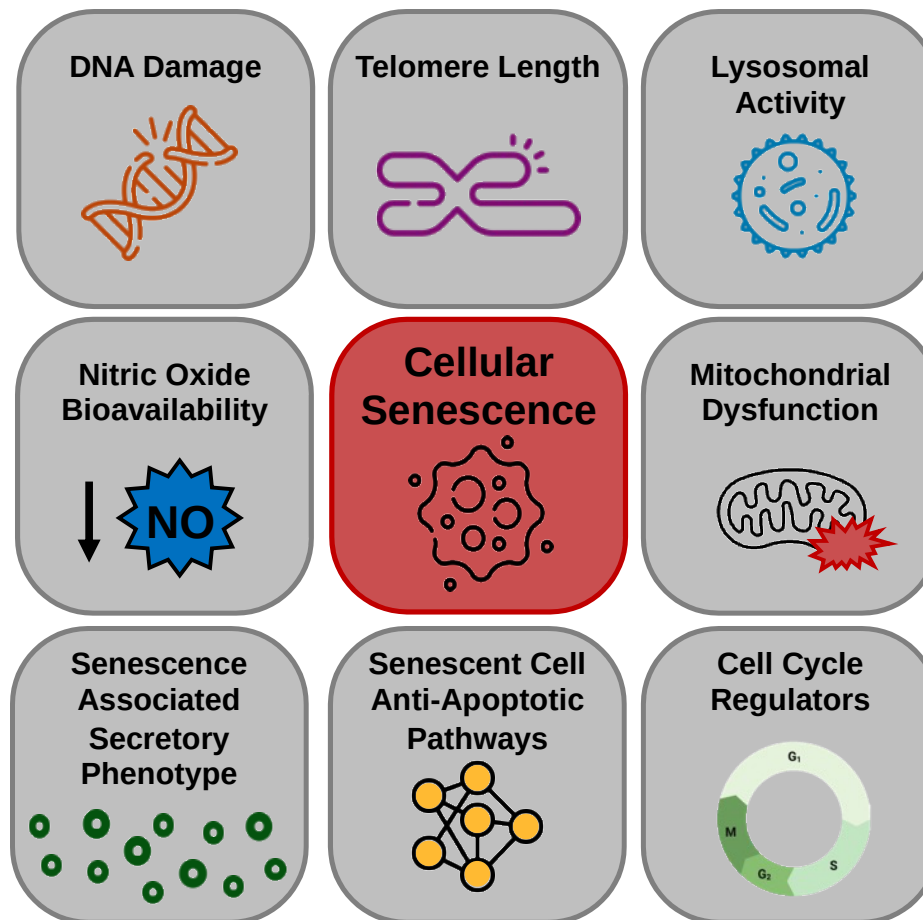


Figure 1. Characteristics of a vascular senescent cell. Vascular cells including endothelial cells and vascular smooth muscle cells undergo phenotypic changes up cellular senescence induction. These senescent cell characteristics are leveraged to identify senescent cells in the vasculature.

Established and emerging mechanisms underlying cellular senescence-induced vascular dysfunction.

Established mechanisms of vascular senescence.

Cellular senescence has recently been causally established as a contributor of age-related vascular dysfunction. Many of the underlying mechanisms of vascular dysfunction and vascular cell senescence overlap providing unique therapeutic targets that improve vascular function by reducing senescent cell burden. Select underlying processes have been established supporting the

mechanisms by which senescent cells contribute to vascular dysfunction and serve as therapeutic targets for senescent cells⁴⁶. Here, we discuss studies which investigate the established mechanisms by which senescent cells contribute to vascular dysfunction then discuss putative mechanisms that have recently emerged as novel means by which senescent cells elicit their detrimental effects on vascular physiology (**Figure 2**).

Mitochondrial dysfunction. Mitochondrial dysfunction, characterized in part by reduced electron transport system function, impaired mitophagic capacity and elevated ROS production, is a key feature of vascular aging⁴⁷. Cellular senescence promotes mitochondrial dysfunction in both an autocrine and paracrine manner as a result of the pro-inflammatory SASP^{48,49}. In senescent cells, mitochondria undergo outer membrane permeabilization which releases mitochondrial DNA into the cytosol⁵⁰. In turn, cytosolic mitochondrial DNA activates the cGAS–STING pathway, a major regulator of the SASP⁵⁰. Mitochondrial-derived ROS contribute to the vascular dysfunction with aging as evidenced by a reduction in mitochondrial ROS, reduced vascular mitochondrial pro-oxidant markers, and higher abundance of mitochondrial antioxidant defense following systemic senescent cell clearance^{13,51}. Although cellular senescence and mitochondrial dysfunction are both considered hallmarks of aging the bidirectional relationship between these processes further highlights the multiplicity of benefits of targeting senescent cells. Given the essential role of the mitochondria in senescent cells and in regulating the SASP, dysfunctional mitochondria may serve as important therapeutic target to mitigate the detrimental effects of cellular senescence and the SASP on vascular function.

Telomere shortening and dysfunction. Telomeres, the DNA sequences and proteins that protect the caps of chromosome, shorten with every replication cycle and are maintained by the enzyme telomerase⁵². Loss of telomere due to excess shortening or insufficient telomerase

expression in an established cause of stress-induced senescence⁵². Endothelial cells specifically have high replicative capacity and as such are prone to undergoing telomere shortening-induced senescence⁵². In addition to shortening, telomere dysfunction is also stimulated by oxidative stress to directly mediate vascular cell senescence⁴⁰. Specifically, endothelial cell telomere dysfunction-induced senescence impairs endothelial function through increased inflammatory signaling and oxidative stress in young mice⁴⁰. Further, telomere length is inversely associated with vascular elasticity and CVD events starting in people in mid-life⁵³. More studies are warranted to determine the direct implications of telomere function/length with arterial stiffening and to extend these findings to people.

Shear stress. Endothelial cells are constantly exposed to shear stress created by blood flow. This unique environmental stimulus has pathophysiological implications, which are regional, as blood flow patterns vary widely across the vascular network. At vascular curvatures, branches, and bifurcations, low and oscillatory shear stress impairs endothelial function and is associated with an inflammatory phenotype. Conversely, high and laminar shear stress tends to be vasoprotective⁵⁴. In relation to cellular senescence, regions of the circulation exposed to low and oscillatory shear stress exhibit elevated markers of cellular senescence and stimuli such as DNA damage and telomere dysfunction when compared to cells that experience high and laminar shear stresses^{40,55-57}. Importantly, these regions are also prone to the development of atherosclerosis^{55,57-59}. Mechanistically, low and oscillatory shear stress induces cellular senescence by increasing cell turnover which is likely exaggerated in the vasculature compared to other cells in the body^{59,60}. However, in regions exposed to low and oscillatory shear stress, endothelial cells are pushed to divide more rapidly, and clonal hyperplasia of VSMC contributes to atherosclerotic plaque development⁶¹⁻⁶³. In endothelial cells of large arteries, the replicative

burden is not mitigated by circulating progenitors as cell regeneration requires mitotic cell division by neighboring cells⁶⁴. Thus, the vasculature is comprised of cells that are long-lived and therefore enduring long-term exposure to low, oscillatory blood flow and senescence-inducing circulating factors. At vascular bifurcations, however, replicative stress may promote states of cellular senescence.

ECM remodeling. As previously mentioned, arterial stiffening is a key pathophysiological process of vascular aging. Senescent vascular cells contribute to arterial stiffness in various capacities including VSMC stiffness and unfavorable ECM remodeling. Particularly, senescent VSMC have a higher secretion of ECM proteins and are stiffer and less organized than the ECM derived from young control VSMC²⁷. Pro-stiffening changes in the ECM of VSMC is mediated by altered collagen deposition and reduced contractility in senescent VSMC²⁷. Sry-box transcription factor 9 (SOX9), a regulator of chondrocyte differentiation, is as a central mediator of ECM stiffness and organization in senescent VSMC²⁷. Mitochondrial-derived ROS also contributes to the vascular dysfunction with aging as evidenced by a reduction in mitochondrial ROS, reduced vascular mitochondrial pro-oxidant markers, and higher abundance of mitochondrial antioxidant defense following systemic senescent cell clearance^{13,51}. Importantly senescent cell clearance results in a reduction in collagen and AGE abundance in the vascular wall of old mice^{13,65}. As such, the vascular ECM is an important contributor to cellular senescence-induced arterial stiffening and a therapeutic target to prevent or reverse unfavorable vascular remodeling.

The circulating milieu. The influence of the circulating milieu (i.e., the collection of bioactive factors in systemic circulation) on physiological function has been studied using a variety of experimental approaches including heterochronic parabiosis (surgical joining of two

circulatory systems)⁶⁶, heterochronic blood exchange (administration of blood/blood factors of one organism into the circulating milieu of another organism)⁶⁷ and *in vitro* or *ex vivo* exposure of cells^{68,69} or organ systems^{13,70}, respectively, to distinct components of the circulation (e.g., blood/plasma/serum). Collectively, these experimental approaches have demonstrated that age-related vascular dysfunction can be transferred via the circulating milieu such that the aged circulating milieu promotes cellular senescence in a variety of cell/tissue types in young animals^{13,67,70}. Notably, age-related vascular dysfunction is prevented in arteries exposed to the circulating milieu collected from old animals following senescent cell clearance⁷¹. Individual circulating factors, including nutrients, metabolites, and hormones are also capable of inducing cellular senescence. A large body of evidence from a variety of models suggests that glucose, amino acids, lipids, and insulin are capable of provoking senescent states (reviewed in detail elsewhere)⁷². Together these results demonstrate that the circulating milieu mediates vascular aging, and this may occur, in part, via cellular senescence and changes in the circulating SASP. However, future studies are warranted to determine a mechanistic role of the circulating SASP factors implicated in vascular dysfunction^{73,74} that are responsible for increasing cellular senescence in the vasculature and transducing the vascular aging phenotype.

Dysregulated Immune Responses. An additional characteristic that predisposes vascular cells to damage is their interaction with the interstitial microenvironment and neighboring cells. Inflammatory immune cells traffic through the vascular network, and endothelial cells mediate extraversion from the circulation to the extravascular space⁷⁵. Endothelial cells can upregulate expression of adhesion molecules and secrete inflammatory chemokines to bind and recruit immune cells, respectively, and inflammatory signals increase endothelial permeability to allow for immune infiltration to neighboring tissues⁷⁵. Similarly, inducing cellular senescence in

VSMC increases vascular immune cell infiltration⁷⁶. Importantly, increased senescent vascular and immune cells manifest in atherosclerotic plaques and exacerbate atherosclerotic pathology via dysregulation of immune responses exerted by the SASP⁷⁷.

Cell-Cell Contact. Unfavorable cell-cell contact manifests in senescent vascular cells as increased expression of adhesion molecules and reduced expression of junction proteins^{78,79}. As previously mentioned, senescent endothelial cells have increased expression of adhesion molecules on the cell surface which attracts neighboring cells or cells in the circulation to local senescent cells⁷⁹. Despite this improved contact, senescent cells leverage this mechanism to promote the senescent program in their microenvironment rather than for the enhancement of favorable communication⁷⁹. In contrast, junction proteins, which are transmembrane proteins that aim to transfer material between cells, are reduced and impair cell-cell communication. Given the key role of endothelial cells in forming barriers to regulate the transport of factors from circulation to peripheral tissues, the dysregulation of junction proteins in senescent vascular cells leads to an impairment of transport of essential nutrients and oxygen⁷⁸. Specifically, dysfunctional barrier function in senescent endothelial cells has been observed in the brain⁸⁰, lung⁸¹, and eye⁸² and may result in dysregulated permeability of factors into these tissues⁷⁸. As such, senescent cells contribute to the established link between increased vascular permeability and vascular dysfunction⁸.

Oxygen Partial Pressure. Finally, it is well established that culturing cells at higher oxygen tensions hastens the onset of cellular senescence, and the partial pressure of oxygen is 97-44%, which is greater in the vascular circulation when compared to the interstitial space of other tissues⁷². Partial pressure of oxygen varies from blood to tissues to cells, and exposure to higher partial pressures of oxygen *in vivo* likely promotes cellular senescence through production

of ROS^{83,84} produced endogenously by vascular cells, or by circulating immune cells, neighboring tissues, or through iatrogenic means.

Potential novel mechanisms of vascular senescence.

As previously mentioned, the accumulation of senescent cells mediates age-related vascular dysfunction, and clearance of excess senescent cells improves vascular function in old mice^{13,85}. However, the direct mechanisms by which senescent cells promote vascular aging are not fully understood. In the following section, we describe potentially novel and understudied mechanisms that could regulate cellular senescence-mediated vascular aging (**Figure 2**).

Although causal evidence may not exist for all these mechanisms in contributing to cellular senescence-mediated vascular aging, the processes warrant mention given that they may provide insight into future experimental considerations.

Circadian misalignment. Circadian rhythms are approximately 24-hour biological oscillations which have evolved in response to predictable environmental cues⁸⁶. All cells, including cells of the vasculature, contain precisely regulated and conserved transcriptional-translational feedback loops which perpetuate the 24-hour oscillations and ultimately regulate fundamental circadian rhythms (also referred to as clock components)⁸⁶. *Bmal1* is the only imperative clock gene required for circadian rhythm regulation⁸⁷ and is considered the master regulator of the circadian clock⁸⁸. Moreover, the BMAL1 protein is a key transcription factor for rhythmic control of chromatin accessibility; thus, influencing system level gene expression⁸⁹. Circadian misalignment is associated with early vascular aging in humans⁹⁰. The physiological oscillations in *Bmal1* transcription are dampened with aging⁹¹ and knockout of *Bmal1* can directly impair vascular function in young mice⁹². Moreover, *in vitro* evidence suggests dampened *Bmal1* oscillations contribute to the accumulation of senescent cells⁹³. Thus, circadian

misalignment, particularly dampened oscillations in *Bmal1* transcription, may be a novel mechanism underlying cellular senescence-induced vascular aging.

Glycolytic byproducts. As mentioned above, mitochondrial dysfunction is a key characteristic of senescent cells. Mitochondrial dysfunction is often accompanied by a compensatory reliance on glycolysis for ATP production⁹⁴, and consistent with this concept, senescent cells demonstrate a preferential reliance on glycolysis⁹⁵. Considering mitochondrial dysfunction is a well-established biological hallmark of aging⁹⁶ and dysfunctional mitochondria are directly implicated in vascular aging⁴⁷, it is warranted to investigate the role of increased glycolytic flux as a mechanism mediating cellular senescence-induced vascular aging. Serum proteomic analyses reveal that increased glycolytic metabolism is likely responsible for circulating SASP-mediated changes in vascular aging¹³. Indeed, increased glycolytic flux is implicated in vascular aging, due in part to increased production of the glycolytic byproduct and AGE precursor, methylglyoxal (MGO)⁹⁷. Under normal physiological conditions, MGO is detoxified via the glyoxalase-1 (GLO-1) pathway; however, GLO-1 activity is reduced with aging, resulting in a greater abundance of MGO⁹⁸. In further support of this concept, MGO has shown to stimulate cellular senescence in human keratinocytes⁹⁹ and in the brain¹⁰⁰, and increasing GLO-1 activity has been demonstrated to reduce senescent cell burden in renal cells¹⁰¹. Moreover, preservation of GLO-1 activity prevents age-related vascular endothelial dysfunction¹⁰². Thus, there is likely merit to investigating the role of glycolysis and MGO/GLO-1 balance in mediating cellular senescence-induced vascular aging.

Membrane contact sites. Changes in membrane contact sites are emerging as biological mechanisms of aging, which has been reviewed in detail elsewhere¹⁰³. Membrane contact sites comprise of physical bridges which allow for exchange of metabolites between organelles,

ultimately regulating intracellular metabolic flux. Moreover, membrane contact sites create subcellular signaling platforms, which allow for local protein modifications and interactions¹⁰³. Two organelles that have garnered attention in the context of membrane contact sites are mitochondria and endoplasmic reticulum (ER), referred to herein as mitochondria-ER contacts (MERCs). MERCs are involved in the regulation of inflammation and redox signaling, and thus have been implicated in age-related pathologies, including adverse vascular remodeling¹⁰³. Exposure of endothelial cells in culture to oxidized low-density lipoprotein, an established inducer of cellular senescence¹⁰⁴, results in an upregulation of MERCs¹⁰⁵. Moreover, artificial tightening of MERCs in human fibroblasts leads to the induction of cellular senescence, a heightened pro-inflammatory secretory profile and increased ROS production¹⁰⁶. Inversely, the deletion of MERC-related proteins in senescent cells has shown to reduce overall senescence burden and favorably modulate the SASP^{107,108}. These results suggest there is a clear relation between MERCs and cellular senescence, but the extent to which MERCs mediate cellular senescence (or vice versa) are not well understood. Furthermore, it remains to be determined if MERCs regulate the SASP and how MERCs and cellular senescence and the SASP interact to mediate vascular aging.

Endothelial-to-mesenchymal transition. The endothelial-to-mesenchymal transition (endoMT) is a process by which endothelial cells lose their characteristic markers, such as VE cadherin, CD31, TIE2, and Von Willebrand factor (VWF), while gaining increased expression of mesenchymal markers, such as alpha 2 smooth muscle actin (α -SMA), fibroblast-specific protein-1 (FSP1), and type I/III collagen, which ultimately leads to a shift toward enhanced contractile properties and reduced NO production¹⁰⁹. The endoMT is commonly provoked by states of excessive inflammation and is highly implicated in vascular aging¹⁰⁹; however, sources

of inflammation mediating the endoMT are not fully understood. Senescent endothelial cells are more susceptible to undergoing endoMT and the pro-inflammatory SASP may drive the endoMT in non-senescent endothelial cells¹¹⁰. An emerging pro-inflammatory SASP factor thought to promote the endoMT is transforming growth factor- β (TGF- β), as it is a common component of the SASP¹¹⁰ and an established inducer of the endoMT¹⁰⁹. Although physiological levels of TGF- β are required for homeostatic endothelial function¹¹¹, excessive production of TGF- β can promote vascular aging^{112,113}. As such, the endoMT may be a mechanism by which cellular senescence and the SASP contribute to vascular aging.

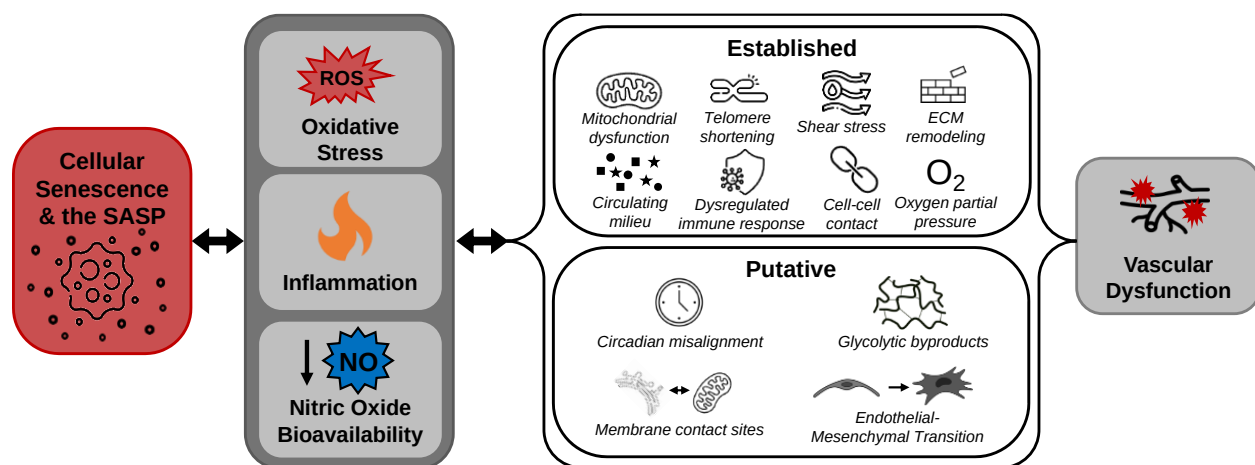


Figure 2. Mechanisms by which cellular senescence contributes to vascular dysfunction.

Excess senescent cell and senescence-associated secretory phenotype (SASP) burden drive macro-mechanistic processes including reactive oxygen species (ROS)-related oxidative stress, chronic inflammation, and reduced nitric oxide (NO) bioavailability. In turn, these fundamental aging processes that promote other established and putative underlying mechanisms that contribute to vascular dysfunction. ECM = extracellular matrix; ER = endoplasmic reticulum.

Groups with high cellular senescence burden.

Cellular senescence increases with chronological age and due to other intrinsic and extrinsic stressors. As such, senescent cells accumulate disproportionately in different individuals and groups and may contribute to vascular health disparities. Conditions of early vascular aging – states in which biological age is greater than chronological age – also often demonstrate increased senescent cell accumulation¹¹⁴. In this section we highlight select groups may experience early vascular aging and a greater senescence burden due to chronological aging, lifestyle choices, genotoxic drugs, genetic mutations, or disease progression¹¹⁴ (**Figure 3**).

Estrogen-deficient postmenopausal women and men with low testosterone. Aging is characterized in part by a decline in gonadal function, which presents in late life as estrogen deficiency in women and low testosterone levels in men, both of which are risk factors for the development of CVD¹¹⁵. Vascular function declines in parallel with gonadal aging in women and emerging evidence suggests this may also be the case for older adult men with low testosterone^{116,117}. However, the mechanisms by which gonadal aging mediates vascular dysfunction are incompletely understood. Physiological levels of estrogen maintain homeostatic redox balance in mitochondria and, as such, estrogen deficiency can increase tonic mitochondrial ROS production. Considering mitochondrial ROS is an established inducer of cellular senescence, it is plausible that vascular aging in estrogen deficient women is mediated by an increase in cellular senescence. Moreover, in a mouse model with accelerated senescent cell accumulation, estrogen deficiency exacerbates vascular aging¹¹⁸, demonstrating an interaction between gonadal status, cellular senescence and vascular aging. Low testosterone levels have long been associated with increased tonic ROS production and chronic low-grade inflammation; therefore, it appears likely that testosterone also regulates cellular senescence. Estrogen- and

testosterone-based hormone replacement is contraindicated for CVD prevention for a variety of reasons; thus, rather than replacing hormone levels to improve vascular function in estrogen deficient postmenopausal women and in men with low testosterone, it may be more beneficial to target the mechanisms (potentially cellular senescence) by which the reductions in these hormones drive vascular aging.

Lifestyle behaviors. Certain detrimental lifestyle behaviors may also lead to cellular senescence accumulation and early vascular aging. In preclinical models, a high-fat diet promotes vascular remodeling and stimulate vascular cell senescence¹¹⁹. This is in agreement with clinical studies observing that exercise-trained mid-life/older adults have less endothelial senescent cell burden than sedentary mid-life/older adults⁴². Additional detrimental lifestyle behaviors that have been linked to increased vascular cell senescence include cigarette smoking¹²⁰ and poor sleep quality¹²¹. However, more studies are needed to establish the mechanisms and implications of the lifestyle behaviors on cellular senescence and early vascular aging and determine if healthy lifestyle behaviors promote healthy vascular aging.

Mental stress. Mental stress has emerged as an important risk factor for the development of CVD^{122,123} as it promotes early vascular aging phenotypes¹²⁴, as reviewed in detail elsewhere^{122,123,125}. In brief, there are many shared biological processes between the effects of mental stress and the mechanisms underlying age-related vascular dysfunction. Mental stress is associated with reduced NO bioavailability, impaired endothelial function, tonic elevations in ROS, and chronic low-grade inflammation¹²⁴, all of which are key mechanisms of vascular aging and features of excess cellular senescence^{11,126}. Preclinical literature suggests that experimentally provoked mental stress can promote cellular senescence in the brain¹²⁷, and patients who experience frequent episodes of mental stress have higher circulating markers of cellular

senescence relative to healthy controls¹²⁸. Post-traumatic stress disorder (PTSD) is a severe type of mental stress and a psychiatric/mental condition affecting approximately 5% of the US population^{129,130}. Individuals most affected by PTSD include military combat veterans, victims of assault and abuse, and those that have experienced major disasters¹³¹. PTSD is associated with reduced overall and healthy lifespan, inferring PTSD accelerates biological aging¹³². As such, the relation between PTSD and the biological mechanisms of aging have gained significant attention over the last decade. Interestingly, a comprehensive review compared SASP markers from numerous studies measuring individual factors in people with PTSD and found multiple lines of investigation suggesting that PTSD may be associated with a pro-inflammatory SASP profile compared to age-matched individuals without PTSD¹³³. However, it remains to be determined if PTSD and mental stress directly increase the burden of cellular senescence in the vasculature and if the accumulation of senescent vascular cells is mechanistically implicated in mental stress-induced vascular aging.

Radiation. Radiation is a well-established inducer of cellular senescence that occurs in diverse settings and experienced most commonly by cancer patients via ionizing radiation during their cancer treatment and by space travelers through high solar and cosmic radiation¹³⁴. During radiation exposure, high-energy rays induce DNA damage leading to the induction of cellular senescence, which has been well documented in endothelial cells and VSMCs^{134,135}. Specifically, exposure of vascular cells to ionizing radiation induces cellular senescence and activates SASP-mediated inflammation¹³⁵. In mice, ionizing radiation exposure has shown to mediate DNA-damage induced vascular cell senescence and inflammation, and accelerated plaque formation in atherosclerotic mice¹³⁶. Moreover, cancer survivors that received cancer treatment with ionizing radiation had 2-fold higher arterial stiffness in comparison to cancer patients who did not receive

ionizing radiation¹³⁷. Together, these data provide evidence that radiation induces vascular cell senescence and promotes early vascular aging.

Chemotherapy treatment. Common forms of cancer, including breast cancer, leukemia and lymphomas are often treated with anthracycline chemotherapeutic agents^{138,139}. Anthracyclines are highly effective at treating the cancer but are considered to be the most CV toxic with patients treated with anthracyclines experiencing early CV-related morbidity and often die prematurely of CVD^{138,139}. A key mechanism by which anthracyclines inhibit cancer growth is through the induction of cellular senescence¹⁴⁰, and cancer survivors previously treated with anthracyclines experience heightened (relative to age- and sex-matched healthy controls) senescent cell burden years into survivorship¹⁴¹⁻¹⁴³. Considering the link between vascular dysfunction and overt CVD, as well as the role of cellular senescence in mediating vascular aging, it is highly plausible that cellular senescence-induced vascular aging is responsible for the early CV-related morbidity and mortality observed in these patients. Doxorubicin, the most commonly administered anthracycline chemotherapeutic agent, is a well-established inducer of cellular senescence and early vascular aging¹⁴⁴. Specifically, young mice that received systemic, genetic clearance of senescent cells following doxorubicin treatment do not develop vascular dysfunction compared to doxorubicin treated mice that did not receive genetic senescent cell clearance¹⁴⁵. Other classes of chemotherapeutic agents are known to elicit cellular senescence and require further investigation into adverse effects on the vasculature¹⁴⁶. Specifically, paclitaxel, a commonly used microtubule inhibitor, has direct adverse effects on the proliferation and migration of senescent vascular cells and contributes to disruption of blood brain barrier integrity^{147,148}. Together, these data suggest that cellular senescence is likely an underlying mechanism of cancer chemotherapy-induced early vascular aging.

Antiretroviral therapy treatment. Although antiretroviral therapies are highly effective at treating the fatal effects of the sexually-transmitted human immunodeficiency virus (HIV) infection, long-term use of these therapies promotes early vascular aging and accelerated senescence accumulation¹⁴⁹. Antiretroviral therapies inhibit the reverse transcription utilized by HIV viruses to infect cells¹⁵⁰. Like chemotherapeutic agents, antiretroviral therapies are considered to mediate stress-induced premature senescence¹⁵¹. Studies in cultured endothelial cells showed that antiretroviral therapies elicits premature senescence associated with inflammation, oxidative stress and altered eNOS activation¹⁵². Additional preclinical studies in mice demonstrated that antiretroviral therapies promote states of cellular senescence by impairing mitochondrial function, defecting nuclear wall structure, and promoting the SASP, which was accompanied by physiological dysfunction¹⁵¹. In people, associations have emerged between HIV infection, antiretroviral therapy, and cellular senescence burden^{150,153}. However, the direct effects of antiretroviral therapies on vascular function of HIV-infected patients are still yet to be determined.

Vascular disease. The presence of clinical vascular disorders and associated risk factors may be another contributor to of early vascular aging by increasing cellular senescence. Increased senescent cell burden has been observed in several clinical vascular disorders including atherosclerosis^{154,155}, pulmonary hypertension^{38,156}, and aortic aneurysms^{157,158}, however it is unclear whether senescent cells are primary or secondary to the onset of these conditions. Given that senescent cell accumulation is only associated with clinical vascular diseases, preclinical models are currently being leveraged to better understand the implications of cellular senescence in these conditions. Below we highlight studies in which senescent cells play a role in vascular pathologies.

Senescent cells have been identified in atherosclerosis, a progressive disease defined by the accumulation of lipid and fibrous elements in the large elastic arteries³⁵. Senescent cells exist in atherosclerotic lesions and contribute to the destabilization of plaques that may promote acute CV events^{44,159}. VSMCs derived from plaques exhibit key senescent cell characteristics including DNA damage and telomere shortening¹⁵⁵. Further, SASP factors have been shown to aggravate atherosclerosis through the recruitment and activation of immune responses and promoting chronic inflammation and oxidative stress¹⁶⁰. Pulmonary arterial hypertension is a proliferative disorder that results mainly from increased vascular cell production and contributes to CVD events and chronic lung diseases³⁸. Cellular senescence contributes to pulmonary vascular remodeling which ultimately manifests in vascular dysfunction in the lung as signified by increased right ventricular systolic pressure^{38,156}. Yet, senescent cells may play a multifaceted role in the stability of the pathology by inhibiting the worsening of the pulmonary hypertension state (discussed further in this review)³⁸. As such, more studies are required to determine the cost-benefit of senescent cell burden in pulmonary arterial hypertension states. Senescent cells in aortic aneurysms, characterized by a persistent, localized dilation (ballooning) of the aorta, contribute to a loss of aortic structural wall integrity, which may lead to severe CVD events^{157,161}. Proteomic analyses have identified cellular senescence as a key pathway in thoracic aortic aneurysms¹⁵⁸. Senescent cells also contribute to aortic aneurysm development through the secretion of proinflammatory cytokines and ROS through the SASP¹⁶¹. Senescent cells accumulate primarily at sites of vascular hypertrophy in pulmonary hypertension models¹⁶² and contribute to the pathology through the recruitment of inflammatory cells by the SASP and by promoting vascular cell growth and migration^{38,156}. Further mechanistic studies are needed to uncover the complex interaction between cellular senescence and vascular disease.

Progeria. Hutchinson-Gilford progeria syndrome (HGPS) is an extremely rare, premature aging disorder caused by the genetic mutation of the *Lmna* gene resulting in a defective nuclear lamina¹⁶³. In patients with HGPS, there is greater senescent cell burden and vascular dysfunction that manifests at young ages, with CVD being the leading cause of the death in this group¹⁶⁴. Mouse models of progeria have been developed to study disease progression and have also been adapted to study premature aging^{12,165}. For example, the ERCC1 knockout mouse model depletes the enzyme responsible for multiple DNA repair mechanisms that protect the nuclear genome, resulting in progeria disease progression¹⁶⁵. At young ages, ERCC1 knockout progeroid mice have increased cellular senescence burden in all tissues comparable to old mice, with the aorta having the greatest burden¹². Like HGPS patients, the ERCC1 knockout progeria model exhibits impaired vascular function even at young ages¹⁶⁶. However, more studies are required to determine if these vascular compromises are directly mediated by cellular senescence burden.

Another progeroid syndrome, Werner syndrome, is defined by a mutation in the *Wrn* gene which encodes the WRN RecQ helicase¹⁶⁷. This helicase has a variety of functions, including DNA replication and repair, telomere maintenance, and epigenetic regulation, such as heterochromatinization¹⁶⁷. Unlike HGPS, Werner syndrome patients develop normally until after puberty after which many age-related phenotypes manifest including hair graying, sarcopenia, skin abnormalities, cataracts, atherosclerotic disease and some cancer with most patients succumbing to one or more malignancies in their fifth decade of life¹⁶⁷. CVD is a major cause of morbidity and mortality in these patients as they demonstrate dyslipidemia, hypertension and atherosclerosis at a very early age¹⁶⁸. One of the most profound phenotypes known is that of accelerated induction of cellular senescence during replicative culture collected from Werner syndrome patients¹⁶⁹. *In vitro* models suggest that macrophages from Werner syndrome patients

co-cultured with endothelial cells and VSMC are sufficient to induce endothelial cell dysfunction and a synthetic VSMC phenotype¹⁷⁰. Future studies are required to further document the direct contribution of vascular cell senescence in vascular dysfunction and CVD progression in Werner syndrome patients and in preclinical models of the disorder.

Down syndrome. Down syndrome is a genetic disorder caused by the presence of all or part of a third copy of chromosome 21 (trisomy 21; +21). It is not only associated with intellectual disability but also with a group of clinical manifestations of premature aging. These include but aren't limited to Alzheimer's disease, cancer, hearing and vision declines and suppressed immune system function. Like progeria conditions, CVD is prevalent in people with Down syndrome and is the leading cause of death¹⁷¹. One key insight is that people with Down syndrome have blunted endothelial function compared to age matched controls¹⁷². Furthermore, initial studies suggest alterations in vascular function may contribute to some of the age-related diseases in this group. Decreased cerebral blood flow has been observed prior to β amyloid accumulation in mouse models of Down syndrome^{173,174}. Moreover, individuals with Down syndrome demonstrate less carotid blood flow and vascular conductance when compared with individuals without Down syndrome, suggesting that alterations in vascular control mechanisms are present in Down syndrome¹⁷⁵. Despite this, little is known about how vascular cell senescence and function is altered with aging in this group. While measures of cellular senescence have not been made in vascular cell types, cellular senescence markers are high in neural progenitor cells¹⁷⁶, primary fibroblasts¹⁷⁷ and primary lymphocytes¹⁷⁸ from people with Down syndrome. Thus, future studies should evaluate vascular cell senescence in people with Down syndrome and further explore if vascular cell senescence is a contributor to vascular dysfunction and elevated CVD risk in people with Down syndrome.

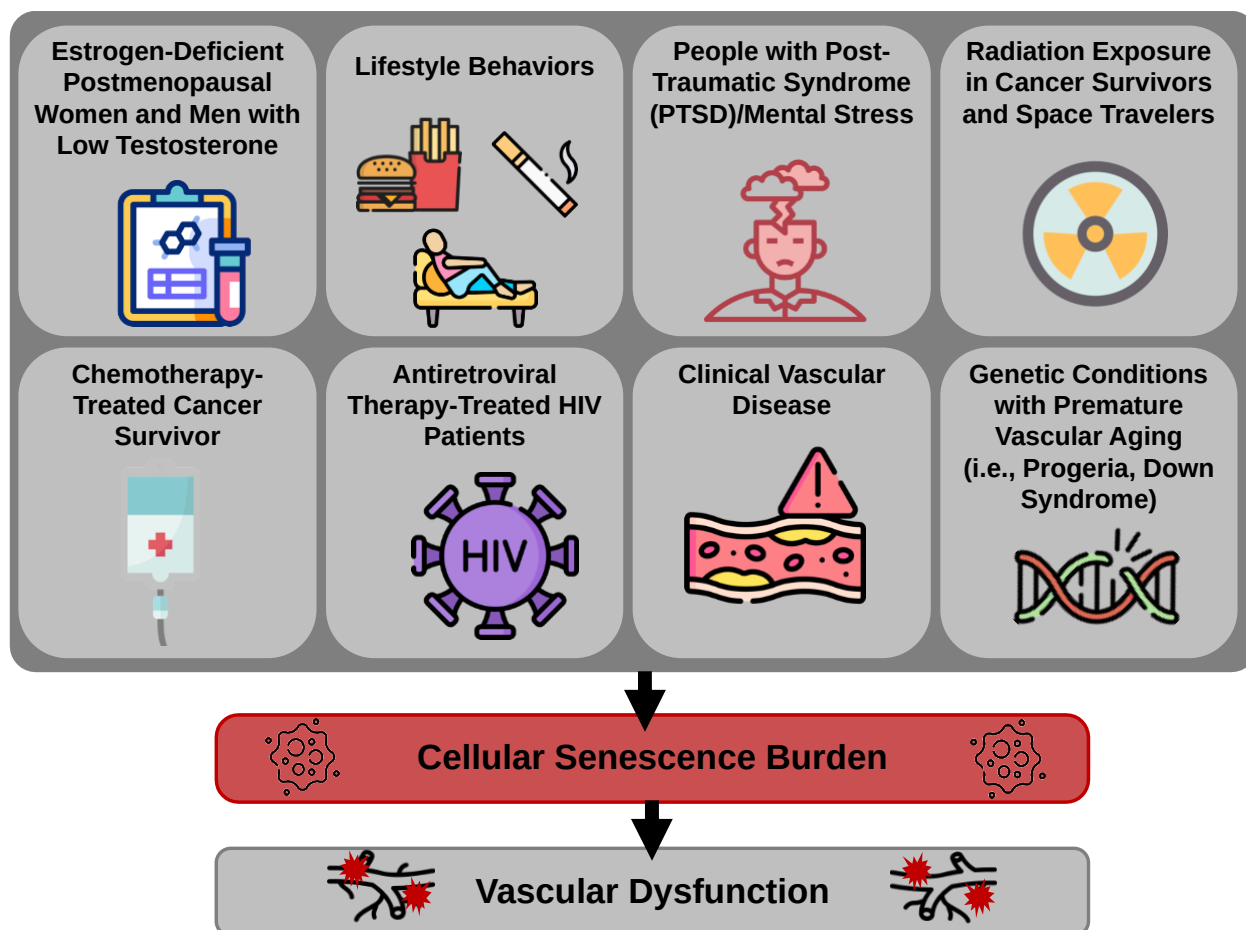


Figure 3. Select groups with high cellular senescence burden and premature vascular aging. Groups that may have high senescent cell burden often experience early vascular aging phenotypes. HIV = human immunodeficiency virus; CVD = cardiovascular disease.

Senotherapeutics and translation to clinical research to improve vascular function.

The select groups discussed above represent groups that experience early vascular aging likely due in part to excess cellular senescence. As such, the groups stand to benefit the greatest from the clearance of excess senescent cells. In the following section, we discuss therapeutic strategies that may be leveraged to reduce vascular cell senescence and the SASP and improve vascular function. Senotherapeutics are a broad class of therapeutic strategies aimed at targeting excess senescent cells and/or the SASP¹⁷⁹. Senotherapeutics often refer to natural or synthetic pharmacological agents but have also been attributed to lifestyle interventions such as aerobic

exercise and weight loss in people^{42,179-181}. There are two main classes of pharmacological senotherapeutics: senolytics and senomorphics¹⁷⁹. Senolytic therapies are interventions that selectively suppress excess senescent cells by targeting SCAP networks⁴⁶, whereas senomorphics are compounds that suppress SASP signaling by modulating senescent cell transcription¹⁸². The use of senolytics or senomorphics in pathological states is challenged and highly dependent on the context (discussed in further detail later in the review)¹⁷⁹. Here, we examine evidence of senotherapeutics treatment in vascular studies (**Table 1**) and discuss benefits and drawbacks of the senotherapeutic strategies. Although several senolytic therapies are currently being tested in clinical trials, most evidence of senolytic activity on vascular function is reported in cell culture and preclinical models.

Senolytics.

Select well-established and experimental senolytic therapies exist that suppress senescent cells by targeting SCAP networks to allow apoptosis to resume in senescent cells such that they are cleared from the tissue¹⁸³. Senolytics therapies are often administered using a “hit-and-run” approach which is also often referred to as an intermittent dosing paradigm¹⁸³. This approach allows high doses of senolytics to be administered such that excess senescent cell burden can be suppressed back to basal levels. Once senescent cells reaccumulate (~2-4 weeks later in mid-life/older adults), another dosing round is administered. This approach is thought to allow for senescent cells to maintain their physiological roles (*i.e.*, cancer suppression) while keeping levels and pathological effects to a minimum^{11,29,32}. Using this dosing approach, senolytic therapies have been shown to reduce age-related senescent cell burden and improve vascular function with chronological aging, early aging, and CVD (**Table 1**).

Navitoclax. Initial studies investigating the role of cellular senescence in vascular dysfunction used one of the first senolytics to be established, navitoclax (also known as ABT-267 and ABT-737). Navitoclax is one of the most potent and broad-spectrum senolytic agents to be identified to date¹⁸⁴. However, due to its chemotherapeutic properties, its translation to healthy mid-life/older adults is limited¹⁸⁵. As such, navitoclax is primarily utilized as a “theoretical” senolytic in basic biological and preclinical studies. Preclinical studies have shown that navitoclax-induced senescent cell clearance improves endothelial function and lowers arterial stiffness in old mice¹³, but may have detrimental effects on mouse models of vascular pathologies including atherosclerosis^{36,186-188} and pulmonary hypertension^{38,156}.

Dasatinib + Quercetin. The combination senolytic treatment, D + Q, is a well-established preclinical intervention that is currently being tested in clinical populations^{189,190}. Although not related to the vasculature, outcomes from the first completed clinical trials using D + Q have shown that it may reduce cellular senescence markers and select circulating SASP factors in patients with idiopathic pulmonary fibrosis¹⁸⁹ and Alzheimer’s disease¹⁹⁰. In the vasculature, D + Q improved vasomotor function and reduced arterial stiffness in old mice⁸⁵. Further, D + Q ameliorated plaques in atherosclerotic mice⁸⁵ and abdominal aortic aneurysms in angiotensin II-induced mice¹⁹¹.

Fisetin. Fisetin is a natural flavonoid with senolytic properties that is currently considered to be the safest compound for translation to mid-life/older adults¹⁹². In old mice, intermittent fisetin supplementation improved endothelial function, lowered arterial stiffness, and ameliorated vascular cell senescence burden⁶⁵. Several clinical trials are underway testing the effects of fisetin supplementation on chronic conditions including two studies that are focusing on vascular-related conditions including fisetin supplementation in healthy mid-life/older adults

on vascular function (NCT06133634) and in people with peripheral artery disease (NCT06399809).

Experimental senolytics. Finally, experimental senolytic interventions, including 25-hydroxycholesterol¹⁹³, FOXO4-DRI (a peptide agonist p53 and peroxisomal membrane protein 4 [PXO4] interaction)³⁸, bis-2-(5-phenylacetamido-1,3,4-thiadiazol-2-yl)ethyl sulfide (BPTES; a glutaminase 1 inhibitor)¹⁹⁴, and a vaccine against glycoprotein nonmetastatic melanoma protein B (GPNMB; a transmembrane protein inhibitor)¹⁹⁵, have demonstrated initial efficacy at reducing senescent cell burden and modulating vascular function or pathological vascular disorders in preclinical models. Preliminary studies of 25-hydroxycholesterol, an endogenous cholesterol metabolite, found that vascular SCAPs were modulated to favorably modulate vascular wall remodeling and reduce arterial stiffening in old mice¹⁹³. Notably, FOXO4-DRI worsened CV outcomes in animal models of pulmonary, arterial hypertension by promoting unfavorable vascular wall remodeling³⁸, whereas BPTES and GPNMB interventions both had favorable effects on plaque pathology in atherosclerotic animals^{194,195}. More testing is required to determine the mechanisms, safety, tolerance, and efficacy of both recognized and experimental senolytic agents in people.

Senomorphics.

Established senomorphic therapies include statin, antidiabetic, and anti-inflammatory compounds, which all represent natural or FDA-approved drugs¹⁷⁹. Senomorphics target SASP-mediated pathways in senescent cells by modulating central inflammatory regulators including the transcription factor NF- κ B and AKT pathway¹⁸². Though senomorphics are less established than senolytics, there may be contexts in which they are safer and more efficacious depending on pathological conditions (discussed in detail further in the review). Given the

limited investigations of these compounds for their senomorphic properties, we will highlight the existing preclinical and cell culture evidence in vascular tissue or cells (**Table 2**).

Statins. Statins have various protective effects on the vasculature including lowering cholesterol levels. However, certain protective effects may also be attributed to the inhibition of endothelial SASP production¹⁹⁶. For example, pitavastatin inhibits endothelial cell senescence through AKT-dependent mechanisms in both hydrogen peroxide-induced senescence in cultured endothelial cells and streptozotocin-diabetic mice¹⁹⁶.

Antidiabetic drugs. Several antidiabetic drugs have been found to have beneficial effects on vascular cell function in addition to their traditional uses on glucose regulation. Of note, it is believed that metformin and glucagon-like peptide 1 (GLP-1) agonist exert some physiological benefits with aging by suppressing SASP signaling. Metformin is a potent suppressor of chronic inflammation that blocks NF- κ B activity in doxorubicin-induced senescent endothelial cells¹⁹⁷. Likewise, agonists of the antidiabetic hormone GLP-1, including exendin-4 may mitigate cellular senescence induction and SASP signaling in senescent endothelial cells¹⁹⁸ and VSMC¹⁹⁹ in culture. In both cell types, exendin-4 inhibits signal transduction system that uses cyclic adenosine monophosphate (cAMP) to activate protein kinase A (PKA)^{198,199}. As such, antidiabetic drugs may promote their beneficial properties in part through senomorphic effects and may be promising candidates for repurposing to more general aging populations.

Anti-inflammatory drugs. Several natural and synthetic anti-inflammatory compounds have capabilities of suppressing circulating SASP factors and modulating SASP signaling in vascular cells. Aspirin is a widely prescribed nonsteroidal anti-inflammatory drug which has also been proposed to have putative senomorphic attributions²⁰⁰. Endothelial cells in culture have delayed replicative senescence, reduced oxidative stress, and enhanced NO bioavailability

following aspirin treatment²⁰⁰. Resveratrol is a natural polyphenol compound found in the skin of grapes that exerts cardioprotective effects by targeting excess oxidized low-density lipoprotein²⁰¹. Resveratrol treatment on replicative senescent endothelial progenitor cells in culture prevents the onset of cellular senescence through activation of AKT pathways²⁰¹. Further, VSMC isolated from old rhesus monkeys have lower SASP production after *in vitro* resveratrol treatment, an effect which is mediated by an inhibition of NF- κ B²⁰². Finally, the immunosuppressor rapamycin and its analogs have been assessed for their senomorphic properties in senescent vascular cells. In old mice, dietary rapamycin supplementation reverses age-related vascular dysfunction while modulating cellular senescence and SASP signaling pathways²⁰³. Rapamycin promotes endoMT during stress-induced premature senescence through the activation of autophagy²⁰⁴. Together, assessing the senomorphic properties of these anti-inflammatory drugs allow for the elucidation of the mechanisms-of-actions and optimal use of these compounds.

Table 1. Investigations of senotherapeutic agents in vascular studies.

Seno-therapeutic (class)	Dosing paradigm	Model	Condition	Senescence/ SASP Outcome	Vascular outcome	Reference
Navitoclax (senolytic)	50 mg/kg/day 1 week on – 2 weeks off – 1 week on	C57Bl/6 mice	Old age	↓ p16	↑ Endothelial function ↓ Arterial stiffness	Clayton et al., 2023
Navitoclax (senolytic)	100 mg/kg/day for 6 weeks or 50mg/kg/day for 9 weeks	Apoe KO mice	Atherosclerosis	↓ <i>Cdkn2a</i> ↓ p21	↓ Plaques stability	Karnewar et al., 2023
Navitoclax (senolytic)	50 mg/kg/day 1 week on – 3 weeks off for 3 cycles	p16-3MR Apoe KO mice	Atherosclerosis	No change <i>Cdkn2a</i> or SASP	↓ Atherosclerotic plaques	Garrido et al., 2022
Navitoclax (senolytic)	100 mg/kg/day for 9 days	p16-3MR Ldlr KO mice	Atherosclerosis	↓ SA- β -Gal and <i>Cdkn2a</i>	↓ Atherosclerotic plaques	Childs et al., 2016

Navitoclax (senolytic)	100 mg/kg/day 1 week on – 3 weeks off for 3 cycles	C57Bl/6 and INK-ATTAC Ldlr KO mice	Atherosclerosis	↓ SA-β-Gal	↓ Atherosclerotic plaques	Childs et al., 2021
Navitoclax (senolytic)	10 mg/kg/day for 8 days	Monocrotaline-treated Wistar rats	Pulmonary arterial hypertension	↓ p16 and p21	↓ RVSP	van der Feen et al. 2020
Navitoclax (senolytic)	10-50 mg/kg/day for 3 weeks	C57Bl/6 and SM22-5-HTT ⁺ mice and Wistar rats	Pulmonary arterial hypertension	↓ p16 and p21 ↓ DNA damage	↑ Vascular remodeling ↓ RVSP	Born et al., 2022
D + Q (senolytic)	5 mg/kg/day (D) + 10 mg/kg/day (Q) once weekly for 2 months	C57Bl/6 Apoe KO mice	Old age and atherosclerosis	↓ DNA damage	↑ Endothelial function ↓ Arterial stiffness ↓ Atherosclerotic plaques	Roos et al., 2016
D + Q (senolytic)	5 mg/kg/day D + 50 mg/kg/day Q 3 days on – 2 weeks off for 3 cycles	C57Bl/6 angiotensin II-induced mice	Old age and abdominal aortic aneurysm	↓ <i>Cdkn1a</i> and SASP	↓ Abdominal aortic aneurysm	Parvizi et al., 2021
Fisetin (senolytic)	100 mg/kg/day 1 week on – 2 weeks off – 1 week on	C57Bl/6 and p16-3MR mice	Old age	↓ p16, <i>Cdkn2a</i> and SCAP	↑ Endothelial function ↓ Arterial stiffness	Mahoney et al., 2024
25-Hydroxycholesterol (senolytic)	50mg/kg/day for 5 days	p16-3MR mice	Old age	↓ p16, <i>Cdkn2a</i> and SASP	↓ Arterial stiffness	Mahoney et al., 2023
FOXO4-DRI (senolytic)	5mg/kg/day 3X per week for 3 weeks	C57Bl/6 and SM22-5-HTT ⁺ mice	Pulmonary arterial hypertension	None reported	↓ RVSP	Born et al., 2022
BPTES (senolytic)	5 µg/kg/day 3X per week for 1 month	C57Bl/6 Apoe KO mice	Old age and atherosclerosis	↓ SA-β-Gal and SASP	↓ Atherosclerotic plaques	Johmura et al., 2026
GPNMB vaccine (senolytic)	Not reported	C57Bl/6 Apoe KO mice	Atherosclerosis	↓ <i>Cdkn1a</i> , <i>Cdkn2a</i> , and SA-β-Gal	↓ Atherosclerotic plaques	Suda et al., 2022
Pitavastatin (senomorphic)	10-100 nmol/L once	HUVEC	H ₂ O ₂ -induced cellular senescence	↓ SA-β-Gal	↑ eNOS, SIRT1, and catalase expression	Ota et al., 2010
Pitavastatin (senomorphic)	3 mg/kg/day daily for entire life	Streptozotocin-treated C57Bl/6 mice	Diabetes	↓ SA-β-Gal	↑ eNOS, SIRT1, and catalase expression	Ota et al., 2010
Metformin (senomorphic)	0.5-5 mM for 24h	HUVEC	Doxorubicin-induced senescence	↓ SASP secretion	↓ Endothelial adhesion molecules	Abdelgawad et al. 2023
Exendin-4 (senomorphic)	10 nmol/l for 72h	HUVEC	H ₂ O ₂ -induced senescence	↓ SA-β-Gal activity	None reported	Oeseburg et al. 2010
Exendin-4 (senomorphic)	0.01-100nM for 72h	Rat aortic VSMC	Angiotensin II-induced senescence	↓ SA-β-Gal activity	↓ ROS levels	Xhao et al. 2014

Aspirin (senomorphic)	100 μ M every 48h until the 12 th passage	HUVEC	Replicative senescence	↓ SA- β -Gal activity ↑ Telomerase activity	↓ ROS levels ↑ NO levels	Bode-Böger et al., 2005
Resveratrol (senomorphic)	10-50 μ M for 72h	HEPC	Replicative senescence	↓ SA- β -Gal activity ↑ Telomerase activity	↑ Cell proliferation and migration	Xia et al., 2008
Resveratrol (senomorphic)	1 μ mol/L for 48h	Rhesus monkey VSMC	Old age	↓ Cytokine profile ↓ SASP profile	↓ Mitochondrial ROS production	Csiszar et al., 2011
Metformin (senomorphic)	0.5-5 mM for 24h	HUVEC	Doxorubicin-induced senescence	↓ SASP secretion	↓ Endothelial adhesion molecules	Abdelgawad et al. 2023
Rapamycin (senomorphic)	14 mg kg ⁻¹ for 6 weeks	B6D2F1 mice	Old age	↓ p19	↑ Endothelial function and ↓ Arterial stiffness	Lesniewski et al., 2017
Rapamycin (senomorphic)	10 nM for 6, 24, 48, or 72h	HCAEC	H ₂ O ₂ -induced senescence	↓ SASP secretion ↓ SA- β -Gal	↑ EndoMT transition	Sasaki et al., 2020

Blue signifies senolytic compounds; green signifies senomorphic compounds. SASP = Senescence-associated secretory phenotype; KO = knockout; HUVEC = Human umbilical vein endothelial cells; HEPC = Human endothelial progenitor cells; VSMC = vascular smooth muscle cell; HCAEC = Human coronary artery endothelial cells; H₂O₂ = Hydrogen peroxide; SA- β -Gal = Senescence-associated β -galactosidase; SCAP = Senescent cell anti-apoptotic pathways; BPTES = bis-2-(5-phenylacetamido-1,3,4-thiadiazol-2-yl) ethyl sulfide; GPNMB = glycoprotein nonmetastatic melanoma protein B; RVSP = Right ventricular systolic pressure. eNOS = Endothelial nitric oxide synthase; ROS = Reactive oxygen species; NO = Nitric oxide; EndoMT = Endothelial-to-mesenchymal.

Senotherapeutic dosing.

Intermittent dosing of senolytic therapies has shown success in suppressing excess senescent cells while maintaining basal physiological function of senescent cells (i.e., cancer suppression, wound healing) in preclinical models^{11,29,32}. However limited studies have assessed the long-term effects of senolytic treatments and the number of treatment cycles has yet to be optimized. Current clinical trials testing senolytic therapies range from 2 to 6 cycles with generally 1-4 weeks of rest period in between dosing periods¹⁸⁹. This dosing paradigm is also optimal for patient burden given that medication is only administered on limited days¹⁸³. Given

the high plasticity of the vasculature (highly adaptable to change given a stimulus), senolytic therapies likely improve vascular function within a short period of time. However, tissues with less plasticity, such as the heart and skeletal muscle, may require longer dosing periods to observe improvements. Thus, future clinical trials should consider senolytic dosing paradigms based off clinical and functional outcomes **(Figure 4)**.

Senomorphics, on the other hand, may require continuous, low-dose administration, given the constant production on the SASP. However, more preclinical studies are required to elucidate the ideal dosing paradigm¹⁷⁹. Further, given the diversity of the SASP and SASP regulators, multiple senomorphic agents may need to be administered for global SASP suppression¹⁸² **(Figure 4)**. Finally, more studies are required to compare senolytic versus senomorphic therapies and to determine the context in which each is more optimal.

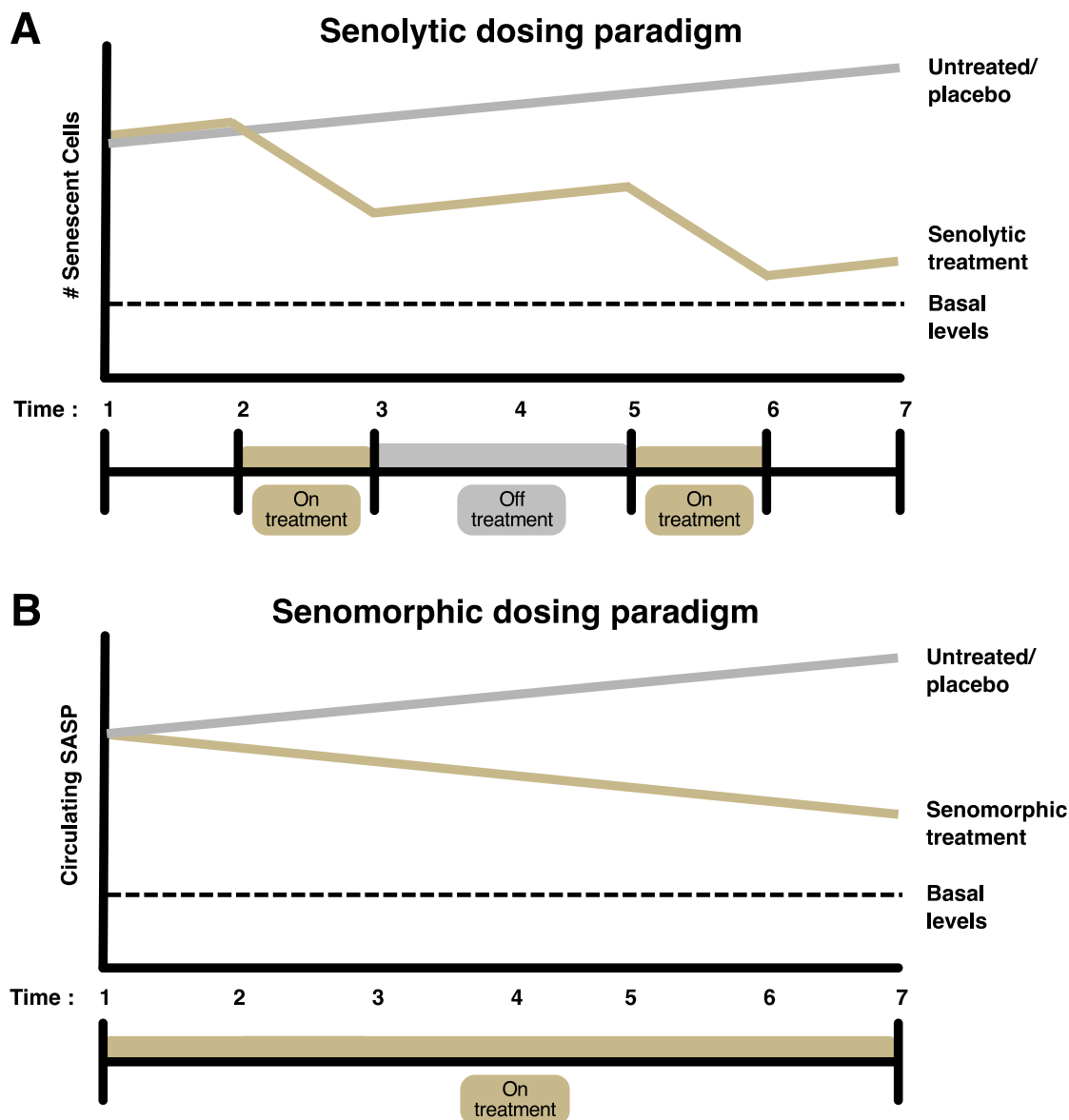


Figure 4. Senotherapeutic dosing paradigms. Senolytic therapies are often administered using an intermittent dosing paradigm composed of 2-6 cycles with generally 1-7 days on a high-dose treatment and 1-4 weeks of rest period (off treatment) in between doing periods allowing for the suppression of excess cellular senescence without interfering with the basal levels and functions of senescent cells (A). Senomorphic therapies are often administered continuously at a low dose to suppress excess circulating SASP and SASP signaling while maintaining basal inflammation levels (B).

Senotherapeutic intervention considerations in vascular disease.

Although it is generally agreed upon that senescent cells contribute to several vascular diseases, senescent cell clearance is contended in select vascular pathologies and disease progression. For example, several studies have investigated the contributions and effects of elimination of senescent cells in animal models of atherosclerosis. Although several studies found favorable effects of senolytic treatment on atherosclerotic plaques^{36,85,188,194,195}, two studies found that senescent cell clearance worsened outcomes including plaque instability and rupture in advanced stages of atherosclerosis^{186,187}. As such, it is hypothesized that senescent cells may contribute to atherosclerotic plaque structure and more strategies are required to understand the optimal timing for senescent cell clearance. Similar controversies exist in models of pulmonary arterial hypertension. A study by van der Feen *et al.* found that senescent cell clearance in a rat model of pulmonary hypertension reversed disease progression by improving vascular hemodynamics¹⁵⁶. Conversely, Born *et al.* found that senolytics in mice and rat models of pulmonary hypertension aggravated disease severity and worsened vascular hemodynamics³⁸.

Finally, several senolytic compounds may have adverse safety profiles that limit their translatability to the clinic. In particular, thrombocytopenia, a condition characterized by low circulating platelet count, has been observed with Navitoclax administration²⁰⁵ and D + Q administration has been reported to have several side effects including hematologic dysfunction, fluid retention and QT prolongation, state of extended electrical interval between the heart's contraction and relaxation²⁰⁶. These findings collectively suggest that senescent cell clearance has variable effects on vascular disease pathogenesis and more studies are required to fully elucidate the benefits and limitations of senotherapeutic interventions in these conditions. Further, senomorphic agents may be more efficacious in contexts where senescent cell

elimination is detrimental and the development of novel senotherapeutic strategies that have more specific targets may be warranted.

Multi-organ benefit of targeting senescence in the vasculature.

The importance of cellular senescence in the vasculature extends beyond its role in CVD. This is because cellular senescence occurs within vascular cells resident to many tissues and organs and affects neighboring non-vascular cells by disrupting blood flow regulation which is crucial for broader physiological implications (**Figure 5**). Thus, studies on vascular cell senescence have been essential to our understanding of the physiological implications of cellular senescence. An increasing number of reports have identified the cells of the vasculature as some of the main cell types that becomes senescent with advancing age as well as in models of genetically and environmentally early aging. Indeed, in these settings, senescent vascular cells arise within nearly all tissues, including the eye^{207,208}, skeletal muscle²⁰⁹, heart²¹⁰, kidney^{211,212}, liver²¹³⁻²¹⁵, pancreas²¹³, testes²¹⁶, brain²¹⁷⁻²²⁰, adipose tissue^{210,221}, and lung^{156,222,223}. In addition to being widespread, endothelial cells are one of the first cell types to become senescent with advancing age²¹⁴, and the cellular senescence markers p16 and p21 increase in the vasculature to a greater extent than any other tissue in both natural aging and progeroid mouse models¹². Cumulatively, these findings support a model by which many diseases, as well as premature and healthy aging, result in vascular cell senescence that occurs early, is robustly induced, and is found within nearly every tissue. As such, an additional important benefit of targeting cellular senescence in the vasculature is that it is likely to have systemic benefits. Vascular cellular senescence directly contributes to a host of cardiometabolic diseases and cancer (reviewed in detail elsewhere⁷²). Recent evidence has shed light onto the mechanisms by which vascular senescence contributes to this multi-system dysfunction. This includes 1) an induction of cellular

senescence in neighboring cells and tissues as well as an increase in local inflammation, and 2) dysregulation of blood flow. Through these mechanisms, inducing cellular senescence in the vasculature appears to act as a rheostat for the physiological age of an organism. This has been demonstrated by manipulation of several known modulators of cellular senescence and the SASP in the vasculature.

Several examples of this phenomenon come from studies using genetically modified mice to specifically manipulate senescent endothelial cells. For example, three independent studies have demonstrated the effects of genetic induction of telomere dysfunction specifically within endothelial cells^{40,224,225}. In endothelial cells, telomere dysfunction stimulated cellular senescence and increased expression of several SASP factors^{40,224,225}. Consequently, cellular senescence and the SASP are elevated in whole artery lysates, perigonadal white adipose tissue and the liver^{40,224,225}. By reducing the replicative capacity of endothelial cells, microvascular density is reduced in the gut⁴⁰ and adipose tissue vascular leak is increased²²⁴. Moreover, insulin mediated production of NO is impaired⁴⁰. In these mice, adipose tissue and the liver display elevated markers of senescence and inflammatory SASP factors and impaired glucose tolerance^{40,224,225}. In addition to the inflammation in metabolically active tissues, alterations in glucose homeostasis may be a consequence of improper nutrient delivery due to a lack of vascularization and blood flow reallocation in response to metabolic stimuli^{40,224,225}. Finally, running endurance and cognitive function are impaired by endothelial telomere dysfunction-induced senescence²²⁴. These data indicate that stimulating cellular senescence in young mice results in a premature aging phenotype that extends beyond changes typically associated with vascular dysfunction.

Mirroring the studies of telomere dysfunction in young mice, endothelial cell specific reductions in the mammalian target of rapamycin (mTOR) reduces cellular senescence in both

whole arteries and endothelial cells, as well as in metabolically active tissues including the adipose tissue and the liver²²⁶. Likewise, inflammatory SASP factors are reduced in these tissues²²⁶. These changes culminate in improvements in glucose tolerance due to a suppression of hepatic gluconeogenesis²²⁶. Importantly, when coupled with findings that show senescent cell clearance in old mice improves vascular function in a multitude of ways^{13,65}, these findings demonstrate the multiplicity of benefits of targeting cellular senescence specifically in the vasculature. It appears that senescent cells in the vasculature may promote cellular senescence and inflammation in a variety of tissues, reduce vessel density, increase vascular permeability, and disrupt proper redistribution of blood in response to metabolic stimuli, all of which act to impair metabolic function and potentially exercise tolerance and cognition. Therefore, clearing senescent vascular cells is likely to be beneficial in conditions that were not originally considered to be driven by cellular senescence in the vasculature.

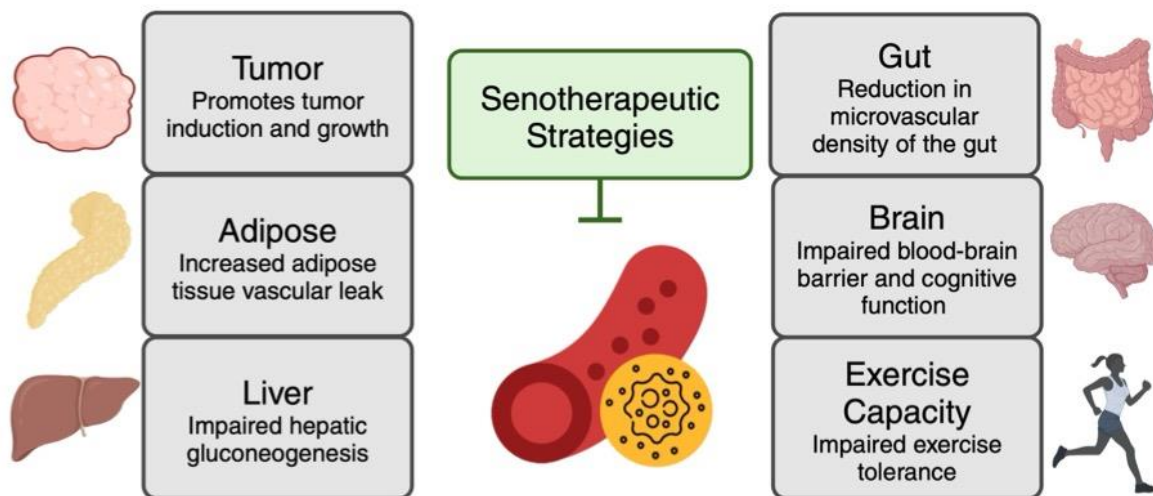


Figure 5. Multi-organ benefit of targeting senescence in the vasculature. Mechanistic studies in preclinical models elucidate the vascular senescent cells have detrimental effects systemically and clearance of senescent vascular cells elicits improvements in several tissues/organs and physiological processes.

Research gaps and future directions.

The role of cellular senescence in vascular aging is an emerging field with promising potential in modulating clinical CV outcomes. Despite the promising evidence in the field, several research gaps exist that represent future directions in CV and cellular senescence research (**Figure 6**). At a fundamental level, the mechanisms underlying vascular cell senescence and mechanisms by which senescent cells contribute to vascular dysfunction are yet to be fully elucidated. Uncovering these mechanisms may reveal novel means to target senescent cells and elucidate the detrimental effects of cellular senescence on vascular function. Next, markers of cellular senescence have only recently allowed researchers to assess cellular senescence levels in clinical population^{31,45}. As such, senescent cell burden has yet to be characterized in many clinical populations but could be a promising therapeutic target for groups that present with high levels of senescent cells. Even in the pathologies in which senescence cells have been identified in, the role of the senescent cells appears to be largely heterogenous, and clearance of senescent cells has variable outcomes. Accordingly, further research exploring and optimizing the dosing, timing, and context of senotherapeutic interventions is essential for translation. Finally, several senotherapeutic strategies exist but the safety, tolerability, and efficacy of these interventions in people is not fully known¹⁸³. Clinical trials of the senotherapeutic compounds discussed in this review are essential as well as the identification of novel strategies that may target specific cell types or affected areas.

Issue	Research Gap	Future Direction
	Mechanisms underlying of vascular cell senescence are incompletely understood	Elucidate novel mechanisms of vascular cell senescence
	Limited senotherapeutic strategies exist and may be unspecific to targeting senescent cells	Develop novel, targeted senotherapeutic strategies
	Most investigations into senotherapeutics exist in cell culture and preclinical models	Test senotherapeutic intervention in clinical trials
	Senescent cells play different roles in different CVD and at different disease stages	Determine when senotherapeutic interventions would be most beneficial in states of CVD
	Senescent cell burden is unknown in most clinical groups	Identify new groups that have high senescent cell burden who may stand to benefit the most from senotherapeutic interventions

Figure 6. Research gaps and future directions. Presented are important research gaps and future directions in the understanding of cellular senescence, vascular function, and cardiovascular disease (CVD). These research gaps require prioritization in ongoing and future studies.

CONCLUSIONS

Aging is the major risk factor for CVD due to the development of vascular dysfunction, particularly arterial stiffening and endothelial dysfunction. In this review, we discussed cellular senescence – a stress response characterized by cell cycle arrest – as a fundamental mechanism of vascular aging that may serve as viable therapeutic targets for the prevention and treatment of vascular dysfunction to reduce the risk of CVD. We reviewed key established and novel mechanisms by which senescent cells contribute to vascular dysfunction. We identified groups that may have high vascular cell senescence burden due to chronological or early vascular aging. Finally, we discussed senotherapeutics – therapies that suppress excess senescent cells or the SASP – and examined their translational potential, dosing paradigm, clinical implications, groups that may stand the most to benefit, and systemic benefits. Together, this review underscores the crucial role of cellular senescence in vascular dysfunction and the need for deeper understanding of the translational use of senotherapeutics in groups with high senescent cell burden.

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CHAPTER III

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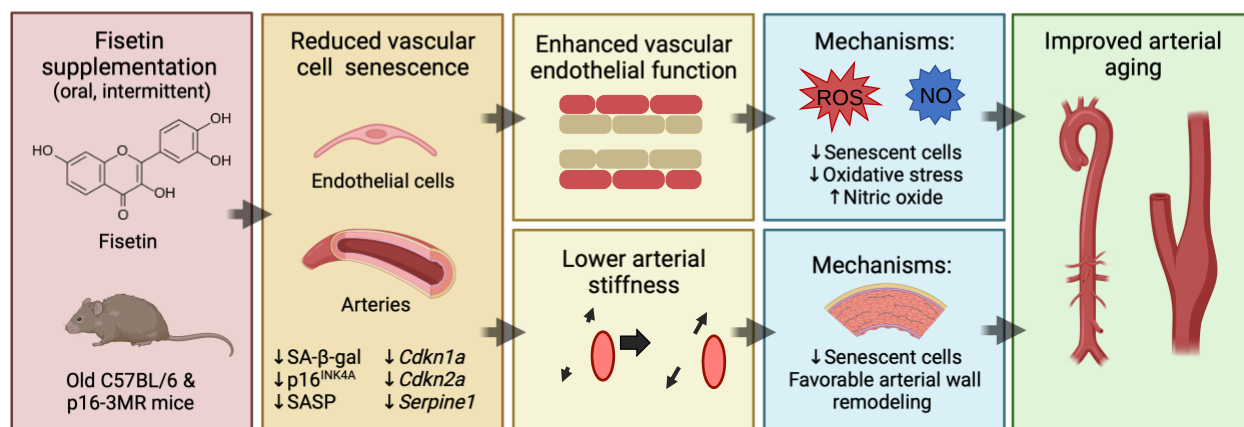
Intermittent Supplementation with Fisetin Improves Arterial Function in Old Mice by Decreasing Cellular Senescence

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GRAPHICAL ABSTRACT. Oral intermittent fisetin supplementation reduces vascular cell senescence to improve vascular endothelial function and reduce aortic stiffness in old mice.

ABSTRACT

Cellular senescence and the senescence-associated secretory phenotype (SASP) contribute to age-related arterial dysfunction, in part, by promoting oxidative stress and inflammation, which reduce the bioavailability of the vasodilatory molecule nitric oxide (NO). In the present study, we assessed the efficacy of fisetin, a natural compound, as a senolytic to reduce vascular cell senescence and SASP factors and improve arterial function in old mice. We found that fisetin decreased cellular senescence in human endothelial cell culture. In old mice, vascular cell senescence and SASP-related inflammation were lower 1 week after the final dose of oral intermittent (1 week on—2 weeks off—1 week on dosing) fisetin supplementation. Old fisetin-supplemented mice had higher endothelial function. Leveraging old p16-3MR mice, a transgenic model allowing genetic clearance of p16INK4A-positive senescent cells, we found that ex vivo removal of senescent cells from arteries isolated from vehicle- but not fisetin-treated mice increased endothelium-dependent dilation, demonstrating that fisetin improved endothelial function through senolysis. Enhanced endothelial function with fisetin was mediated by increased NO bioavailability and reduced cellular- and mitochondrial-related oxidative stress. Arterial stiffness was lower in fisetin-treated mice. Ex vivo genetic senolysis in aorta rings from p16-3MR mice did not further reduce mechanical wall stiffness in fisetin-treated mice, demonstrating lower arterial stiffness after fisetin was due to senolysis. Lower arterial stiffness with fisetin was accompanied by favorable arterial wall remodeling. The findings from this study identify fisetin as promising therapy for clinical translation to target excess cell senescence to treat age-related arterial dysfunction.

KEYWORDS: aging, arterial stiffness, cellular senescence, endothelial function, nutraceutical, vascular dysfunction

ABBREVIATIONS: ACh, Acetylcholine; AGEs, Advanced glycation end products; CuZn, Copper zinc; CVD, Cardiovascular diseases; EDD, Endothelium dependent dilation; GCV, Ganciclovir; HAECs, Human aortic endothelial cells; HUVECs, Human umbilical vein endothelial cells; L-NAME, N(gamma)-nitro-L-arginine methyl ester; Mn, Manganese; NADPH, Nicotinamide adenine dinucleotide phosphate; NO, Nitric oxide; PWV, Pulse wave velocity; ROS, Reactive oxygen species; SA- β -gal, Senescence-associated beta galactosidase; SASP, senescence- associated secretory phenotype; SNP, Sodium nitroprusside; SOD, Superoxide dismutase; TEMPOL, 4-hydroxy-2,2,6,6-tetramethylpiperidin-1-oxyl.

INTRODUCTION

Advancing age is the primary risk factor for cardiovascular diseases (CVDs), which remain the leading cause of morbidity and mortality worldwide¹. Arterial dysfunction, characterized by vascular endothelial dysfunction and large elastic artery stiffening², is a key antecedent to the development of clinical CVDs with aging.

Age-related declines in endothelial function, as shown by reduced endothelium-dependent dilation (EDD), are mediated primarily by lower bioavailability of nitric oxide (NO)³⁻⁶. Age-associated arterial stiffening, as indicated by an increase in aortic pulse wave velocity (PWV), is largely due to increased aortic intrinsic mechanical wall stiffness^{7,8} and arterial wall remodeling, featuring collagen deposition (fibrosis), elastin degradation and crosslinking of these structural proteins by advanced glycation end-products (AGEs)^{9,10}. Reduced NO-mediated endothelial function and arterial stiffening are driven by excess reactive oxygen species (ROS)-related oxidative stress, a key source being dysfunctional mitochondria¹¹, and chronic low-grade inflammation¹². Thus, identifying therapies that reduce oxidative stress and inflammation to improve arterial function with aging is an important biomedical research goal¹³⁻¹⁵.

Cellular senescence is a cellular stress response characterized by essentially irreversible cell cycle arrest^{16,17}. Homeostatic levels of cellular senescence aids in wound healing and inhibits tumorigenesis in part through the production of the senescence-associated secretory phenotype (SASP) — a collection of cytokines, chemokines, and inflammatory mediators — which activate immune responses¹⁷⁻¹⁹. Senescent cells accumulate in excess with aging leading to adverse cellular effects in part via the SASP, which contributes to a pro-inflammatory and pro-oxidative environment^{17,20}. Cellular senescence is a mechanism of age-related arterial dysfunction^{21,22} and increased cellular senescence in vascular cells from older adults is associated with impaired arterial function²³. Thus, reducing cellular senescence and the SASP in the vasculature may be a promising therapeutic approach for improving arterial function in old age.

Clinical trials of senolytics — compounds that clear senescent cells — are underway, but the impact and safety of current synthetic senolytic therapies being tested are not fully understood^{24,25}. Fisetin is a flavonoid found in a variety of commonly consumed foods, appears to be safe in humans and has high translational potential²⁶⁻³². Intermittent fisetin treatment – to clear excess senescent cells, but not interfere with the homeostatic functions of cellular senescence (*i.e.*, wound healing) – reduces cellular senescence in select tissues and extends lifespan without adverse effects in old mice²⁷. However, previous studies have limited physiological outcomes and it is unknown if intermittent fisetin supplementation reduces vascular cell senescence and improves arterial function with aging.

The purpose of this study was to assess the efficacy of fisetin for reducing cellular senescence and the SASP in vascular cells and arteries and improving arterial function in old mice. We first sought to identify concentrations of fisetin that selectively reduce cellular senescence in endothelial cells in culture. Using a dose of fisetin based on our findings in cell

culture, we set out to determine if oral, intermittent fisetin treatment reduced vascular senescent cell burden and SASP-related inflammation in old mice. We then assessed if fisetin could improve endothelial function and arterial stiffness by decreasing cellular senescence. Lastly, we determined if fisetin improved endothelial function by increasing NO bioavailability and reducing total and mitochondrial ROS, and if reductions in arterial stiffness were accompanied by favorable arterial wall remodeling.

METHODS

Cell Culture Experiments.

HUVECs and HAECs (Lonza, Basel, Switzerland) were cultured at 37°C and 5% CO₂ to ~80% confluency in Endothelial Cell Growth Media (EGM)-2 media (Lonza) supplemented with an additional 2% fetal calf serum (FCS; Sigma-Aldrich Corp., St. Louis, MO), 100µg/mL penicillin and 172µg/mL streptomycin (Gibco, Gaithersburg, MD). Non-senescent (control) cells were passaged 3-6 times, and senescent cells were grown to passage 15 to induce replicative senescence, as previously shown³³. Senescent cells were treated with fisetin in EGM-2 for 48hr and cell assays were immediately performed in triplicate following the treatment period³⁴. Experimental details on the cell viability assay, SA-β-gal staining, assessment of ROS levels, and gene expression are provided in the **Supplemental Data**.

Ethical Approval and Animal Studies.

All mouse studies and procedures were reviewed and approved by the Institutional Animal Care and Use Committee at the University of Colorado Boulder (Protocol No. 2618). All procedures adhered to the guidelines set forth by the National Institutes of Health's Guide for the Care and Use of Laboratory Animals³⁵.

Male C57BL/6N (wildtype) mice were obtained from the National Institute of Aging colony (maintained by Charles River, Wilmington, MA). Wildtype mice were allowed to acclimate to our facilities for 4 weeks before beginning the study. Male and female p16-3MR mice were bred and aged in our mouse colony at the University of Colorado Boulder. These mice carry a trimodal fusion protein (3MR) under the control of the p16^{INK4A} promoter which allows for selective genetic clearance of excess p16^{INK4A}-positive senescent cells by administering the antiviral agent GCV¹⁸. For the duration of the study, all mice were single housed at our animal facility with a 12hr:12hr light-dark cycle and allowed ad libitum access to an irradiated, fixed, and open rodent chow (Inotiv/Envigo 7917, stored at room temperature) and drinking water.

For the intervention period, old (27 mo.) wildtype and p16-3MR mice were assigned to receive vehicle (10% EtOH, 30% PEG400 and 60% Phosal 50 PG) or fisetin (100 mg/kg/day in vehicle). For wildtype mice, 18 mice received vehicle and 19 mice received fisetin, and for p16-3MR mice, 27 mice received vehicle and 35 mice received fisetin. Treatment was administered via oral gavage using an intermittent dosing paradigm involving one week of daily active dosing, two weeks of no intervention and then another one week of active dosing, which has previously been shown to reduce senescent cell burden in select tissues in old mice back to young/basal levels²⁷. Mice were sacrificed one to two weeks after the final dose. Treatment groups were randomized and matched for baseline body weight and aortic PWV. Throughout the intervention period, 1 old wildtype mouse and 11 p16-3MR mice died as a result of expected age-related attrition, which resulted in a final sample size of old wildtype vehicle, n = 17; old wildtype fisetin, n = 19; old p16-3MR vehicle, n = 22; and old p16-3MR fisetin, n = 29.

In Vivo Aortic Stiffness and Arterial Blood Pressure.

Aortic PWV was assessed *in vivo* one week before (pre) and one week after (post) the intervention, as previously described^{4,9,36,37}. Briefly, mice were anesthetized via inhaled isoflurane (1.0-2.5%) and positioned supine on a warmed heat platform with paws secured to electrocardiogram (ECG) leads. Two Doppler probes were then placed at the transverse aortic arch and the abdominal aorta. Three repeated 2-second ultrasound tracings were recorded, and average pre-ejection time (i.e., time between the R-wave of the ECG to the foot of the Doppler signal) was determined for each location. Aortic PWV was then calculated as: aortic PWV = (physical distance between the two probes) / ($\Delta\text{time}_{\text{abdominal}}$ minus $\Delta\text{time}_{\text{transverse}}$) and reported in centimeters per second. Blood pressure was measured pre- and post-intervention on three consecutive days using a non-invasive tail-cuff method (CODA; Kent Scientific, Torrington, CT), as we have described previously^{4,9,36,37}.

Endothelial Function.

Endothelium-dependent dilation (EDD) in response to increasing doses of acetylcholine (Sigma Aldrich, Cat. No. A6625) and endothelium-independent dilation in response to increasing concentrations of the exogenous NO donor sodium nitroprusside (SNP; Sigma Aldrich, Cat. No. 13755-38-9) were measured in isolated carotid arteries, as previously described^{4,36-38}. Further details, including all pharmacological agents and approaches used for pharmacodissection experiments, are provided in the **Supplemental Data**.

Statistical Analyses.

Detailed descriptions of all statistical analyses performed are provided in the Data Supplement. Data are presented as mean $\pm$ SEM in text, figures, and tables unless specified otherwise. Statistical significance was set to $\alpha=0.05$. All statistical analyses were performed using Prism, version 9 (GraphPad Software, Inc, La Jolla, CA).

RESULTS

Fisetin Reduces Cellular Senescence in Endothelial Cells.

Initially, we performed *in vitro* cell culture experiments in human umbilical vein endothelial cells (HUVECs) to determine the effect of fisetin on senescent and non-senescent (control) endothelial cells. HUVECs were brought to senescence via replicative exhaustion (an *in vitro* model of an aged-like state)^{33,39}. Cell viability of senescent and control HUVECs were measured in response to increasing concentrations of fisetin to determine the concentrations of fisetin that reduced the viability of senescent cells without affecting viability of control cells. We found that fisetin had an overall higher potency against senescent HUVECs compared to control cells (IC₅₀ of fisetin 7.0 ± 0.4 vs. $3.4 \pm 0.3 \mu\text{M}$ on control and senescent cells, respectively; **Figure 1A**). Moreover, we observed a reduction in viability of senescent HUVECs with $1 \mu\text{M}$ fisetin, whereas no significant effects on viability were observed on control HUVECs at this concentration (viability at $1 \mu\text{M}$ fisetin: control cells, $97 \pm 1\%$ vs. senescent cells, $91 \pm 1\%$, $P=0.003$; **Figure 1A**). Based on these observations, we selected $1 \mu\text{M}$ as the maximal concentration of fisetin for the remainder of the cell culture experiments, as this concentration appeared to selectively influence senescent cell viability without affecting non-senescent cells.

To determine if the decrease in viability of senescent cells with fisetin treatment was associated with reductions in cellular senescence, we assessed the senescence-associated β -galactosidase (SA- β -gal) signal — an established hallmark of cellular senescence⁴⁰ — in control and senescent HUVECs exposed to increasing concentrations of fisetin up to $1 \mu\text{M}$. We found that senescent HUVECs had 13.5-fold higher SA- β -gal signal relative to non-senescent control HUVECs ($P=0.002$) and that fisetin reduced the SA- β -gal signal in a concentration-dependent manner up to $1 \mu\text{M}$ (**Figure 1B** and **Figure S1**).

After establishing the effect of fisetin in senescent HUVECs, we aimed to extend our understanding of the senolytic effects of fisetin in arterial cells using human aortic endothelial cells (HAECs). Accordingly, in HAECs, we measured gene expression of *Cdkn2a* and *Cdkn1a*, which encode the cyclin-dependent kinase inhibitor proteins p16^{INK4A} and p21^{Cip1/Waf1}, respectively. Compared to control cells, these cyclin-dependent kinase inhibitors are observed at increased levels in most senescent cells and are largely responsible for the irreversible cell cycle arrest associated with cellular senescence^{18,41}. Senescent HAECs demonstrated 17-fold ($P<0.0001$) and 4-fold ($P<0.0001$) higher expression of *Cdkn2a* and *Cdkn1a*, respectively, compared to control HAECs (**Figure 1C**). In replicative senescent HAECs, 1 μ M fisetin lowered the expression of *Cdkn2a* by 71% ($P<0.0001$) back towards levels of control cells but had no effect on the relatively modest increase in *Cdkn1a* expression, suggesting that fisetin may preferentially act upon the p16^{INK4A}, rather than the p21^{Cip1/Waf1}, transcription factor cascade to eliminate senescent endothelial cells (**Figure 1C**).

Collectively, we observed a reduction in viability of senescent endothelial cells treated with 1 μ M fisetin, but no effects on viability of control cells treated with this concentration. In senescent endothelial cells treated with fisetin, we found decreases in select markers of cellular senescence. Thus, our *in vitro* experiments demonstrate that fisetin can selectively target excess cellular senescence in vascular endothelial cells.

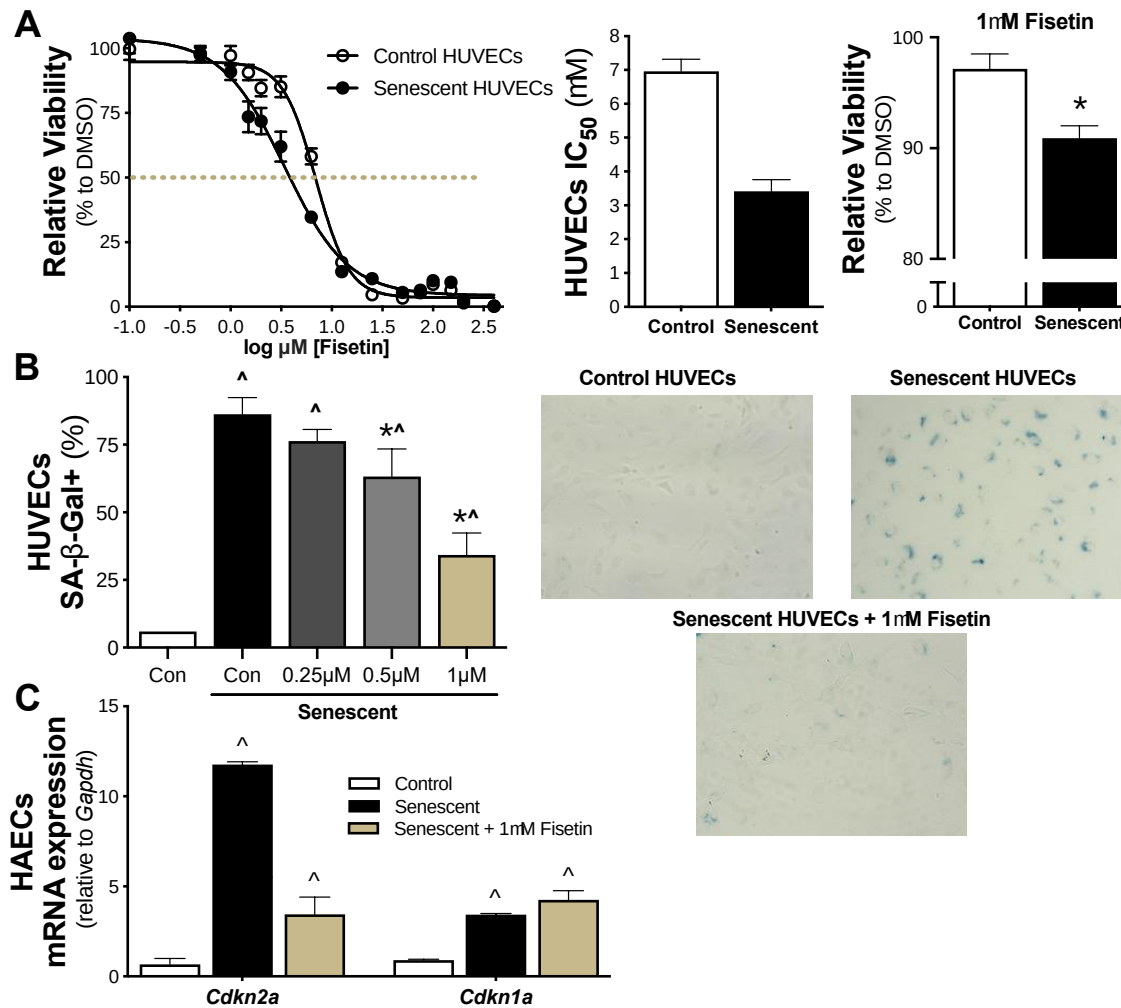


Figure 1. Fisetin suppresses cellular senescence in endothelial cells. Cell viability in replicating (control) and senescent human umbilical vein endothelial cells (HUVECs) to increasing doses of fisetin (dashed line represents the IC_{50}). IC_{50} values of control and senescent HUVECs with 95% confidence intervals. Viability of control and senescent HUVECs at 1 μM fisetin ($n = 4-8$) (A). Senescence- associate beta galactosidase (SA- β -Gal) signal in control and senescent HUVECs to increasing doses of fisetin ($n = 3$) and representative images, right (B). mRNA gene expression of cellular senescence markers *Cdkn2a* and *Cdkn1a* in control, senescent, or 1 μM fisetin- treated senescent human aortic endothelial cells (HAECs) ($n = 3$) (C). Values represent mean $\pm$ SEM. * $P < 0.05$ senescent versus senescent + treatment; $^{\wedge}P < 0.05$ versus control.

Animal Characteristics.

We next sought to extend the senolytic effects of fisetin to *in vivo* arterial aging by assessing cellular senescence in intact arteries and arterial function of aged mice intermittently supplemented with fisetin. For the intervention, old male wildtype received vehicle (10% EtOH, 30% PEG400 and 60% Phosal 50 PG) or fisetin (100 mg/kg/day in the vehicle). After establishing the effects of fisetin in male wildtype mice, we used the same intervention regimen to look at sex differences and dissect the role of cellular senescence in old male and female p16-3MR mice^{18,42}. Since we did not observe sex differences in any of the outcomes in p16-3MR mice, male and female data were combined in the results. The dose of fisetin was chosen based on the *in vivo* translation of the 1 μ M concentration used in our cell culture experiments, as peak plasma levels of fisetin reach ~1 μ M following oral ingestion of fisetin at a dose of 100 mg/kg/day in mice^{43,44}. Treatment was administered via oral gavage using an intermittent dosing paradigm – one week on; two weeks off; one week on, as previously described²⁷ (**Figure 2A**). Functional and biochemical measurements were assessed one week after the final dose.

Blood pressure, total body mass, mass of key tissues, and carotid artery and aorta characteristics at time of euthanasia in the vehicle- and fisetin-treated wildtype mice are presented in **Table 1** (p16-3MR animal characteristics are presented in **Supplemental Table 1**). Fisetin did not affect any of these characteristics in either mouse model ($P>0.05$).

Table 1. Wildtype animal characteristics.

Characteristics	Vehicle	Fisetin
<i>n</i>	15	19
Body mass, g	29.3 ± 0.5	28.0 ± 0.7
Heart mass, mg	152 ± 4	165 ± 6
Quadriceps mass, mg	271 ± 11	280 ± 11
Visceral adipose mass, mg	324 ± 31	292 ± 44
Liver mass, g	1.6 ± 0.1	1.4 ± 0.1
Spleen mass, mg	82.4 ± 7.6	82.7 ± 8.5
Carotid artery		
<i>Resting diameter, μm</i>	428 ± 15	412 ± 13
<i>Maximal diameter, μm</i>	526 ± 11	519 ± 9
Aorta		
<i>Diameter, μm</i>	416 ± 10	415 ± 8
<i>Intima media thickness, μm</i>	40 ± 2	39 ± 2
Systolic blood pressure, mmHg		
<i>Pre</i>	96 ± 2	94 ± 2
<i>Post</i>	94 ± 2	89 ± 2
Diastolic blood pressure, mmHg		
<i>Pre</i>	73 ± 2	73 ± 2
<i>Post</i>	75 ± 2	71 ± 2

Values represent mean ± SEM.

Fisetin Targets Cellular Senescence and the SASP in the Vasculature of Old Mice.

To establish the senolytic effects of fisetin in the vasculature of old mice, we assessed markers of cellular senescence in mouse aorta lysates. We found that aortic gene expression of the cellular senescence markers *Cdkn2a* (-85%, $P=0.010$), *Cdkn1a* (-48%, $P=0.006$), and *Serpine1* (-81%, $P=0.011$) were lower in old fisetin-supplemented p16-3MR mice compared to vehicle-treated p16-3MR mice, whereas no statistical differences were observed in *Lmnbl* expression (+45%, $P=0.213$), a contributor of chromatin stability known to be reduced in

senescent cells⁴¹ (**Figure 2B**). Given that we observed a general reduction in vascular cell senescence transcripts in our initial gene expression screening, we validated these differences by measuring vascular protein abundance of p16^{INK4A}. We found that old wildtype mice supplemented with fisetin had 36% lower aortic p16^{INK4A} protein abundance ($P=0.007$; **Figure 2C** and **Figure S2A**) compared to old vehicle-treated mice. Fisetin supplementation also lowered the aortic expression of several SASP factors, including pro-inflammatory cytokines (*Tnf α* ; $P=0.107$), chemokines (*Ccl2*, *Cxcl2*; $P=0.032$ and $P=0.126$, respectively), growth factors (*Vegf*; $P=0.172$), and proteinases (*Mmp3*, *Plat*; $P=0.045$ and $P=0.037$, respectively) (**Figure 2D**). These results suggest that intermittent fisetin supplementation can decrease cellular senescence and modulate SASP factors in the vasculature of old mice.

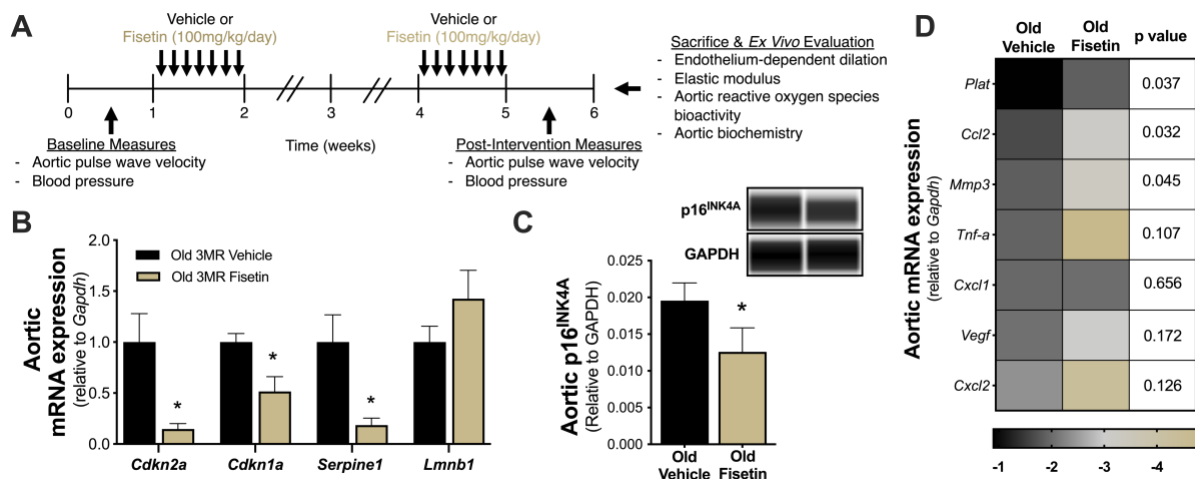


Figure 2. Fisetin reduces cellular senescence in the vasculature of old mice. Study design used in the animal model (**A**). Aortic mRNA gene expression of cellular senescence markers in old p16-3MR (3MR) mice ($n = 12-15$) (**B**). Aortic protein abundance of p16^{INK4A} in wildtype mice with representative virtual blot bands ($n = 8$) (**C**). Aortic mRNA gene expression of senescence-associated secretory phenotype factors ($n = 14-15$) (**D**). Color intensities represent log2-fold changes. Values represent mean $\pm$ SEM. * $P < 0.05$ vehicle versus fisetin.

Fisetin Improves Endothelial Function in Old Mice by Suppressing Cellular Senescence, Enhancing NO Bioavailability and Abolishing Oxidative Stress and its Associated Suppression of EDD.

EDD. To determine whether fisetin improves endothelial function in advanced age, we assessed EDD *ex vivo* in carotid arteries excised from old wildtype vehicle- and fisetin-supplemented mice in response to increasing doses of acetylcholine (ACh). EDD was 20% greater in old mice supplemented with fisetin relative to old vehicle-treated mice (peak EDD: vehicle, $81 \pm 3\%$ vs. fisetin, $97 \pm 1\%$, $P < 0.001$; **Figure 3A-B**), suggesting that intermittent fisetin supplementation improves vascular endothelial function in old age.

Cellular Senescence-Mediated Suppression of EDD. Following initial EDD measurements, we aimed to acquire causal evidence that reductions in cellular senescence with fisetin mediate improvements in endothelial function using “pharmaco-dissection” techniques. To accomplish this, we utilized the p16-3MR transgenic mouse model which allows for genetic clearance of p16^{INK4A}-positive senescent cells with GCV *in vivo* and *ex vivo*¹⁸. Leveraging genetic senolysis models in combination with senolytic agents is considered the reference standard for assessing the causal impact of cellular senescence on physiological function^{18,22,42}. Thus, to selectively clear senescent cells, we incubated carotid arteries isolated from old vehicle- and fisetin-supplemented p16-3MR mice with 5 μ M GCV and assessed EDD prior to and after the incubation period. Prior to GCV, we observed that vehicle-treated mice had lower initial EDD, and senescent cell clearance with GCV incubation increased EDD, indicating that an accumulation of cellular senescence impaired endothelial function in these arteries (peak EDD: ACh alone, $76 \pm 4\%$ vs. with GCV, $90 \pm 3\%$, $P = 0.005$). In contrast, mice that received fisetin supplementation had high initial EDD and incubation with GCV did not further improve EDD in

these arteries (peak EDD: ACh alone, $93 \pm 1\%$ vs. with GCV, $92 \pm 3\%$, $P=0.71$; **Figure 3C** and **Figure S3A**). These findings suggest reductions in cellular senescence in arteries underlie improvements in EDD with intermittent fisetin supplementation.

NO Bioavailability. The production and subsequent bioavailability of the potent vasodilatory molecule NO is diminished in senescent endothelial cells and with aging⁴⁵. Thus, we next assessed the effects of fisetin on NO-mediated endothelial function. To accomplish this, we assessed EDD prior to and following inhibition of NO production with the NO synthase inhibitor L-NAME. Incubation with L-NAME abolished group differences in EDD (peak EDD: vehicle, $37 \pm 5\%$ vs. fisetin, $44 \pm 5\%$, $P=0.350$; **Figure 3D** and **Figure S3B**), suggesting that improvements in endothelial function with fisetin were due to greater NO bioavailability. Consistent with this idea, peak NO-mediated dilation (ACh alone [-] ACh with L-NAME) was 40% greater in fisetin- vs. vehicle-supplemented mice (vehicle, $43 \pm 4\%$ vs. fisetin, $60 \pm 5\%$, $P<0.001$; **Figure 3E**). Next, to determine if improvements in vasodilation were a result of enhanced smooth muscle sensitivity to NO, we measured endothelium-independent dilation as the vasodilatory response to sodium nitroprusside (SNP). We found no differences among groups (peak response to SNP, vehicle, $98 \pm 1\%$ vs. fisetin, $97 \pm 1\%$, $P=0.856$) indicating that fisetin improves vasodilation in aged mice in an endothelium-specific manner (**Figure 3F**). Collectively, these data indicate that intermittent supplementation with fisetin enhances endothelial function in old mice as a result of greater NO bioavailability and not via increased vascular smooth muscle sensitivity to NO.

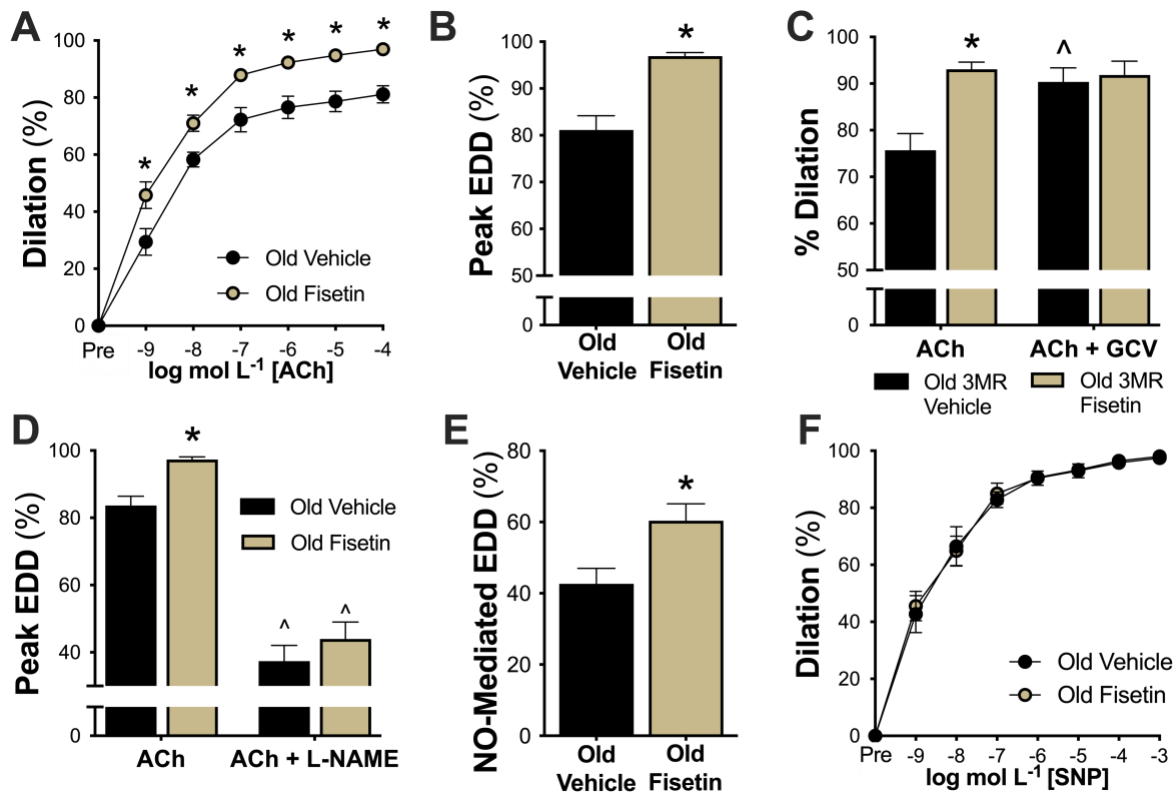


Figure 3. Fisetin improves endothelial function in old mice by suppressing cellular senescence and increasing nitric oxide (NO) bioavailability. Endothelium-dependent dilation (EDD) in isolated carotid arteries in response to acetylcholine (ACh; 1×10^{-9} to 1×10^{-4} M; $n = 12-18$) (A-B). EDD in response to ACh in the presence or absence of ex vivo genetic elimination of cellular senescence via ganciclovir (GCV; $5 \mu\text{M}$; 180 min preincubation; $n = 9-11$) (C). NO-mediated EDD in response to ACh in the presence or absence of the NO-synthase inhibitor, L-NAME (0.1 mM ; 30 min preincubation; $n = 15$) (D-E). Endothelium-independent dilation to increasing doses of the NO donor, sodium nitroprusside (SNP; 1×10^{-10} to 1×10^{-4} M; $n = 9$) (F). Values represent mean $\pm$ SEM. * $P < 0.05$ old vehicle versus old fisetin; ^ $P < 0.05$ versus ACh alone.

Whole-Cell Oxidative Stress. Age-related reductions in NO are mediated in part by an excessive production of ROS, which react with NO reducing its bioavailability⁴⁶. Excess cellular senescence is associated with an increased production of ROS⁴⁷. As such, we next determined if the favorable effects of fisetin on endothelial function in old mice were associated with reduced whole-cell oxidative stress. To do so, we first assessed ROS levels in aortic ring sections. Old fisetin-supplemented mice had 1.7-fold lower aortic ROS levels relative to old vehicle-

supplemented mice (vehicle, 8173 ± 1243 vs. fisetin, 4703 ± 455 arbitrary units [AU], $P=0.028$; **Figure 4A**). To investigate the mechanisms responsible for lower ROS levels with fisetin supplementation, we measured the protein abundance of the pro-oxidant enzyme NADPH oxidase, a key source of endothelial cell ROS production⁴⁸, and the cytosolic antioxidant CuZn superoxide dismutase (SOD), an important ROS scavenger⁴⁹, in aortic lysates. Fisetin-supplemented mice had 30% lower NADPH oxidase abundance (vehicle, 0.086 ± 0.008 vs. fisetin, 0.060 ± 0.006 chemiluminescence units [CU], $P=0.027$; **Figure 4B** and **Figure S2B**) but no differences in CuZnSOD abundance were observed between groups (vehicle, 0.919 ± 0.131 vs. fisetin, 1.116 ± 0.276 CU, $P=0.525$; **Figure 4C** and **Figure S2C**). This suggests that fisetin may facilitate the reduction in tonic oxidative stress in old mice by reducing key sources of ROS which, once lowered, were in balance with cellular antioxidant status. To determine if this reduction in whole-cell ROS levels played a direct role in the enhanced endothelial function of fisetin-supplemented mice, we assessed EDD in isolated carotid arteries with and without the presence of the ROS scavenger TEMPOL. We found that pre-incubation with TEMPOL increased EDD in old vehicle-treated mice (peak EDD: ACh alone, $84 \pm 4\%$ vs. with TEMPOL, $93 \pm 2\%$, $P=0.011$) towards levels observed in old fisetin-supplemented mice with ACh alone. In contrast, the addition of TEMPOL had no effect on EDD in old fisetin-supplemented mice (peak EDD: ACh alone, $99 \pm 1\%$ vs. with TEMPOL, $98 \pm 1\%$, $P=0.487$; **Figure 4D** and **Figure S3C**). Taken together, these observations indicate that fisetin improved EDD in old mice by ameliorating tonic ROS-related suppression of EDD.

Mitochondrial Oxidative Stress. Mitochondria are a primary source of excess ROS with aging and mitochondrial-derived ROS levels are higher in senescent cells⁵⁰. Thus, we aimed to determine if fisetin supplementation reduced mitochondrial-derived ROS in the vasculature of

old mice. Fisetin-supplemented old mice had 2.6-fold lower aortic mitochondrial ROS levels relative to old vehicle-supplemented mice (vehicle, 6603 ± 1956 vs. fisetin, 2527 ± 440 AU, $P=0.011$; Figure 4E). To determine the mechanisms contributing to lower mitochondrial ROS levels with fisetin supplementation, we measured protein abundance of the activated form of the mitochondrial pro-oxidant marker and master regulator of mitochondrial ROS production, phosphorylated p66^{SHC} (p-p66^{SHC})⁴⁷, and the major mitochondrial antioxidant enzyme, MnSOD⁴⁹, in aortic lysates. Old fisetin-supplemented mice had ~40% lower arterial abundance of p-p66^{SHC} (vehicle, 0.049 ± 0.008 vs. fisetin, 0.030 ± 0.003 CU, $P=0.034$; **Figure 4F** and **Figure S2D**) whereas no difference was observed in total p66^{SHC} abundance (vehicle, 0.050 ± 0.001 vs. fisetin, 0.050 ± 0.001 CU, $P=0.961$; **Figure S2E** and **Figure S4**). MnSOD abundance was ~100% higher in old fisetin-supplemented mice relative to old mice supplemented with the vehicle (vehicle, 0.202 ± 0.018 vs. fisetin, 0.410 ± 0.101 CU, $P=0.046$; **Figure 4G** and **Figure S2F**). These data suggest that fisetin may reduce mitochondrial-mediated oxidative stress by suppressing a mitochondrial pro-oxidant enzyme and upregulating mitochondrial antioxidant defenses. To determine if lower mitochondrial ROS had a functional role in the enhanced EDD observed in old fisetin-supplemented mice, we assessed EDD in isolated carotid arteries with and without prior incubation with the mitochondrial ROS scavenger MitoQ. We found that pre-incubation with MitoQ increased EDD in the old vehicle-treated mice (peak EDD: ACh alone, $86 \pm 5\%$ vs. with MitoQ, $99 \pm 1\%$, $P=0.025$) to levels observed in the fisetin-supplemented mice with ACh alone, indicating that old vehicle-treated mice experienced tonic inhibition of EDD by excess arterial mitochondrial ROS. Importantly, we did not observe an additional increase in EDD with MitoQ in the fisetin-supplemented group, indicating that fisetin enhanced endothelial function in old mice by ameliorating mitochondrial ROS-related suppression of EDD (peak

EDD: ACh alone, $97 \pm 1\%$ vs. with MitoQ, $97 \pm 1\%$, $P=0.487$; **Figure 4H** and **Figure S3D**).

Collectively, these data suggests that fisetin improved EDD in old mice by ameliorating mitochondrial ROS-related suppression of EDD.

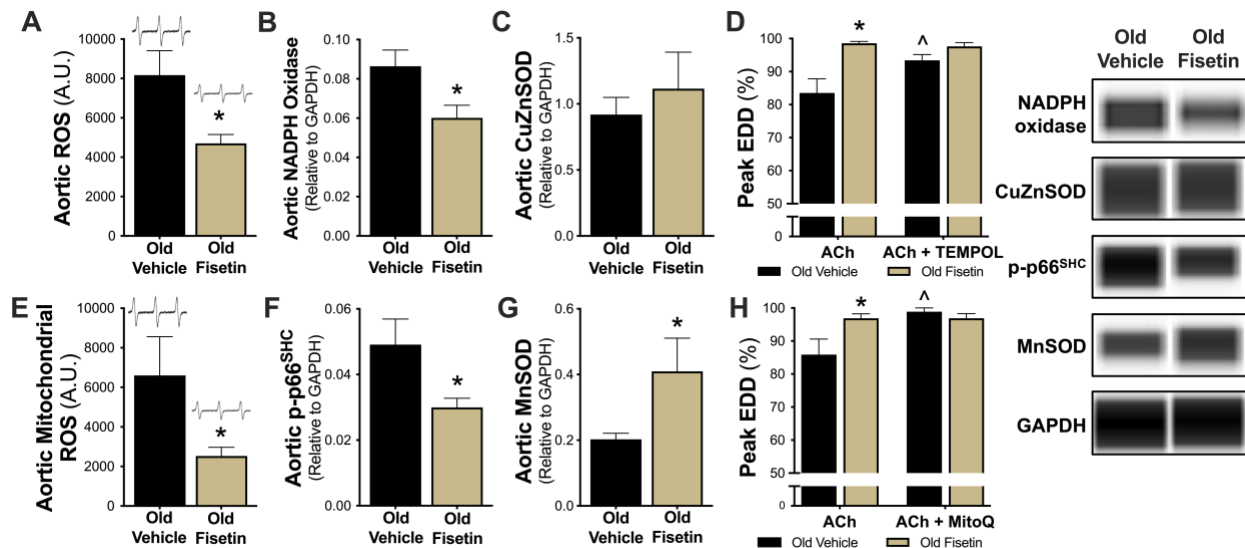


Figure 4. Fisetin improves endothelial function by ameliorating whole- cell and mitochondrial oxidative stress. Whole- cell aortic reactive oxygen species (ROS) levels (n = 6–8) (**A**). Aortic protein abundance of NADPH oxidase (n = 8) (**B**) and CuZn superoxide dismutase (SOD; n = 11–12) (**C**) with representative virtual blot bands, right. Endothelium-dependent dilation (EDD) in isolated carotid arteries in response to acetylcholine (ACh) in the presence or absence of the SOD mimetic, TEMPOL (1 mM, 60 min preincubation; n = 7–8) (**D**). Aortic mitochondrial ROS levels (n = 6–10) (**E**). Aortic protein abundance of phosphorylated p66SHC (p- p66SHC; n = 11–12) (**F**) and MnSOD (n = 11–12) (**G**) with representative virtual blot bands, right. EDD in carotid arteries in response to ACh in the presence or absence of the mitochondrial-specific antioxidant, MitoQ (1 μ M; 60 min preincubation; n = 5–7) (**H**). Values represent mean $\pm$ SEM. * $P < 0.05$ old vehicle versus old fisetin; ^ $P < 0.05$ versus ACh alone.

Fisetin Reduces Aortic Stiffness in Old Mice: Effects on Intrinsic Mechanical Wall Stiffness and Components of the Arterial Wall.

Aortic PWV. To determine whether intermittent supplementation with fisetin lowers aortic stiffness in old mice, we serially assessed aortic PWV, an established analogous measure to the reference standard carotid-femoral PWV in humans, at baseline and after treatment in both the vehicle- and fisetin-supplemented mice. Aortic PWV was matched at baseline in the two

groups ($P=0.586$). Fisetin supplementation lowered aortic PWV by ~20% (pre-, 425 ± 7 cm/s vs. post-, 335 ± 6 cm/s, $P<0.001$), whereas no significant change was observed over time in the vehicle-supplemented group (pre-, 425 ± 7 cm/s vs. post-, 433 ± 5 cm/s, $P=0.403$; **Figure 5A**). These effects occurred without obvious changes in systolic or diastolic (tail cuff) blood pressures (**Table 1** and **Table S1**), which is in line with previous studies suggesting that age-related aortic stiffness can be reduced independent of changes in blood pressure^{9,51}.

Intrinsic Mechanical Wall Stiffness (Elastic Modulus). After establishing reductions in aortic PWV with intermittent fisetin supplementation *in vivo*, we sought to determine whether structural modifications to the arterial wall may have contributed to this effect. To do so, we measured the elastic modulus of aortic rings isolated from old vehicle- and fisetin-supplemented mice. Elastic modulus, defined as the association between the change in stress on the arterial wall in response to an increase in strain (stretch), is a key mechanism underlying increases in PWV with aging^{13,14}. Because this measure is conducted *ex vivo*, it cannot be influenced by humoral factors or blood pressure, allowing us to directly assess the influence of intermittent fisetin supplementation on arterial wall mechanical stiffness. **Figure 5B** shows a representative stress-strain curve of the aortic elastic modulus from old vehicle- and fisetin-supplemented mice. Aortic elastic modulus was ~20% lower in old fisetin- vs. vehicle-supplemented mice (vehicle, 3342 ± 246 kPa vs. fisetin, 2736 ± 170 kPa, $P=0.012$; **Figure 5B**) indicating that fisetin reduced the intrinsic mechanical wall stiffness of the aorta.

Cellular Senescence-Mediated Intrinsic Mechanical Wall Stiffness. To determine if fisetin reduced aortic wall stiffness by targeting cellular senescence, we used a “pharmacodissection” approach, as performed previously by our laboratory^{9,52}. Here, elastic modulus was measured in aorta rings isolated from old vehicle- and fisetin-supplemented p16-3MR mice after

a 48hr incubation in standard culture (control) media with or without 5 μ M GCV. In aorta rings from old vehicle-treated mice, GCV-incubated aortic rings had a lower elastic modulus compared to aorta rings incubated in control media, indicating that cellular senescence burden contributes to the intrinsic wall stiffness of the aorta (media only, 4653 $\pm$ 482 kPa vs. GCV, 3036 $\pm$ 231 kPa, $P=0.009$; **Figure 5C**). After incubation in control media, fisetin-supplemented mice had lower initial elastic modulus compared to old vehicle-treated mice (vehicle, 4653 $\pm$ 482 kPa vs. fisetin, 3537 $\pm$ 289 kPa, $P=0.054$) and there was no further reduction following genetic senescent cell clearance with GCV, suggesting that fisetin effectively cleared excess vascular senescent cells to lower aortic stiffness (fisetin, media only, 3537 $\pm$ 289 kPa vs. GCV, 3346 $\pm$ 278 kPa, $P=0.640$; **Figure 5C**). Taken together, these data suggest that cellular senescence influences arterial stiffness in old mice and that fisetin can modulate large elastic artery stiffening by reducing cellular senescence-mediated increases in intrinsic mechanical wall stiffness.

We next used a complimentary approach to further establish senolysis as a primary mechanism of reductions in arterial stiffness with fisetin in old mice. We assessed elastic modulus in aorta rings isolated from old vehicle- and fisetin-supplemented wildtype mice after *ex vivo* incubation with the well-established synthetic senolytic ABT-263^{53,54}. As with genetic senescent cell clearance, we found that clearance of senescent cells with ABT-263 lowered the elastic modulus in aorta rings isolated from vehicle-treated mice (vehicle, media only, 4860 $\pm$ 344 kPa vs. ABT-263, 3210 $\pm$ 210 kPa, $P=0.002$; **Figure 5D**). ABT-263 exposure did not further reduce the elastic modulus in aorta rings obtained from fisetin-supplemented mice (fisetin, media only, 2977 $\pm$ 216 kPa vs. ABT-263, 3323 $\pm$ 199 kPa, $P=0.329$; **Figure 5D**). Collectively, these results implicate cellular senescence as an underlying mechanism of large elastic artery stiffening

in aged mice and suggest that fisetin decreases aortic intrinsic mechanical wall stiffness by reducing cellular senescence.

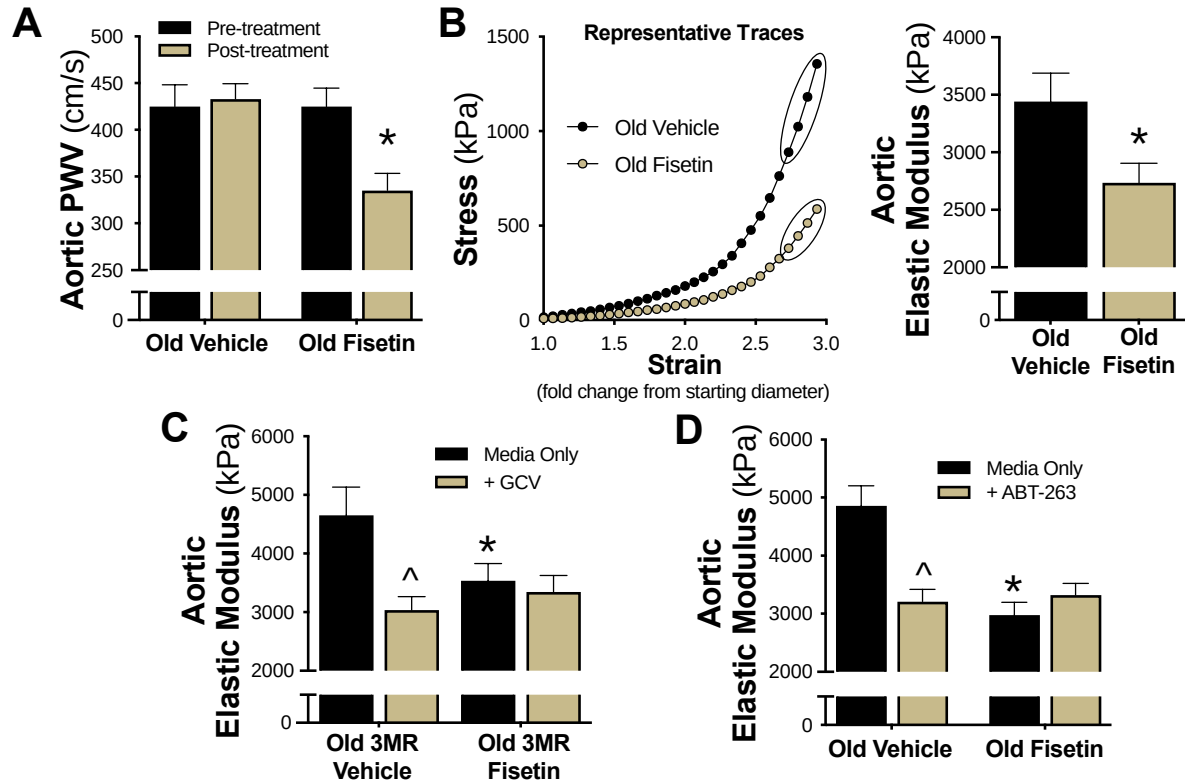


Figure 5. Fisetin reduces arterial stiffness in old mice by improving aortic intrinsic mechanical wall stiffness. Aortic pulse wave velocity (PWV) was measured pre- and post-intervention (n = 9–10) (**A**). Representative stress–strain curve for determination of ex vivo intrinsic mechanical wall stiffness which was assessed by elastic modulus in aortic rings (calculated as the slope of the final four points in the stress–strain curve) (n = 7) (**B**). Elastic modulus was assessed after ex vivo genetic (GCV) (n = 8–10) (**C**) and pharmacological (ABT-263) (n = 9–10) (**D**) suppression of cellular senescence in aortas obtained from old p16-3MR (3MR) and wild-type mice, respectively. Values represent mean $\pm$ SEM. * P < 0.05 old vehicle versus old fisetin; ^ P < 0.05 versus media only.

Fisetin-Induced Remodeling of the Arterial Wall. To gain insight into the downstream mechanisms by which fisetin reduces arterial stiffness, we measured the aortic abundance of AGEs and structural proteins using immunoblotting. The abundance of AGEs, which increases aortic stiffening by forming crosslinks in structural proteins¹⁰, was 36% lower in aortas from fisetin- vs. vehicle-supplemented mice (vehicle, 0.022 ± 0.004 vs. fisetin, 0.014 ± 0.001 CU, $P=0.030$; **Figure 6A** and **Figure S2G**). We also observed a reduction in the major arterial isoform of collagen (collagen-1), a protein that provides stiffness to the aortic wall (vehicle, 1.379 ± 0.271 vs. fisetin, 0.772 ± 0.123 CU, $P=0.070$; **Figure 6B** and **Figure S2H**). In contrast, whole aorta protein abundance of α -elastin, the primary structural protein conferring the elasticity of the arterial wall, was unaffected by fisetin treatment (vehicle, 0.012 ± 0.001 vs. fisetin, 0.013 ± 0.001 CU, $P=0.379$; **Figure 6C** and **Figure S2I**). To gain insight into the location of these structural changes within the arterial wall, we performed immunohistochemistry on aorta sections. Reductions in the abundance of AGEs and collagen-1 appeared to be specific to the adventitial region of the aortic wall, which is the primary load-bearing component of the arteries under physiological conditions (**Figure 6D-E**). Together, these data suggest that intermittent fisetin supplementation may reduce aortic stiffness in part by favorably modulating components of the arterial wall.

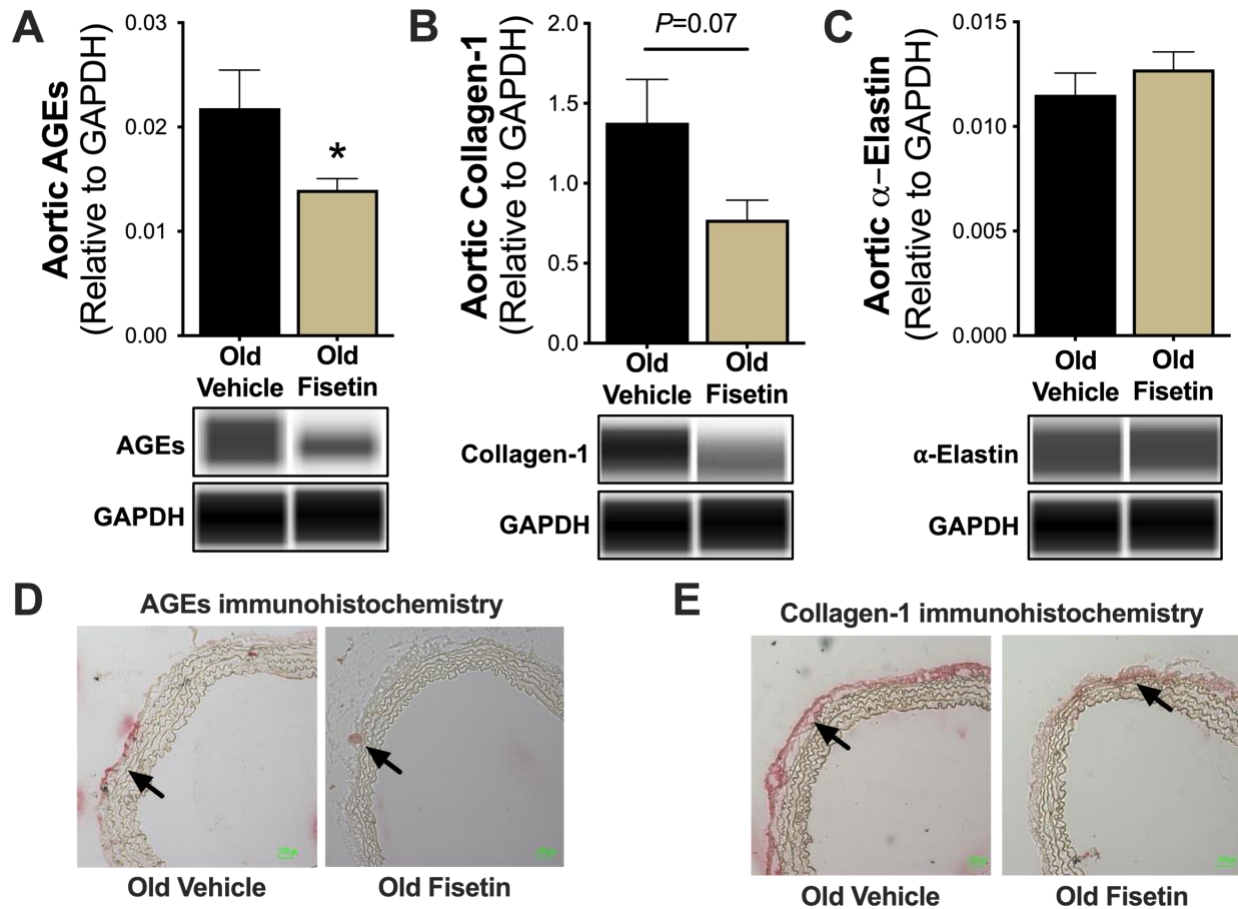


Figure 6. Fisetin reduces aortic intrinsic mechanical wall stiffness by favorably remodeling components of the arterial wall. Aortic protein abundance of advanced glycation end products (AGEs) (n = 9) (A), collagen- 1 (n = 9) (B), and α- elastin (n = 9) (C) with representative virtual blot bands. Representative immunohistochemical staining of aortic AGEs (D) and collagen- 1 (E). Arrows denote protein accumulation in the medial- adventitial layer. Values represent mean ± SEM. * $P < 0.05$ old vehicle versus old fisetin.

DISCUSSION

Our findings demonstrate that fisetin reduces vascular cell senescence and improves arterial function in old mice. Using cell culture models, we first identified a concentration of fisetin that reduced cellular senescence in late-passaged endothelial cells without affecting non-senescent cells. In old mice, we next found that oral intermittent fisetin supplementation lowered markers of cellular senescence and SASP factors in the vasculature. Fisetin supplementation also improved two key manifestations of age-related arterial dysfunction, vascular endothelial dysfunction and arterial stiffness, at least in part, by decreasing cellular senescence. Further, we found that fisetin improved vascular endothelial function by increasing NO bioavailability and reducing whole cell and mitochondrial ROS bioactivity and that reductions in arterial stiffness were accompanied with favorable remodeling of the arterial wall. These findings identify oral intermittent fisetin supplementation as a promising therapeutic strategy for clinical translation to improve arterial function with aging.

Fisetin and vascular cell senescence.

Cellular senescence and the SASP increase with advancing age in the vasculature^{6,55-57}. Fisetin has been shown to reduce cellular senescence in fibroblasts and endothelial cells brought to senescence via ionizing radiation in culture⁵⁸ or primary fibroblasts from progeroid mice²⁷. In vivo, fisetin supplementation reduced markers of cellular senescence in various tissues in old mice²⁷ and other mammalian species⁵⁹. Here, we have extended these observations and showed that fisetin at a concentration of 1 μ M reduced viability of HUVECs brought to senescence via replicative exhaustion, with minimal effects in non-senescent cells. This concentration is lower than previous concentrations shown to reduce senescent cell viability in culture⁵⁸, an observation likely related to differences in the induction of cellular senescence. We further

showed that 1 μ M fisetin reduced multiple markers of cellular senescence in venous and arterial cells *in vitro*, establishing the senolytic effects of this concentration.

We then translated our findings to an *in vivo* setting using an oral dose of 100mg/kg/day given that pharmacokinetic studies demonstrate peak plasma levels reach $\sim$ 1 μ M following oral ingestion of fisetin at this dose⁴⁴. We selected an established intermittent dosing paradigm for oral supplementation to remove excess senescent cells but not chronically inhibit cellular senescence activation and interfere with its principal physiological roles, and to distinguish between the senolytic effects of fisetin and any off-target/acute effects of the compound on arterial function²⁷. Consistent with our findings in endothelial cell culture, oral intermittent fisetin supplementation *in vivo* lowered vascular cell senescence and SASP markers in old mice.

Fisetin and vascular endothelial function.

Vascular endothelial dysfunction is a key manifestation of age-related arterial dysfunction and a major antecedent to overt CVDs, including atherosclerosis and occlusive stroke¹. In the current study, we showed that intermittent supplementation with the emerging senolytic fisetin improved endothelial function in old wildtype mice. To demonstrate a contributing role for reductions in cellular senescence in mediating improvements in endothelial function with fisetin, we also conducted fisetin supplementation experiments in the p16-3MR transgenic mouse model, to show that improvements in endothelial function with fisetin were mediated, in part, by a suppression of vascular cell senescence.

Age-associated endothelial dysfunction is largely mediated by reduced NO bioavailability, which is driven mainly by excess ROS, a key source being dysregulated mitochondria^{3,60-63}. Vascular cell senescence is associated with diminished NO bioavailability and increased whole cell and mitochondrial ROS-related oxidative stress⁴⁵. Further,

mitochondrial dysfunction, including increased ROS production, is a key feature of senescent cells⁶⁴. We showed that fisetin-induced improvements in endothelial function were due to improvements in NO bioavailability and a reduction in whole cell- and mitochondrial ROS-related suppression of endothelial function. The reductions of whole cell and mitochondrial ROS appeared to be associated with a combination of a reduction in pro-oxidant signaling molecules (*e.g.*, NADPH oxidase and phosphorylated-p66^{SHC}) and an increase in antioxidant enzyme defenses (*e.g.*, MnSOD). Collectively, these findings demonstrate that oral intermittent fisetin supplementation can target excess cellular senescence to improve NO bioavailability and reduce oxidative stress to ultimately improve vascular endothelial function in old age.

Fisetin and large elastic artery stiffness.

Increases in large elastic artery stiffness (*i.e.*, increased aortic PWV and intrinsic mechanical wall stiffness) with aging is a major independent predictor of age-associated clinical CVD⁶⁵, impaired glucose tolerance⁶⁶, kidney dysfunction⁶⁷, cognitive impairment⁶⁸ and diabetic retinopathy⁶⁹. Fisetin supplementation reduced arterial stiffness (aortic PWV) in old wildtype mice by reducing aortic intrinsic mechanical wall stiffness. Using the p16-3MR mouse model, we showed that reductions in excess cell senescence with fisetin contributed to decreased aortic intrinsic mechanical wall stiffness to reduce arterial stiffness in old mice.

Pathological remodeling of the arterial wall with aging includes: (1) greater deposition of collagen-1, the main load-bearing protein in the aortic wall; (2) fragmentation and degradation of elastin, the primary structural protein that confers elasticity to the arteries; and (3) increased formation and expression of AGEs, which together further promoting greater arterial stiffness^{9,10}. Fisetin supplementation tended to reduce levels of collagen-1 and AGEs in the adventitial regions of the arterial wall of old mice. The combination of lower collagen-1 and AGEs levels

following fisetin supplementation, despite no changes in elastin, could confer reductions in aortic stiffness. Together, these findings provide evidence that oral intermittent fisetin supplementation leads to favorable remodeling of the arterial wall, which is associated with lower aortic stiffness, and suggests that fisetin may be a viable therapeutic strategy to translate to older adults to lower arterial stiffness.

STUDY LIMITATIONS

It is possible that some of our observed effects were independent of the senolytic actions of fisetin, as fisetin modulates a variety of cell signaling pathways and has antioxidant and anti-inflammatory properties^{26,28}. However, we observed reductions in multiple markers of cellular senescence and the SASP with fisetin, our measurements in mice were made one week after the final dose of fisetin (i.e., long after the compound is cleared from circulation⁴³), and, as cellular senescence and the SASP promote arterial inflammation and oxidative stress^{38,45}, fisetin-induced senolysis is likely an upstream, contributing mechanism by which fisetin exerts antioxidant and anti-inflammatory effects²⁷. We also utilized genetic clearance of excess senescent cells to demonstrate that reductions in vascular cell senescence contribute to improvements in arterial function with fisetin treatment in old mice. We recognize that there are limitations to the p16-3MR mouse model, such as elimination of non-senescent p16^{INK4A} positive cells and limited ability to target senescent cells that do not express p16^{INK4A}¹⁸. Nonetheless, senescent endothelial cells highly express p16^{INK4A} and, as such, the p16-3MR mouse model is a highly relevant model to study the role of cellular senescence in arterial aging²².

CONCLUSIONS

Overall, our findings provide the first evidence that oral intermittent fisetin supplementation reverses vascular endothelial dysfunction and large elastic artery stiffness through the suppression of excess cellular senescence, inflammation, and oxidative stress. Although fisetin is found in a variety of fruits and vegetables, fisetin-rich diets are not feasible due to content variations in food. Nonetheless, fisetin is commercially available as a dietary supplement and is reported to be safe for human consumption²⁸⁻³². As such, our preclinical findings provide the necessary proof-of-concept evidence of efficacy supporting the need for future clinical studies assessing the potential of fisetin to treat arterial dysfunction in older adults and to potentially reduce CVD risk with aging.

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SUPPLEMENTAL METHODS

Cell Culture Experiments.

Cell Viability Assay. Cell viability was assayed on a BioTek Synergy Neo2 using CellTiterGlo (Promega, Madison, WI, Cat. No. G7570) according to the manufacturer's instructions. Briefly, cells were plated in a white-bottom 384 well plate (Corning Inc., Corning, NY) and treated with the indicated concentrations of fisetin, as described¹. Viability was assessed 48hr after treatment and normalized to the DMSO control. IC₅₀ values were calculated using GraphPad Prism. All values are the mean of 4-8 replicate measurements.

Senescence-associated β -galactosidase (SA- β -gal) staining. SA- β -gal staining was performed using the Senescence Detection Kit (Abcam, Cambridge, UK, Cat. No. ab65351) according to the manufacturer's instructions. Briefly, after overnight seeding, late passaged human umbilical vein endothelial cells (HUVECs) were treated with 1 μ M fisetin in media for 48hr. Following the incubation period, early passaged (control), late passaged (senescent), and senescent cells treated with 1 μ M were washed, fixed, and incubated with the X-gal solution overnight. Images were captured using bright-field microscopy at 20X magnification and quantified using ImageJ, as described².

Assessment of Reactive Oxygen Species (ROS) levels. Oxidative stress was detected in cells using the CellROX Deep Red ROS probe (Invitrogen, Waltham, MA) according to the manufacturer's instructions. Briefly, cells were plated in a high content image plate with 0.2mm glass bottom (Corning, Kennebunk, ME) coated with bovine gelatin substrate (Attachment Factor Solution; Cell Applications, Inc., San Diego, CA) to facilitate cell adhesion. After a 24hr incubation, attached cells were washed once with Hanks' balanced salt solution (HBSS) and treated with 1 μ M fisetin for 4hr. Attached cells were then washed twice with HBSS and

incubated with 9.2 μ M CellROX Deep Red ROS probe for 60 minutes. After dye removal and three washes with HBSS, cells were left in 100 μ L of HBSS for subsequent imaging. Microscopy was carried out on an EVOS FL microscope (Invitrogen, Carlsbad, CA) set at 20X magnification. Cells were selected for imaging based on characteristic endothelial cell and nuclear morphology. Cell images were analyzed with Fiji (National Institutes of Health) and image intensity was averaged over the entire cell area³.

Cell and Aortic Gene Expression. mRNA gene expression was measured in human aortic endothelial cells (HAECs) and segments of thoracic aorta following mechanical homogenization. RNA was extracted using the RNeasy mini kit (Qiagen, Hilden, Germany). cDNA was synthesized using the iScript cDNA synthesis kit (Bio-Rad Laboratories, Hercules, CA). Transcripts of cellular senescence and senescence-associated secretory phenotype (SASP) markers (primer sequences reported in **Table S2**) were analyzed using a StepOnePlus Real-Time PCR System (Applied Biosystems, Waltham, MA) in 96-well plates and the Taqman OpenArray (Applied Biosystems, Waltham, MA) was used as a master mix, as described⁴. SimpleSeq DNA sequencing (Quintara Biosciences, Cambridge, MA) was used to validate PCR products.

Animal Experiments.

Animals and Experimental Design. Male C57Bl/6N (wildtype) mice were obtained from the National Institute of Aging colony (maintained by Charles River, Wilmington, MA). Wildtype mice were allowed to acclimate to our facilities for 4 weeks before beginning the study. Male and female p16-3MR mice were bred and aged in our mouse colony at the University of Colorado Boulder. These mice carry a trimodal fusion protein (3MR) under the control of the p16^{INK4A} promoter which allows for selective genetic clearance of p16^{INK4A}-positive (p16⁺) senescent cells by administering the antiviral agent ganciclovir (GCV)⁵. For the

duration of the study, all mice were single housed at the University of Colorado Boulder animal facility with a 12hr:12hr light-dark cycle and allowed ad libitum access to an irradiated, fixed, and open rodent chow (Inotiv/Envigo 7917, stored at room temperature).

For the intervention period, wildtype and p16-3MR mice were randomly assigned to receive vehicle (10% Ethanol, 30% PEG400 and 60% Phosal 50 PG) or fisetin (100 mg/kg/day in vehicle). For wildtype mice, 18 mice received vehicle and 19 mice received fisetin, and for p16-3MR mice, 27 mice received vehicle and 35 mice received fisetin. Treatment was administered via oral gavage using an intermittent dosing paradigm – one week on; two weeks off; one week on. Mice were sacrificed one to two weeks following the final dose to rule out any acute effects of the compound as the terminal half-life of fisetin is ~3.1 hours in plasma⁶. Animals were administered vehicle or fisetin daily during the treatment period. Treatment groups were matched for baseline body weight and aortic PWV. Throughout this intervention period, 1 old wildtype mouse and 11 p16-3MR mice died as a result of typical age-related attrition, which resulted in a final sample size of old Vehicle, n = 17; old Fisetin, n = 19; old 3MR Vehicle, n = 22; and old 3MR Fisetin, n = 29.

In Vivo Aortic Stiffness and Arterial Blood Pressure. Aortic stiffness was assessed using the reference standard non-invasive *in vivo* measure, aortic PWV, one week before (pre) and one week after (post) the intervention, as previously described⁷⁻¹⁰. Briefly, mice were placed under light isoflurane anesthesia (1.0-2.5%) and positioned supine on a warmed heat pad. Front- and hind-limb paws were then secured to corresponding ECG electrodes. Two Doppler probes were then placed on the skin at the transverse aortic arch and the abdominal aorta. Once clear R-waves were registered, three repeated 2-second ultrasound tracings were recorded and average pre-ejection time (*i.e.*, time between the R-wave of the ECG to the foot of the Doppler signal)

was determined for each location. To calculate aortic PWV, the distance between the two probes was divided by the difference between the transverse aortic arch and abdominal aorta pre-ejection times ($\text{time}_{\text{abdominal}} - \text{time}_{\text{arch}}$) and is reported as centimeters/second (cm/s). To examine the potential contribution of changes in arterial blood pressure to any treatment-related differences in aortic PWV, systolic and diastolic blood pressures were assessed using a CODA noninvasive tail-cuff system (Kent Scientific, Torrington, CT), as we have described previously⁷⁻¹⁰. Briefly, the pressure measurements from 20 collection cycles (following 5 acclimation cycles) for 3 consecutive days were averaged per mouse at each timepoint.

Vascular endothelial function: endothelium-dependent dilation (EDD), cellular senescence-mediated suppression of EDD, NO-mediated EDD, whole-cell and mitochondrial-specific ROS suppression of EDD, and endothelium-independent dilation (EID). Upon completion of *in vivo* measures, mice were euthanized via cardiac exsanguination under inhaled isoflurane anesthesia, and carotid arteries were immediately excised. Vasodilatory function was measured via *ex vivo* carotid artery EDD and EID in response to increasing doses of acetylcholine (ACh) and sodium nitroprusside (SNP), respectively, as described previously⁷⁻¹⁰. In brief, after vessels were pre-constricted with phenylephrine (PE; 2mM; Sigma-Aldrich, Cat. No. P6126), EDD was assessed by measuring increases in luminal diameter in response to increasing concentrations of ACh (1×10^{-9} to 1×10^{-4} M; Sigma-Aldrich, Cat. No. A6625) with and without *ex vivo* coadministration of GCV (5 μ M, 180min pre-incubation; Sigma Aldrich, St. Louis, MO, Cat. No. G2536), tonic ROS scavenger TEMPOL (1mM, 60min pre-incubation; Sigma-Aldrich, Cat. No. 2226-96-2), mitochondrial-specific ROS scavenger MitoQ (1 μ M, 60min pre-incubation; Antipodean Pharmaceuticals, Inc., Menlo Park, CA, USA), or the NO synthase inhibitor *N*^G-nitro-L-arginine methylester (L-NAME, 0.1mM, 30min pre-incubation;

Sigma-Aldrich, Cat. No. N5751). NO-mediated dilation was then calculated as the difference between maximum EDD to ACh alone and in the presence of ACh and L-NAME:

$$\text{NO-mediated dilation (\%)} = \text{Maximum dilation}_{\text{ACh}} - \text{Maximum dilation}_{\text{ACh+L-NAME}}$$

Following EDD and pharamo-dissections of EDD, EID was assessed by measuring the increase in luminal diameter in response to increasing concentrations of SNP, an exogenous NO donor (1×10^{-10} to 1×10^{-4} M; Sigma-Aldrich, Cat. No. 13755-38-9). All dose response data are presented as percent dilation relative to maximum diameter to account for differences in baseline vessel diameter.

Aortic ROS Production. Whole-cell aortic ROS production was assessed using the spin probe 1-hydroxy-3-methoxycarbonyl-2,2,5,5-tetramethylpyrrolidine (CMH; Enzo Life Sciences, Farmingdale, NY, Cat. No. ALX-430-078) and mitochondrial-specific ROS production was assessed using the spin probe 1-hydroxy-4-[2-triphenylphosphonio-acetamido]-2,2,6,6-tetramethylpiperidine (mitoTEMPO-H; Enzo Life Sciences, Cat. No. ALX-430-171-M005) by electron paramagnetic resonance (EPR) spectrometry, as previously described⁷⁻¹⁰. In short, two 1-mm aortic rings were washed in warm physiological saline solution and incubated in Krebs/HEPES buffer, consisting of 99mM NaCl, 4.7mM KCl, 1.87mM CaCl₂, 1.2mM MgSO₄, 25mM NaHCO₃, 1.03mM KH₂PO₄, 20mM Na-HEPES, 11.1mM glucose, 0.1mM diethylenetriaminepenta-acetic acid, 0.0035mM sodium diethyldithiocarbamate, and Chelex (Sigma-Aldrich, Cat. No. C7901), containing 0.5mM CMH or mitoTEMPO-H at 37°C for 60min. Samples were analyzed using aMS300 Xband EPR spectrometer (Magnettech, Berlin, Germany) with the following instrument parameters: B0-Field, 3350G; sweep, 80G; sweep time, 60s; modulation, 3000mG; MWatten, 7dB; gain, 500.

Aortic Intrinsic Mechanical Wall Stiffness (Elastic Modulus) and Cellular Senescence-Mediated Changes in Aortic Intrinsic Mechanical Wall Stiffness. Aortas were promptly excised from the mice following carotid artery excision, rinsed with cold physiological saline solution, and cleared of any remnant perivascular adipose and connective tissue. To measure *ex vivo* aortic stiffness, two thoracic aorta samples (~1 mm in length) were cut and used to determine intrinsic mechanical stiffness via pin myography as we have previously described⁷⁻⁹. In short, aorta samples were placed in heated (37°C) baths filled with calcium-free, phosphate-buffered saline (PBS). The samples were then mounted on two wire prongs, followed by three rounds of pre-stretching. Once pre-stretching was complete, aortic ring diameter was increased until 1mN of force was reached and incrementally increased by 5 μ m every 3 min thereafter until failure. The force corresponding to each stretching interval was recorded and used to calculate stress and strain. A stress-strain curve was then generated using the following equations:

where d is diameter and d_i is initial diameter.

where L is one-dimensional load, H is intima media thickness, and D is vessel length.

The elastic modulus of the stress-strain curve was determined as the slope of the linear regression fit to the final four points of the stress-strain curve, as previously reported by our laboratory⁷⁻¹⁰. To assess the stiffening role of cellular senescence and the beneficial effects of fisetin on intrinsic mechanical stiffness, wildtype aorta rings were pre-incubated with 2.5 μ M ABT-263, a synthetic senolytic, and p16-3MR aortas were pre-incubated with 5 μ M GCV, which are concentrations that have been shown to reduce the abundance of senescent cells *ex vivo*^{11,12}, for 48hr prior to the following measurements. Aortic intima media thickness and diameter were assessed as we have described previously⁷⁻¹⁰. Briefly, aortic rings (1 mm) were frozen in optimal cutting temperature solution and stored at -80°C until the time of sectioning. Aortic sectioning

was performed on a cryostat (7 μm ; Leica CM300, Leica Biosystems, Wetzlar, Germany) at -22°C and sections were visualized, and images were captured with a bright-field microscope. Aortic intima media thickness and diameter were calculated using ImageJ software.

Immunohistochemistry in Mouse Aortas. The localization of collagen-1 and advanced glycation end-products (AGEs) were assessed using ~1mm sections of mouse thoracic aorta which were collected at the time of sacrifice, frozen in OCT compound, and stored at -80°C until the time of sectioning. Samples were later sectioned (7 μm ; Leica CM300, Leica Biosystems, Wetzlar, Germany) and plated in poly-L-lysine coated slides, fixed in 4% paraformaldehyde and stored at -80°C until further use in immunohistochemical analyses. Upon staining, the slides were rehydrated (PBS with 50 mM glycine), washed with PBS, permeabilized (0.1% Triton X-100), and incubated for 20 minutes with 2.5% normal horse serum blocking buffer following 3 washes with PBS. Slides were then incubated with primary antibodies at room temperature for 1 hour, washed 3 times with PBS, incubated with horseradish peroxidase-conjugated secondary antibody for 30 minutes, treated with 1:1 3,3'-Diaminobenzidine (DAB):H₂O₂ for approximately 30 minutes as the samples began to darken, and then immediately washed in PBS and cured overnight with gelvatol. Collagen-1 and AGEs primary antibodies (antibody descriptions reported in **Table S3** were used for aortic protein targets, as previously described^{9,10}. The slides were imaged using a Nikon Eclipse TS100 under identical conditional and analyzed using ImageJ.

Aortic Protein Abundance. Protein abundance was measured in segments of thoracic aorta following mechanical homogenization in radioimmunoprecipitation assay lysis buffer supplemented with protease and phosphatase inhibitors (1mM sodium orthovanadate, 1X complete mini protease inhibitor cocktail tablet [Roche, Mannheim, Germany, Cat. No.

11836153001], 1 mM phenylmethylsulphonyl fluoride, 1:100 Phosphatase Inhibitor Cocktail [Sigma-Aldrich, Cat. No. P2850], 5mM sodium fluoride, and 5mM sodium pyrophosphate). Total protein content was quantified using a bicinchoninic acid assay (Thermo Fisher Scientific, Cat. No. 23225). Next, abundance of AGEs, α -elastin, collagen-1, p16^{INK4A}, total and phosphorylated p66^{SHC}, nicotinamide adenine dinucleotide phosphate (NAPDH) oxidase, CuZn superoxide dismutase (SOD), MnSOD, and Glyceraldehyde 3-phosphate dehydrogenase (GAPDH) (antibody descriptions reported in **Table S3**) were determined by loading 20ng/mL of aortic protein per capillary in a 25-lane (capillary) automated Western blot quantitative analyzer (WES, ProteinSimple, San Jose, CA), according to the manufacturer's guidelines, as described previously ⁷⁻¹⁰, following the validation of these antibodies in test aorta lysates. Secondary antibodies were provided by the manufacturer and used according to the manufacturer's guidelines. A grayscale analysis of the band intensities was then performed to quantify protein abundance using Compass software (ProteinSimple), with target proteins expressed relative to a loading control (GAPDH). Data files may be accessed at DOI: 10.6084/m9.figshare.24061176.

Statistical Analyses. Power calculations were performed using G*power 3.1 (RRID: SCR_013726) for our primary outcome variable, aortic PWV. Previously, our laboratory has obtained effect sizes of 1.35 when comparing aortic PWV between treatment groups. With this effect size, N=6 mice per condition were required to achieve 99% statistical power. Additional mice were studied in each group to ensure sufficient PWV traces were obtained and to account for age-related attrition.

Statistical analyses were conducted using GraphPad Prism version 9.4.0 (GraphPad Software, Inc., San Diego, CA, USA; RRID:SCR_002798). Data were assessed for statistical outliers (ROUT test; Q = 1%), and outliers were excluded from final analyses. All variables were

assessed using unpaired t-test Statistical significance was set to $\alpha=0.05$. Data are presented as mean $\pm$ SEM.

SUPPLEMENTAL TABLES

Table S1. p16-3MR Animal Characteristics

Characteristics	Vehicle	Fisetin
<i>n</i>	15	18
Body mass, g	27.9 $\pm$ 0.7	27.9 $\pm$ 1.0
Heart mass, mg	177 $\pm$ 6	186 $\pm$ 8
Quadriceps mass, mg	179 $\pm$ 9	175 $\pm$ 9
Visceral adipose mass, mg	266 $\pm$ 45	233 $\pm$ 42
Liver mass, g	1.4 $\pm$ 0.1	1.6 $\pm$ 0.1
Spleen mass, mg	109 $\pm$ 27	100 $\pm$ 18
Carotid artery		
<i>Resting diameter, μm</i>	428 $\pm$ 11	427 $\pm$ 9
<i>Maximal diameter, μm</i>	496 $\pm$ 10	493 $\pm$ 7
Aorta		
<i>Diameter, μm</i>	386 $\pm$ 6	412 $\pm$ 13
<i>Intima media thickness, μm</i>	37 $\pm$ 2	33 $\pm$ 1
Systolic blood pressure, mmHg		
<i>Pre</i>	95 $\pm$ 3	96 $\pm$ 2
<i>Post</i>	98 $\pm$ 3	96 $\pm$ 2
Diastolic blood pressure, mmHg		
<i>Pre</i>	65 $\pm$ 3	66 $\pm$ 2
<i>Post</i>	69 $\pm$ 3	67 $\pm$ 2

Values represent mean $\pm$ SEM.

Table S2. Primer sequences.

Gene	Species	Forward primer	Reverse primer
<i>Cdkn2a</i>	Human	GAGCAGCATGGAGCCTTC	CCGTTTTTCGACCCTGAGAG
<i>Cdkn1a</i>	Human	TCACTGTCTTGTACCCTTGTGC	TTTGCTCCTGTGCGGAAC
<i>Gapdh</i>	Human	ATGTTTCGTCATGGGTGTGAA	GGTGCTAAGCAGTTGGTGGT
<i>Cdkn2a</i>	Mouse	CCCAACGCCCCGAACT	GCAGAAGAGCTGCTACGTGAA
<i>Cdkn1a</i>	Mouse	TTGCCAGCAGAATAAAAGGTG	TTTGCTCCTGTGCGGAAC
<i>Serpine1</i>	Mouse	TGGAAGGGCAACATGACCAG	TCAGGCATGCCCAACTTCTC
<i>Lmnbl</i>	Mouse	GAGCCCCAAGAGCATCCAAT	CTGAGAAGGCTCTGCACTGT
<i>Cxcl2</i>	Mouse	CCTGGTTCAGAAAATCATCCA	CTTCCGTTGAGGGACAGC
<i>Vegf</i>	Mouse	AAAAACGAAAGCGCAAGAAA	TTTCTCCGCTCTGAACAAGG
<i>Cxcl1</i>	Mouse	CTGGGATTACCTCAAGAACATC	CAGGGTCAAGGCAAGCCTC
<i>Tnf-α</i>	Mouse	ATGAGAAGTTCCCAAATGGC	CTCCACTTGGTGGTTTGCTA
<i>Mmp3</i>	Mouse	CTCGTGGTACCCACCAAGTC	CGCCAAAAGTGCCTGTCTTT
<i>Ccl2</i>	Mouse	CACTCACCTGCTGCTACTCA	GCTTGGTGACAAAACTACAGC
<i>Plat</i>	Mouse	AAGCATGAGGCATCGTCTCC	ATGCATCGTGGAGGTCTTGG
<i>Gapdh</i>	Mouse	AAGGTCATCCCAGAGCTGAA	CTGCTTCACCACCTTCTTGA

Table S3. Antibodies.

Protein	Assay	Species	Dilution factor	Company	Catalog no.
Advanced glycation end-products (AGEs)	IHC	Rabbit	1:200	Abcam	ab23722
Collagen-1	IHC	Rabbit	1:200	Sigma Aldrich	234167
AGEs	Wes	Rabbit	1:250	Abcam	ab23722
α -Elastin	Wes	Rabbit	1:20	Invitrogen	PA572440
Collagen-1	Wes	Rabbit	1:10	Sigma Aldrich	234167
p16 ^{INK4A}	Wes	Rabbit	1:20	Invitrogen	PA5-20379
p66 ^{SHC}	Wes	Rabbit	1:10	Sigma Aldrich	AV50490
phosphorylated-p66 ^{SHC}	Wes	Mouse	1:10	Abcam	AB54518
Nicotinamide adenine dinucleotide phosphate (NAPDH) oxidase	Wes	Goat	1:200	BD Bioscience	610912
CuZn Superoxide dismutase (SOD)	Wes	Goat	1:500	R&D Systems	AF3787
MnSOD	Wes	Goat	1:50	R&D Systems	AF3419
Glyceraldehyde 3-phosphate dehydrogenase (GAPDH)	Wes	Rabbit	1:200	Cell Signaling	14C10

IHC: Immunohistochemistry; WES: Simple Western's WES Instrument.

SUPPLEMENTAL FIGURES

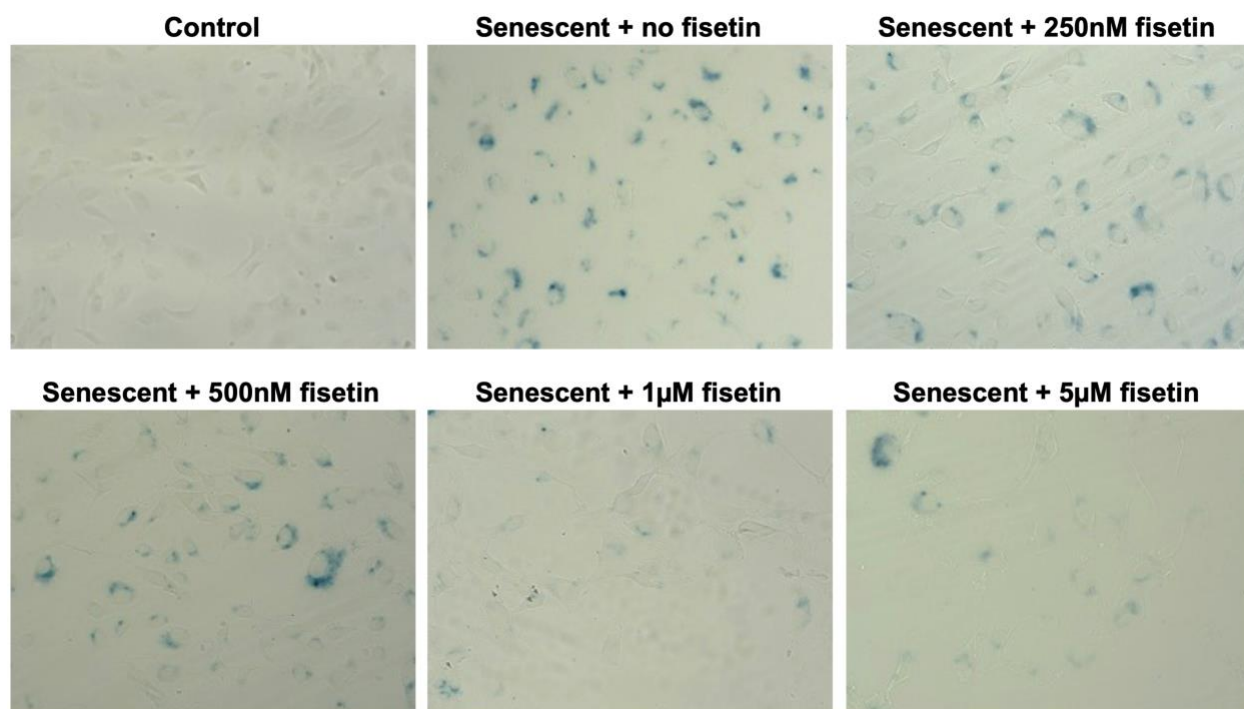


Figure S1. Representative images of senescence-associated beta galactosidase (SA-β-Gal) signal in non-senescent (control) and senescent human umbilical vein endothelial cells to increasing doses of fisetin.

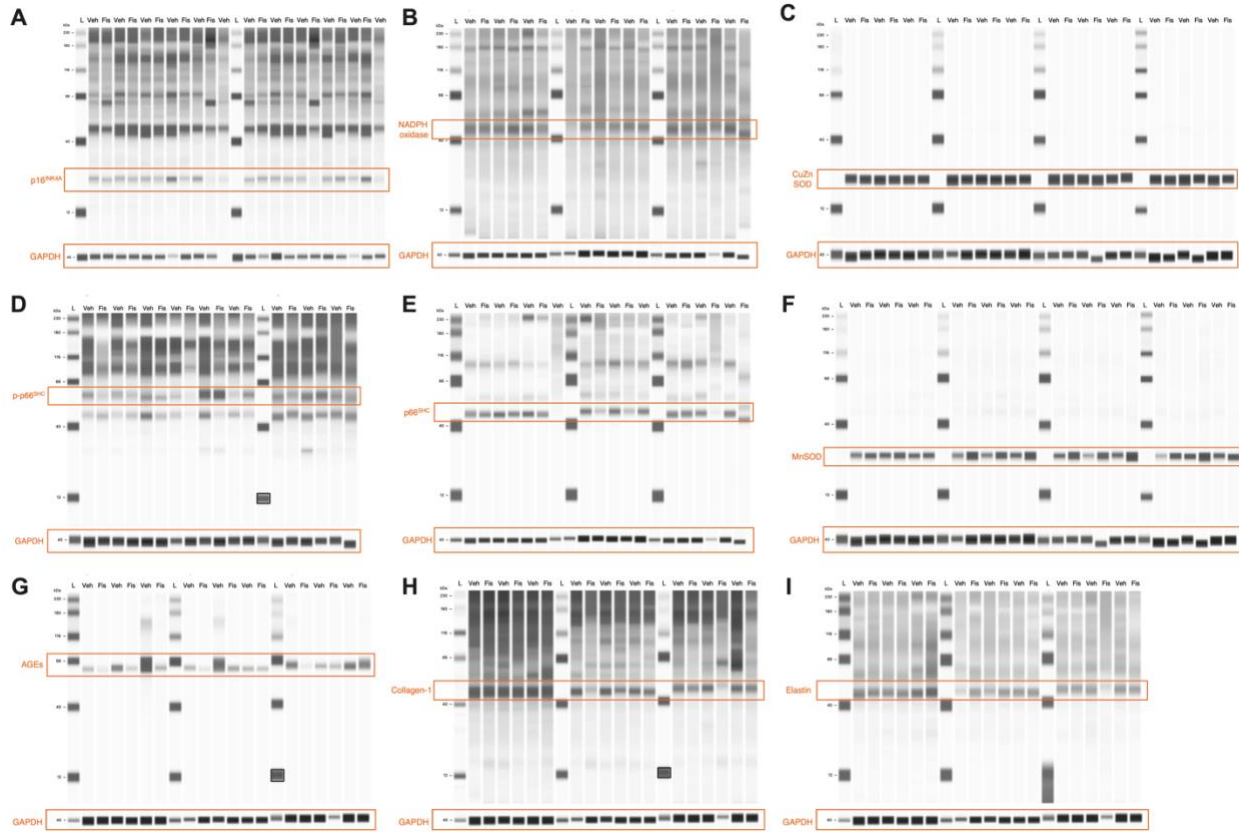


Figure S2. Representative Western Blots in aorta lysates from old mice treated with vehicle (Veh) or fisetin (Fis) for the following protein targets: p16^{INK4A} (n=8) **(A)**, NADPH oxidase (n=8) **(B)**, CuZn superoxide dismutase (SOD; n=11-12) **(C)**, phosphorylated (p) p66^{SHC} (n=11-12) **(D)**, total p66^{SHC} (n=8-10) **(E)**, MnSOD (n=11-12) **(F)**, advanced glycation end-products (AGEs; n=9) **(G)**, collagen-1 (n=9) **(H)**, and elastin (n=9) **(I)**. Individual blots are separated by the protein ladder (L) and all proteins were normalized to GAPDH.

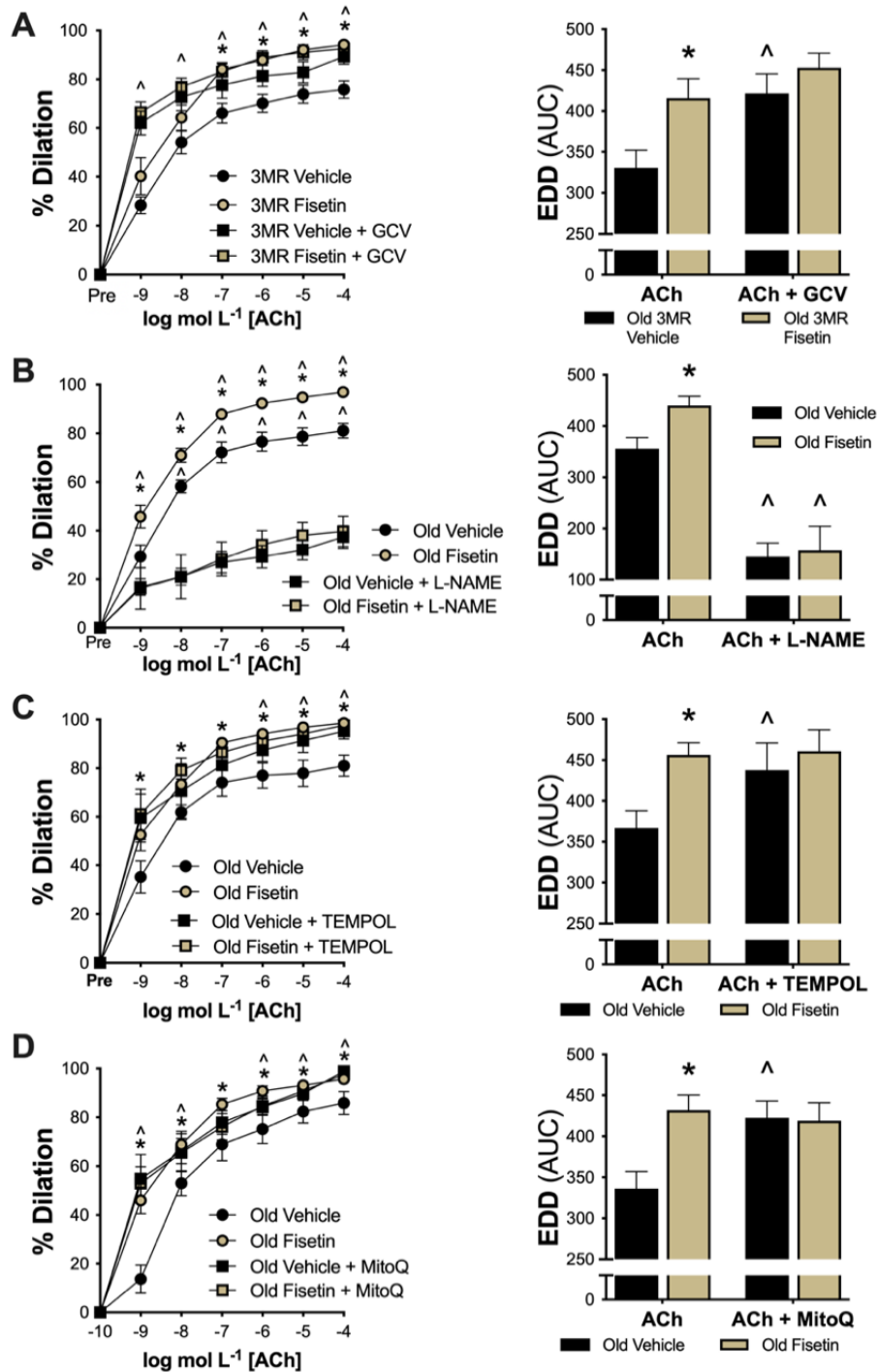


Figure S3. Endothelium-dependent dilation (EDD) curves and area under the curve (AUC) in isolated carotid arteries in response to acetylcholine (ACh; 1×10^{-9} to 1×10^{-4} M; $n=12-18$) and in the presence or absence of ganciclovir (GCV; $5\mu\text{M}$; 180min pre-incubation; $n=9-11$) (**A**), L-NAME (0.1mM ; 30min pre-incubation; $n=15$) (**B**), TEMPOL (1mM , 60min pre-incubation; $n=7-8$) (**C**), and MitoQ ($1\mu\text{M}$; 60min pre-incubation; $n=5-7$) (**D**). Values represent mean $\pm$ SEM. * $P<0.05$ old vehicle vs. old fisetin; ^ $P<0.05$ vs. ACh alone.

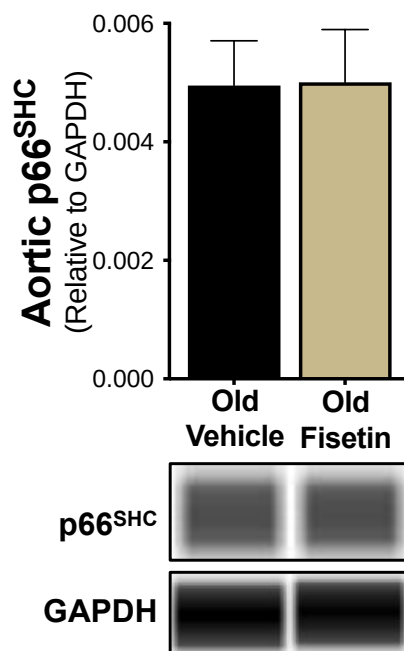


Figure S4. Aortic protein abundance to total p66^{SHC} with representative virtual blot bands. n=8-10/group. Values represent mean $\pm$ SEM.

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CHAPTER IV

Manuscript in preparation, to be submitted for publication at *Nature Aging*

Vascular Cell Senescence and Endothelial Dysfunction: The Role of the Senescence-Associated Secretory Phenotype and CXCL12

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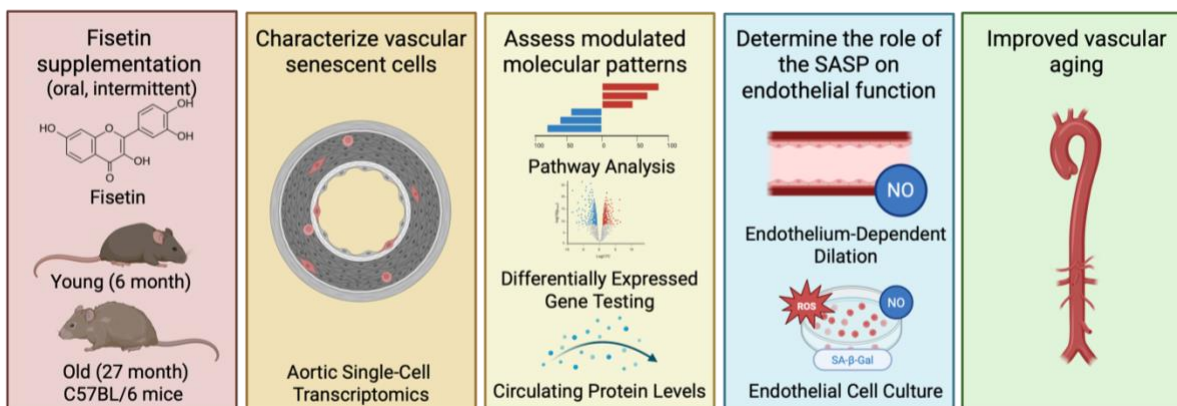
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GRAPHICAL ABSTRACT



ABSTRACT

Cellular senescence and the senescence-associated secretory phenotype (SASP) contribute to age-related endothelial dysfunction, in part, by promoting oxidative stress (a key source being mitochondria) and inflammation, which reduce the bioavailability of the vasodilatory molecule nitric oxide (NO). In the present study, we leveraged aortic single-cell transcriptomics to characterize specific vascular senescent cells and SASP profiles in young (6mo) vs. old (27mo) mice. *In vivo* supplementation with the natural senolytic compound fisetin allowed for the assessment of senescent cell signatures with aging and senolytic intervention. Both aging and senolytic treatment had the greatest effect on endothelial cell populations. As such, we focused on senescent-specific molecular changes in endothelial cells. Senescent endothelial cells demonstrated increased transcription of genes related to cardiovascular pathology and high expression of SASP factors. Of the identified SASP factors that had higher expression with aging, *Cxcl12* was the most highly upregulated with old age and downregulated with senolytic treatment and this pattern persisted in circulating CXCL12 protein levels in plasma. We next sought to determine the causal role of the circulating SASP milieu and CXCL12 in regulating endothelial function with aging and fisetin supplementation. *Ex vivo* exposure of isolated arteries and cultured endothelial cells to plasma from young, old and old fisetin-supplemented mice revealed that the aged circulating SASP milieu directly induced endothelial dysfunction through induction of cellular senescence induction, reduced NO bioactivity, and increased mitochondrial oxidative stress. The circulating SASP milieu-mediated impairments were in part transduced by CXCL12 and prevented by fisetin supplementation. These findings establish the circulating SASP milieu as an underlying mechanism of age-related endothelial dysfunction and identify

CXCL12 as a novel mediator and therapeutic target of cellular senescence-induced endothelial dysfunction with aging.

INTRODUCTION

Advancing age is the primary risk factor for cardiovascular diseases (CVDs), which remain the leading cause of morbidity and mortality worldwide^{1,2}. Vascular dysfunction, characterized in part by impaired vascular endothelial function, is a key antecedent to the development of clinical CVDs with aging³. Cellular senescence is an antagonistic cellular “hallmark” of aging that directly contributes to age-related vascular dysfunction⁴⁻⁷. Basal levels of senescent cells mediate stress responses that contribute to tumor suppression, wound healing and tissue regeneration^{5,7}. However, states of excess cellular senescence promote the development of cancer, chronic inflammation and declines in physiological function and tissue aging^{5,7}. Senescent cells elicit a multifactorial stress response, in part, via the production of the senescence-associated secretory phenotype (SASP)⁸. The SASP can induce cellular and tissue level dysfunction by driving excess superoxide-related oxidative stress (a key source being from the mitochondria), chronic inflammation, and induction of cellular senescence in neighboring cells/tissues (a process termed senescence-induced senescence)⁸. Unique to the vasculature, the SASP from senescent endothelial cells may have systemic effects given the direct contact between the endothelium and the circulating milieu (*i.e.*, bioactive molecules in the bloodstream), a key regulator of endothelial function with aging^{9,10}. Conversely, vascular endothelial cells may also be more susceptible to entering a senescent state given the constant interaction of the endothelium with the circulating SASP milieu^{9,11}. However, vascular cell-specific origins of senescent cells and the SASP are not well characterized. As such, gaining a greater understanding of the cellular senescence- and SASP-related molecular mechanisms may

be exploited as translational diagnostic biomarkers, and may serve as putative therapeutic targets for novel interventions.

In addition to the SASP, characteristic properties of senescent cells include upregulation of the genes *Cdkn1a* and *Cdkn2a* and activation of senescent cell anti-apoptotic pathways (SCAPs)^{9,12}. Senolytics – compounds that selectively target and deactivate SCAPs to enable senescent cell apoptosis^{12,13} – have demonstrated viable efficacy for suppressing excess vascular cell senescence and improving vascular function in old mice^{6,14,15}. Though several senolytic interventions exist, the naturally occurring senolytic compound fisetin has emerged as a potent senolytic in endothelial cells and a promising candidate for clinical translation^{14,16,17}. Importantly, fisetin supplementation is currently being tested in several clinical trials in healthy mid-life/older adults and people with chronic disorders. Despite considerable clinical senolytic testing underway, the understanding of the mechanisms by which senolytic compounds target specific cell types across tissues and the SASP is limited. As such, in the present study, we sought to characterize vascular senescent cells and the SASP in the vasculature from young (6mo) and old (27mo) mice and leveraged *in vivo* fisetin supplementation to explore the molecular, cellular, and physiological mechanisms modulated with aging and senolytic treatment.

The central goal of this study was to: 1) characterize specific cell types within the vasculature that undergo cellular senescence and increased SASP burden with aging; 2) assess the role of the circulating SASP milieu in mediating age-related endothelial dysfunction; 3) identify candidate factors in the circulating SASP milieu that are altered with aging; and 4) determine the causal role of candidate SASP factors in mediating endothelial dysfunction with aging. We first uncovered that endothelial cells are the predominant cell type that become

senescent in old age and can be effectively eliminated with fisetin supplementation. Within senescent endothelial cells, we identified the gene *Cxcl12* to be the most highly upregulated in old age and ameliorated with fisetin supplementation, and observed similar patterns in circulating concentrations of CXCL12 with aging and fisetin supplementation. Finally, we revealed a causal role of the circulating SASP milieu in inducing endothelial dysfunction mediated in part by CXCL12. Overall, these findings establish senescent endothelial cells as promising senolytic targets, demonstrate the SASP as an underlying mechanism of age-related endothelial dysfunction, and identify CXCL12 as a specific circulating SASP factor that mediates vascular cell senescence and endothelial dysfunction with aging that can be modulated with senolytic intervention.

RESULTS

Whole aortic single-cell RNA sequencing identifies senescent cell population with aging.

To gain insights into transcriptomic changes that occur with aging, we leveraged single-cell RNA sequencing (scRNAseq) to compare aortas from young (6mo) and old (27mo) mice. scRNAseq workflows allow for the characterization of gene profiles from distinct cell types of the aorta to identify cell type-specific transcriptomic changes with aging¹⁸. Unsupervised clustering of the merged samples was visualized using uniform manifold approximation and projection (UMAP) to reveal 29 clusters representing 11 distinct cell types in the aorta (**Figure 1A**). Endothelial cells encompassed the largest group of cells (46%) followed by fibroblasts (22%), B cells (10%), macrophages (7%), T cells (6%), vascular smooth muscle cells (VSMC; 4%), neutrophil-like cells (2%), neutrophil cells (1%), erythrocytes (1%), mesothelial cells (<1%) and neural cells (<1%) (**Figure 1B**). Cluster cell type was determined using a

combination of the top 50 genes distinguishing the cluster from others and preestablished cell-specific markers (**Figure 1C**).

To assess age-related changes in aortic cell populations, we performed cell composition analysis between young and old mice (**Figure 1D**). We found that the endothelial cell proportion increased with age ($p < 0.001$) and the T cell proportion decreased with age ($p = 0.002$) (**Figure 1D**). Further, we observed trending differences in VSMC ($p = 0.069$), B cells ($p = 0.109$), neural ($p = 0.146$), and neutrophil-like ($p = 0.127$) (**Figure 1D**). These changes in cell populations were consistent with age-related morphological changes that occur during intima-media thickening and dysregulation of immune cell infiltration into the aorta¹⁹. Together, these population changes are aligned with physiological changes that occur in the vasculature with aging^{19,20}.

We next performed whole-aorta pseudo-bulk RNAseq to determine the general transcriptomic changes in old vs. young mice. Volcano plot analysis of differentially expressed genes (DEGs) in the whole aorta revealed that 5625 genes were upregulated and 2690 genes were downregulated with aging (**Figure 1E**). Reactome pathway analysis revealed cellular senescence as the most highly modulated pathway with aging, as well as pathways related to immune dysregulation and cell cycle arrest (**Figure 1F**). Overall, the pseudo-bulk sequencing assessment revealed a pro-aging transcriptomic shift in the aortic genes and pathway of old mice that is expected with chronological aging.

Given the changes in cellular senescence pathways observed in aortas from old mice, we next sought to identify and characterize senescent cells in the aorta. To do so, we leveraged module scoring of the holistic senescent cell signature, SenMayo²¹, to identify cells expressing high levels of cellular senescence and SASP transcripts. We found that SenMayo predicted

senescent cell populations in several cell types including endothelial cell, macrophage, and neutrophil-like cell populations (**Figure 1G**). As such, we next sought to characterize the senescent cell populations in the endothelial cell population given that: *a*) it had the largest vascular senescent population; *b*) we observed age-related differences in the population composition; and *c*) it is highly involved in age-related vascular endothelial dysfunction.

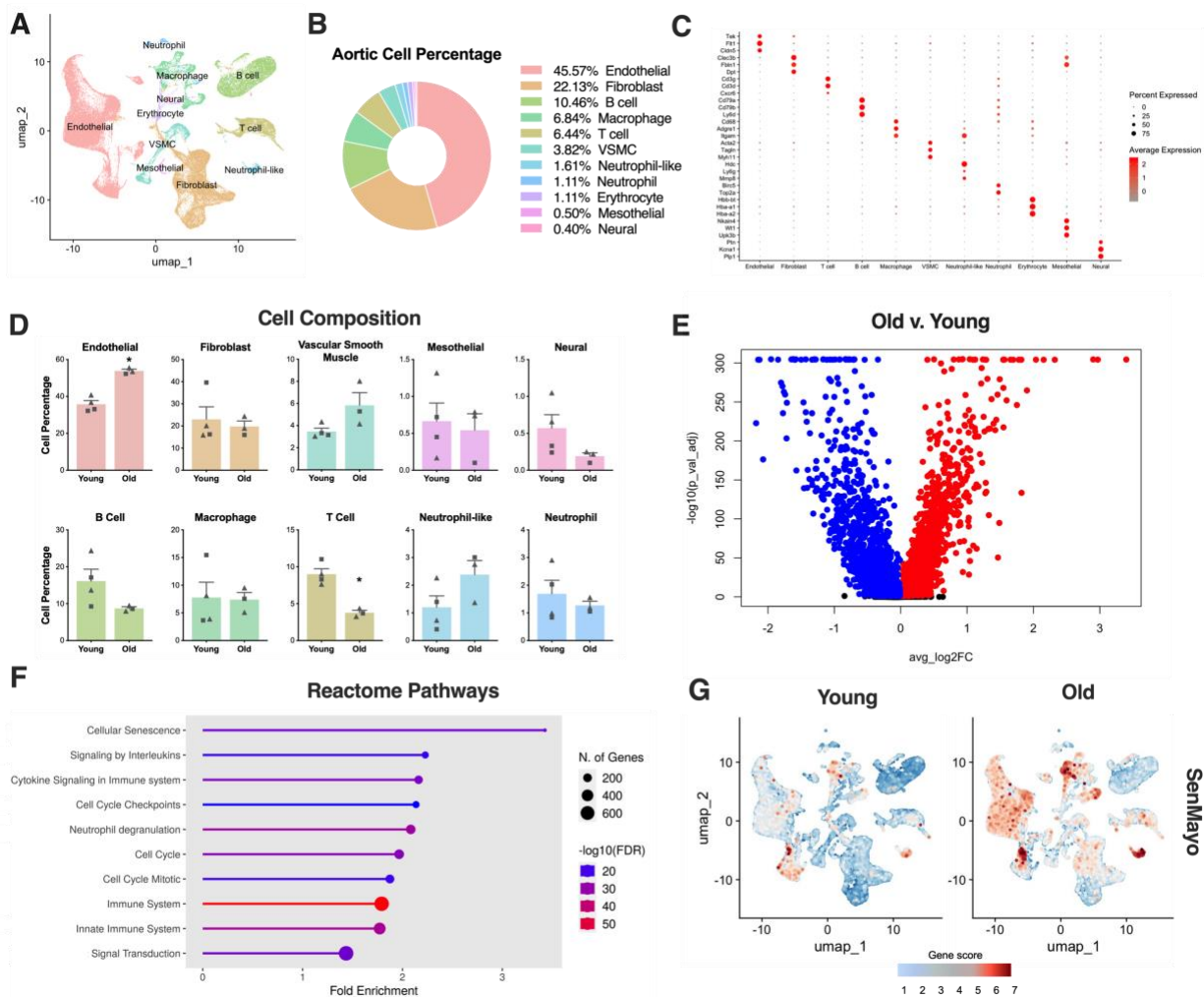


Figure 1. Whole aortic single-cell RNA sequencing identifies senescent cell population with aging. Uniform manifold approximation and projection (UMAP) (**A**) and cell type percentage (**B**) of aortic cells from all the samples color-labeled by cell type. Dot plot analysis of cell type-specific markers used to identify aortic cell types (**C**). Aortic cell composition by cell type (**D**) and volcano plot representation of differentially expressed gene (**E**) in old vs. young samples. Reactome pathway analysis of whole-aortas upregulated genes (**F**). Module gene scoring using SenMayo signature to identify senescent cells (**G**). * $p < 0.05$ vs. young; triangles represent females, squares represent males.

Senescent endothelial cells increase in the aorta with aging and are a putative target to reverse vascular aging.

To begin to specify the effects of aging and cellular senescence on the aortic endothelial cell population, we merged all endothelial cell clusters and reperformed unsupervised clustering which revealed 12 unique endothelial subclusters (**Figure 2A**). We leveraged the endothelial-specific senescent signature EndoSen²² to more precisely identify senescent endothelial cells. According to EndoSen module scoring, senescent endothelial cells were present in endothelial subcluster 4, 7, 9, and 10 with the most intense expression being in subcluster 10 (**Figure 2B**). Moreover, endothelial subcluster 10 had the highest expression of the canonical cellular senescence marker *Cdkn1a* compared to the other subclusters (**Figure 2C**). As such, we focused on age-related changes occurring in this potential senescent subcluster.

With merged data, we first looked at the top 50 DEGs that distinguished the putative senescent subcluster 10 from other endothelial subclusters (**Figure 2D**). Endothelial subcluster 10 expressed genes had roles involved in extracellular matrix remodeling, stress signaling, cell cycle maintenance, and high levels of SASP factors. We performed pathway analysis on the DEGs of senescent endothelial subcluster 10 and found that the identified pathways were related to cardiovascular pathology (*i.e.*, atherosclerosis and cardiomyopathy), cellular senescence and cancer (which share common pathways), and overall dysregulated signaling pathways (another key feature of senescent cells) (**Figure 2F**). Together, these data suggest an increase of senescent endothelial cells with aging that may be a putative target for improving vascular health with aging.

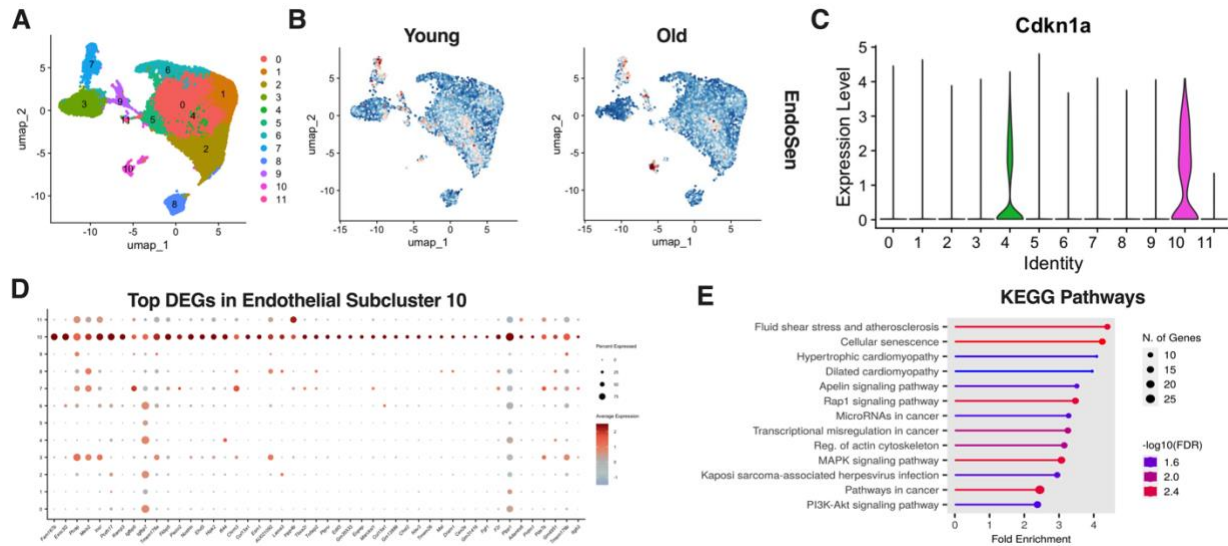


Figure 2. Senescent endothelial cells increase in aortas with aging and are a putative target to reverse vascular aging. Uniform manifold approximation and projection (UMAP) of reclustered aortic endothelial cells from all the samples (A). Module gene scoring using EndoSen signature to identify senescent endothelial cells (B). Violin plot presentation of the canonical cellular senescence marker *Cdkn1a* across clusters on endothelial cells (C).

Dot plot analysis of the top 50 most differentially expressed genes (DEGs) in endothelial subcluster 10 (D). KEGG pathway analysis of DEGs in endothelial subcluster 10 in old vs. young sample (E).

Oral intermittent fisetin supplementation ameliorates senescent vascular cells with aging.

Given the prevalence of senescent cells in aortas with aging, we next sought to determine if we could reduce senescent cell burden using senolytic therapy. Senolytics are compounds that selectively eliminate excess senescent cells^{13,23}. Although several senolytic therapies are currently being advanced in preclinical and clinical studies, we opted to use the natural senolytic fisetin given its efficacy on senescent endothelial cells¹⁷, favorable effects on vascular function with aging¹⁴ and encouraging safety profile for clinical translation¹⁶. To study the effects of fisetin supplementation on aortic gene expression, we added a group of young and old fisetin-treated mice to our transcriptomic analysis (**Figure 3A**). Importantly, we have previously shown that oral, intermittent fisetin supplementation eliminates excess senescent cells¹⁴. Given that

young animals have minimal cellular senescence burden^{11,24}, we included the young fisetin-treated group to ensure minimal effects of fisetin supplementation in young, healthy animals.

To assess fisetin-related changes in aortic cell populations, we first performed cell composition analysis between the treatment conditions (**Figure 3B-C**). We found that in young mice, fisetin supplementation did not change the cell composition of any of the cell types suggesting minimal effects of fisetin on healthy, young animals (**Figure 3B-C**). In old mice, age-related differences in endothelial and T cells persisted, even after fisetin supplementation (**Figure 3B-C**). However, fisetin supplementation mitigated trending increases in VSMC proportions ($p=0.025$), suggesting favorable effects on age-related intima-media thickening (**Figure 3B-C**). We next assessed whether fisetin supplementation had favorable effects on senescent cell burden in old aortas. Using the SenMayo senescence signature, we performed module gene scoring across the treatment conditions. Marginal differences were observed in the young groups suggesting minimal effects of fisetin on young, healthy mice (**Figure 3D**). In aortas from old mice, fisetin supplementation appeared to have the greatest effects on the endothelial cell population, reinforcing the senolytic effects of fisetin on senescent endothelial cells (**Figure 3D**). To assess the senescent cell genes that were modulated with aging and fisetin supplementation, we looked at the SenMayo genes that were upregulated with aging (old vehicle vs. young vehicle) and downregulated with fisetin supplementation (old fisetin vs. old vehicle). We found that these genes were highly involved in the SASP, cell cycle arrest maintenance, and proinflammatory pathways (**Figure 3E**). Together, these data suggest that fisetin supplementation has minimal effects on healthy young mice that have low levels of cellular senescence, whereas fisetin supplementation in old mice results in shifts towards young aortic cell composition and a reduction in whole-aorta cell senescence.

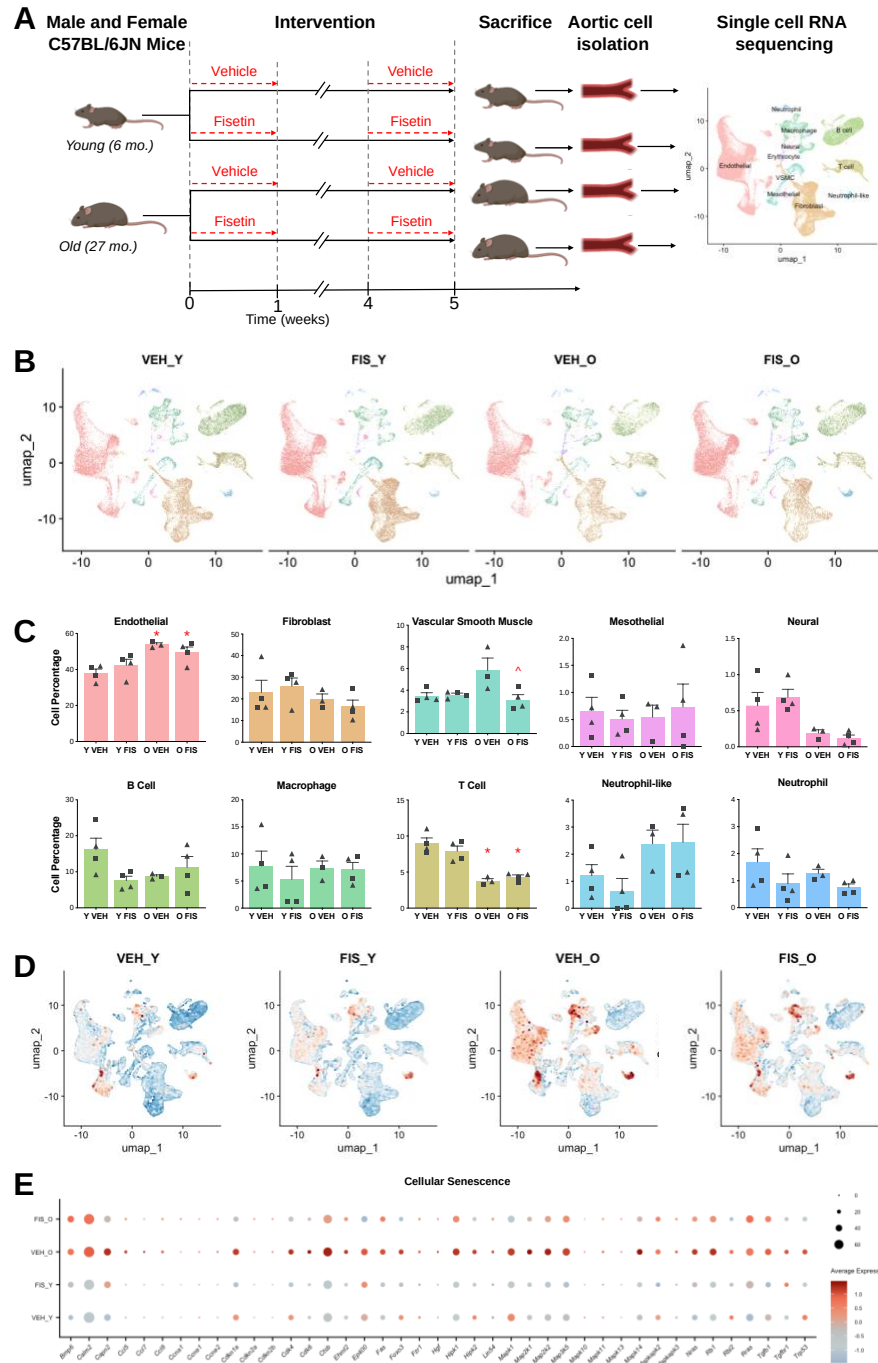


Figure 3. Fisetin supplementation ameliorates vascular cell senescence in the aorta with aging. Schematic of study design used in animal model resulting in 4 groups: young vehicle (VEH_Y), young fisetin (FIS_Y), old vehicle (VEH_O) and old fisetin (FIS_O) treated mice (A). Uniform manifold approximation and projection (UMAP) (B) and aortic cell composition (C) by cell type across treatment conditions. UMAP of module gene scoring using SenMayo signature to identify senescent cells across treatment conditions (D). Dot plot analysis of differentially expressed senescent cell markers across treatment conditions (E). * $p < 0.05$ vs. Y VEH, $^{\wedge}p < 0.05$ vs. O VEH; triangle represent females, squares represent males.

Fisetin supplementation ameliorates senescent endothelial cells in the aorta with aging.

We next sought to assess the effects of fisetin supplementation on aortic senescent endothelial cells. To identify senescent endothelial cell populations, we used a combination of differences in endothelial cell composition and senescent signature module testing. Endothelial cell composition revealed endothelial subcluster 10 as a cluster that was upregulated with age and downregulated, back towards young levels, in old mice that received fisetin supplementation (**Figure 4A-B**). As validated above, we used the endothelial-specific senescent signature, EndoSen, to distinguish senescent populations. Leveraging EndoSen, we found that in the young endothelial populations, there was a small number of senescent cells and fisetin supplementation had marginal effects on the young tissue, consistent with whole-aorta observations (**Figure 4C**). As detected previously, in old samples, we found senescent cells existed, especially in subcluster 10 (**Figure 4C**). Fisetin supplementation appeared to diminish the subcluster 10 back towards intensities observed in the young mice (**Figure 4C**). In combination, the endothelial cell proportions and EndoSen signature implicate endothelial subcluster 10 as a senescent cell population targeted by fisetin.

To further characterize the senescent endothelial subcluster 10, we performed DEG testing between groups to identify gene targets that are modulated with age and fisetin supplementation. We first compared the young groups and found no genes that were significantly changed, suggesting minimal effects of fisetin supplementation on young, healthy endothelial cells (**Figure 4D**). To identify age-related differences, we assessed DEGs in senescent endothelial cells between young vs. old vehicle-treated mice. We found that 278 genes were upregulated, and 135 genes were downregulated, suggesting a large impact of aging on the transcriptome of senescent endothelial subcluster 10 (**Figure 4D**). To assess the effects of fisetin

supplementation on this subpopulation, we compared subcluster 10 between the old vehicle and old fisetin groups. In this comparison we found a greater number of genes to be downregulated (177) than upregulated (34), suggesting a shift back towards young levels (**Figure 4D**). Finally, we compared young vehicle vs. old fisetin subcluster 10 to determine if the transcripts were more similar to that of the old vehicle. We found that no genes were significantly modulated between these groups, suggesting that endothelial cells from old mice that received intermittent fisetin supplementation have a similar transcriptomic profile to that of endothelial cells from young mice (**Figure 4D**).

We next identified the top 50 DEGs that distinguished the old vehicle group from the other treatment conditions (**Figure 4E**). The analysis represents genes that were most overexpressed with aging and downregulated with fisetin supplementation in endothelial subcluster 10 (**Figure 4E**). We found that the genes largely encoded canonical senescence markers (*Cdkn1a*) and several SASP factors including *Cxcl12*, *Ccl7*, *Plat*, *Ctsd*, *Serpine2*, *Igfbp5*, *Tgfb1*, *Igfbp3*, *Igfbp4*, *Fgf1*, and *Plaur* (**Figure 4F**). In summary, these data reveal that fisetin supplementation effectively targets senescent endothelial subcluster 10 and reduces the SASP factors expressed by this subpopulation.

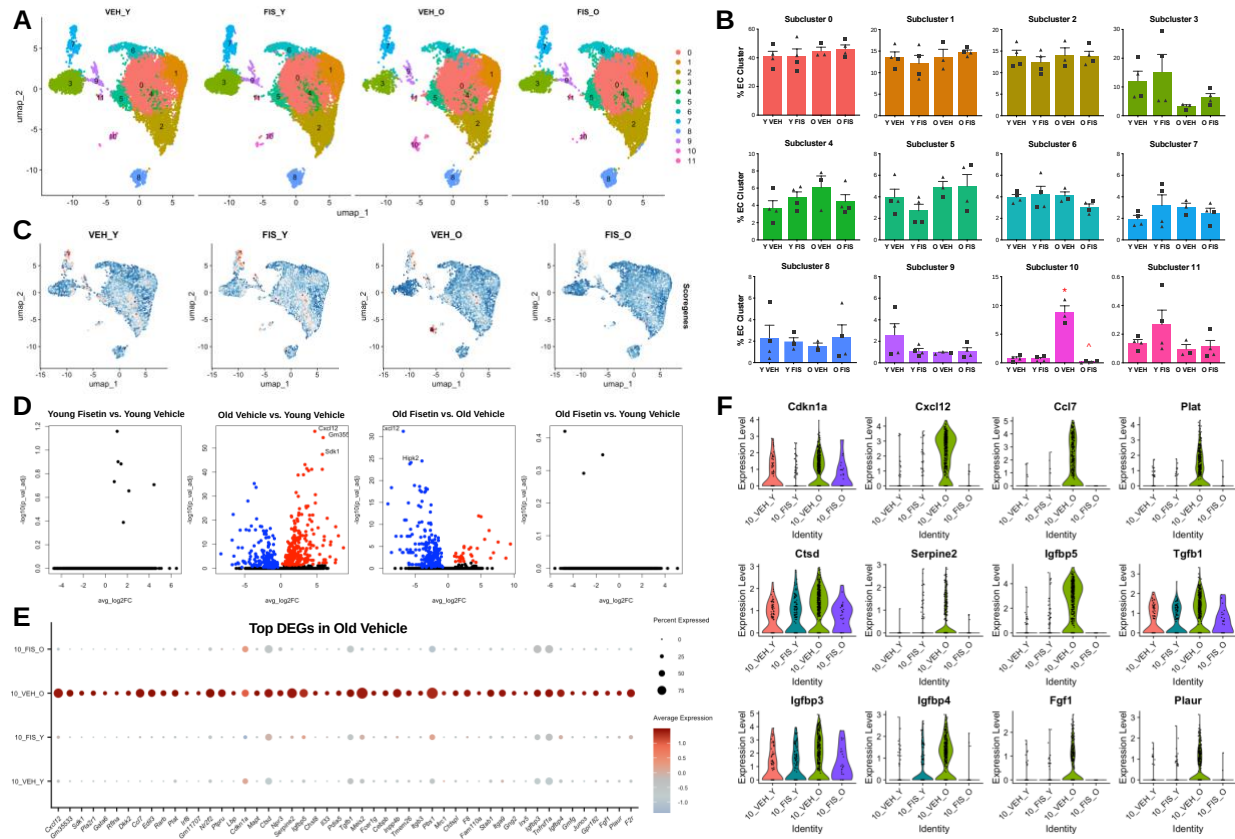


Figure 4. Fisetin supplementation reverses aortic endothelial cell senescence in old mice.

Uniform manifold approximation and projection (UMAP) (A) and subcluster percentage (B) of aortic endothelial cells from the young vehicle (Y VEH), young fisetin (Y FIS), old vehicle (O VEH) and old fisetin (O FIS) treated mice. Module gene scoring using EndoSen signature to identify senescent endothelial cells across treatment conditions (C). Volcano plot representation of differentially expressed genes (DEGs) between different treatment conditions in endothelial subcluster 10 (D). Dot plot analysis of top 50 DEGs between O VEH and other treatment conditions (E). Violin plots of the canonical cellular senescence marker, *Cdkn1a*, and senescence-associated secretory phenotype genes across treatment conditions (F). *p<0.0001 vs. Y VEH, ^p<0.0001 vs. O VEH; triangle represent females, squares represent males.

Fisetin supplementation mitigates circulating SASP abundance in old mice.

The SASP represents a collection of pro-inflammatory molecules that have varied roles in modulating physiological function^{8,25}. Our next aim was to identify candidate SASP factors that may impair endothelial function and be favorably modulated by fisetin supplementation to prevent endothelial dysfunction. Provided the changes observed in SASP factors transcripts from

senescent endothelial cells (**Figure 4F**), we sought to validate that these genes were being translated into proteins and secreted into the circulation. We selected SASP proteins to measure based on: *a*) the magnitude of difference in endothelial gene expression of factors that were unfavorably modulated with aging and normalized toward young levels with fisetin supplementation; and *b*) for biological plausibility of influencing endothelial function.

Leveraging the targeted Luminex platform, we measured candidate SASP factors in plasma from treated animals including: CCL4, CCL7, CCL8, CXCL12, FGF Basic, GDF15, IGFBP3, PAI1, S100A9, TIMP4. Of these factors, no age- or fisetin-related effects were observed in CCL4, CCL7, IGFBP3 or TIMP4 abundance. An age-related increase was observed in CCL8, FGF Basic, and S100A9 between young vehicle and old vehicle-treated mice. Finally, PAI1 and CXCL12 were both higher with aging (old vehicle vs. young vehicle), and this age-related increase was ameliorated by fisetin supplementation (old fisetin vs. old vehicle).

PAI1, a platelet protein that inhibits fibrin degradation²⁶, is a well-established marker of cellular senescence²⁷ and validates our senescent cell clearance. CXCL12 is a chemokine that regulates angiogenesis²⁸ in the vasculature and is implicated in atherosclerotic progression²⁹. Given the novelty and putative biological plausibility of CXCL12 in regulating endothelial function (whereas PAI1 has primary effects on platelets), we next sought to investigate its causal implication in regulating endothelial function.

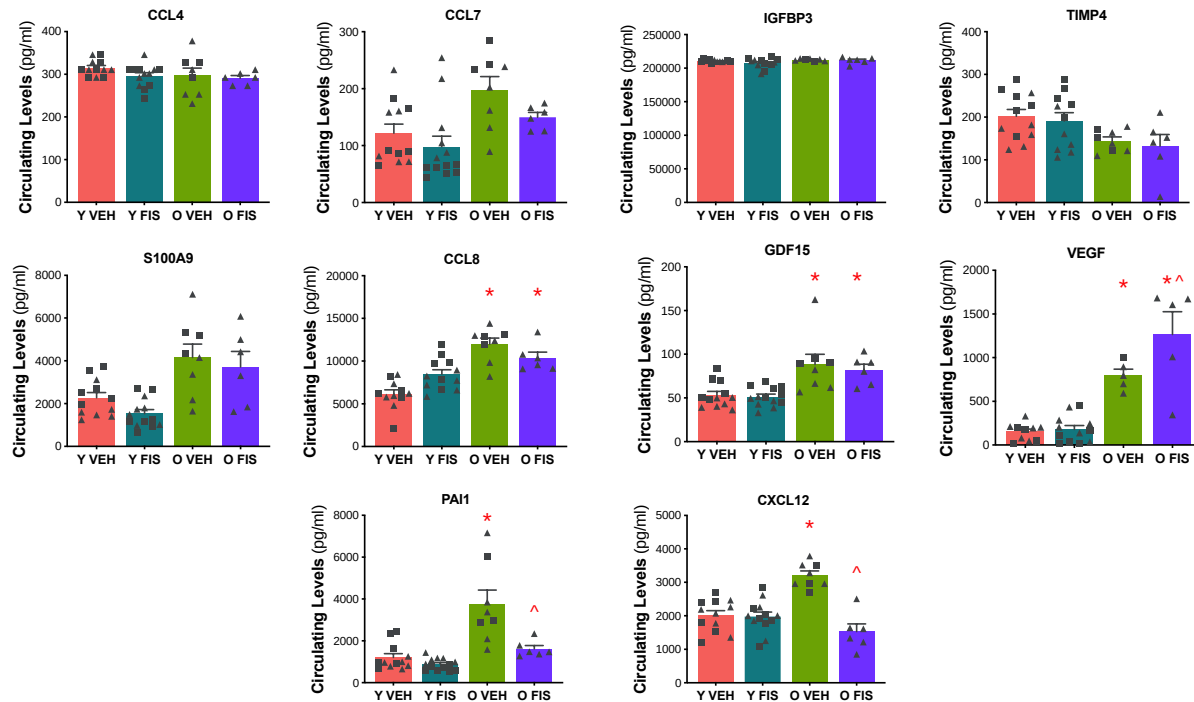


Figure 5. Fisetin supplementation modulates circulating senescence-associated secretory phenotype (SASP) factors in plasma from old mice. Protein abundance of key factors that were favorably modulated in aortic senescent endothelial cells that were upregulated in old vehicle (O VEH)-treated mice, compared to young (Y) VEH mice and ameliorated in old fisetin (O FIS)-treated mice. *p<0.0001 vs. Y VEH, ^p<0.0001 vs. O VEH; triangle represent females, squares represent males.

Fisetin supplementation attenuates plasma-induced endothelial dysfunction with aging.

Endothelial cells line the luminal wall of arteries and have direct contact with the circulating milieu, including secreted SASP factors^{19,30}. Given the observation that select circulating SASP factors were altered with aging and fisetin supplementation, we next sought to assess the functional role of the circulating SASP on age-related endothelial dysfunction and the specific effects of CXCL12 as a candidate circulating SASP factor that may be responsible for transducing the effect of the circulating milieu on endothelial function. To do so, we leveraged our recently developed *ex vivo* plasma exposure model where isolated carotid arteries from young, intervention-naïve mice are exposed *ex vivo* to sex-matched plasma prior to assessing

endothelium-dependent dilation (EDD)¹⁰. Importantly, this model allows us to causally determine the effects of the circulating milieu (*i.e.*, the SASP and other bioactive molecules in circulation) on endothelial function. In combination, we performed plasma exposure on isolated arteries (aortic rings) and cultured human aortic endothelial cells to elucidate underlying mechanisms and verify effects directly on endothelial cells. To elucidate the specific role of CXCL12 in isolated arteries and endothelial cells, we added recombinant mouse CXCL12 to plasma that exhibited low CXCL12 levels (*i.e.*, plasma from young and old fisetin-treated mice). Importantly, CXCL12 levels were added back such that the average concentration of CXCL12 matched that of the plasma from the old vehicle-treated group. In cultured endothelial cells, we used the selective neutraligand of CXCL12, LIT-927^{31,32}, to test the effects of CXCL12 inhibition in plasma with high CXCL12 levels (*i.e.*, old vehicle plasma). Given that we observed minimal effects of fisetin supplementation on the gene expression of young mice, we opted to only use the young vehicle mice as a young reference group. As such, the functional measurements have presented with 3 plasma exposure conditions: young vehicle, old vehicle, and old fisetin.

Circulating SASP- and CXCL12-induced endothelial dysfunction. We first tested the effects of the plasma from the 3 identified groups on endothelial function using carotid artery EDD following intraluminal plasma perfusion. EDD was measured as a response to increasing doses of acetylcholine and is indicative of the endothelial response to the vasodilatory molecule nitric oxide (NO). Peak EDD was lower in carotid arteries exposed to plasma from old control mice compared to young control mice (old vehicle plasma, $76 \pm 3\%$ v. young plasma, $92 \pm 1\%$; $p < 0.0001$), suggesting age-related impairments in plasma-induced endothelial function (**Figure 6A**). Peak EDD was higher in carotid arteries exposed to plasma from old fisetin-treated mice

compared to old vehicle-treated mice (old fisetin plasma, $87 \pm 2\%$ v. old vehicle plasma, $76 \pm 3\%$, $p=0.001$) (**Figure 6A**), suggesting that fisetin supplementation prevents plasma-induced endothelial dysfunction with aging. The addition of CXCL12 to young plasma impaired peak EDD ($80 \pm 2\%$) compared to young plasma alone (-12% , $p=0.002$) (**Figure 6A**). Likewise, CXCL12 added to old fisetin plasma reduced peak EDD ($68 \pm 3\%$) compared to old fisetin plasma alone (-22% , $p<0.0001$) and had comparable peak EDD to plasma from old vehicle-treated mice (-8% , $p=0.029$) (**Figure 6A**). No differences in EID were observed between any of the groups or conditions ($p=0.342$), indicating that differences observed in response to plasma exposure occurred in an encompium-dependent manner (**Figure 6B**). Together, these data suggest that the circulating SASP milieu directly mediated endothelial dysfunction, and its detrimental effects are in part transduced by CXCL12.

Circulating SASP- and CXCL12-induced vascular cell senescence. One approach by which the SASP elicits its adverse effects is through an endocrine induction of cellular senescence^{6,25}. As such, to investigate the mechanisms by which the circulating SASP modulates endothelial function, we sought to assess SASP-induced senescence by measuring senescent cell burden in arteries exposed to plasma. To do so, we measured senescence-associated β -galactosidase (SA- β -Gal) intensity, which is the most established marker of senescent cell burden^{33,34}. We found that when normalized to total cell count there were ~ 7 -fold more SA- β -Gal positive cells in aortas exposed to old versus young plasma ($p<0.0001$) (**Figure 6C**). SA- β -Gal positive cells were ~ 6 -fold lower in aortas exposed to plasma from old fisetin-treated mice compared to old control mice ($p<0.0001$) (**Figure 6C**).

Similar to arteries, endothelial cells exposed to old vehicle plasma had 5.6-fold greater SA- β -Gal positive cells than young plasma ($p < 0.0001$) (**Figure 6D**). SA- β -Gal positive cells were ~2.2-fold lower in aortas exposed to plasma from old fisetin-treated mice compared to old control mice ($p < 0.0001$) (**Figure 6D**). Exogenous CXCL12 induced cellular senescence in endothelial cells exposed to young (3.4-fold, $p < 0.0001$) and old fisetin (1.8-fold, $p < 0.0001$) plasma (**Figure 6D**). Further, CXCL12 inhibition with LIT-927 reduced cellular senescence in endothelial cells exposed to old vehicle by 1.8-fold ($p < 0.0001$) (**Figure 6D**).

In addition to SA- β -Gal, we confirmed SASP and CXCL12 effects on cellular senescence via endothelial gene expression of the cellular senescence biomarkers *Cdkn1a* and *Cdkn2a*, which encode the cell cycle inhibitor proteins p21 and p16, respectively. We found that HAECs exposed to plasma from old vehicle mice had a 45-fold increase in *Cdkn1a* expression compared to young plasma ($p < 0.0001$) and this age-related increase was ameliorated by a 7.5-fold decrease compared to old fisetin plasma ($p < 0.0001$ v. old vehicle) (**Figure 6E**). The addition of CXCL12 to plasma increased *Cdkn1a* expression by 12-fold in HAECs exposed to young plasma ($p < 0.0001$) and by 2-fold in HAECs exposed to old fisetin plasma ($p < 0.0001$) (**Figure 6E**). The addition of LIT-927 to plasma reduced *Cdkn1a* expression by 2.4-fold in HAECs exposed to old vehicle plasma ($p < 0.0001$) (**Figure 6E**). Similarly, we found that HAECs exposed to plasma from old vehicle mice had a 7.2-fold increase in *Cdkn2a* expression compared to young plasma ($p < 0.0001$) and this age-related increase was ameliorated by a 5.5X-fold decrease compared to old fisetin plasma ($p < 0.0001$) (**Figure 6F**). The addition of CXCL12 to plasma increased *Cdkn2a* expression by 2.5-fold in HAECs exposed to young plasma ($p < 0.0001$) and tended to increase expression in HAECs exposed to old fisetin plasma (2.4, $p = 0.052$) (**Figure 6F**). The addition of LIT-927 to plasma reduced *Cdkn2a* expression by 2.6-fold in HAECs exposed to old

vehicle plasma ($p < 0.0001$) (**Figure 6F**). In combination, these data indicate that the circulating SASP milieu directly induces vascular and endothelial cell senescence and that these age-related inductions are prevented by fisetin supplementation. Further, CXCL12 partially mediates circulating SASP-induced vascular and endothelial cellular senescence.

Circulating SASP- and CXCL12-induced NO and Superoxide Bioactivity. Age-related endothelial dysfunction is primarily mediated by reduced NO bioavailability, which is another characteristic of senescent endothelial cells³⁵. As such, we sought to assess NO bioactivity in response to acetylcholine in cultured endothelial cells following plasma exposure and CXCL12 protein manipulation to assess the direct influence of CXCL12 in mediating endothelial NO bioactivity. Endothelial cells exposed to old vehicle plasma exhibited 31% reduced NO bioactivity compared to young plasma ($p < 0.0001$) (**Figure 6G**). This age-related NO bioactivity impairment was prevented in endothelial cells exposed to old fisetin plasma (+22%, $p < 0.0001$) (**Figure 6G**). Exogenous CXCL12 impaired NO bioactivity in endothelial cells exposed to young (-25%, $p < 0.0001$) and old fisetin (-15%, $p < 0.0001$) plasma (**Figure 6G**). Further, CXCL12 inhibition with LIT-927 increased NO bioactivity in endothelial cells exposed to old vehicle by 10% ($p < 0.0001$) (**Figure 6G**).

Reduced NO bioactivity may occur as a result of excessive superoxide from various sources including dysfunctional mitochondria, as excessive superoxide scavenges NO to reduce its bioavailability³⁶. As such, we sought to determine whether the circulating SASP induced increased mitochondrial superoxide levels and if this manifestation was mediated by CXCL12. To do so, we next measured mitochondrial-derived superoxide bioactivity in isolated arteries (**Figure 6H**) and cultured endothelial cells (**Figure 6H**) following plasma exposure and CXCL12 protein manipulation. Arterial mitochondrial superoxide levels were higher in aortas exposed to

plasma from old control mice compared to young control mice (old vehicle plasma, 8292 ± 823 arbitrary units [AU] v. young control plasma, 2387 ± 347 ; $p < 0.0001$) (**Figure 6H**).

Mitochondrial superoxide levels were lower in arteries exposed to plasma from old fisetin-treated mice compared to old vehicle-treated mice (old fisetin plasma, 3443 ± 509 AU v. old vehicle plasma, 8294 ± 823 AU, $p < 0.0001$) (**Figure 6H**). Exogenous CXCL12 induced mitochondrial superoxide bioactivity in arteries exposed to old fisetin plasma (7701 ± 982 $p = 0.001$), but not young plasma (3550 ± 674 , $p = 0.763$) (**Figure 6H**).

Similar to arteries, endothelial cells exposed to old vehicle plasma had 14% greater mitochondrial superoxide bioactivity than young plasma ($p < 0.0001$) (**Figure 6I**). Mitochondrial superoxide bioactivity was 6% lower in endothelial cells exposed to plasma from old fisetin-treated mice compared to old control mice ($p < 0.0001$) (**Figure 6I**). Exogenous CXCL12 induced mitochondrial superoxide bioactivity in endothelial cells exposed to young plasma (+5%, $p = 0.029$), but not old fisetin plasma (**Figure 6I**). Further, CXCL12 inhibition with LIT-927 reduced mitochondrial superoxide bioactivity in endothelial cells exposed to old vehicle plasma by 10% ($p < 0.0001$) (**Figure 6I**). Together these data implicate that the circulating SASP milieu impairs NO bioactivity and promotes mitochondrial superoxide. Further, these SASP-mediated impairments are driven by CXCL12.

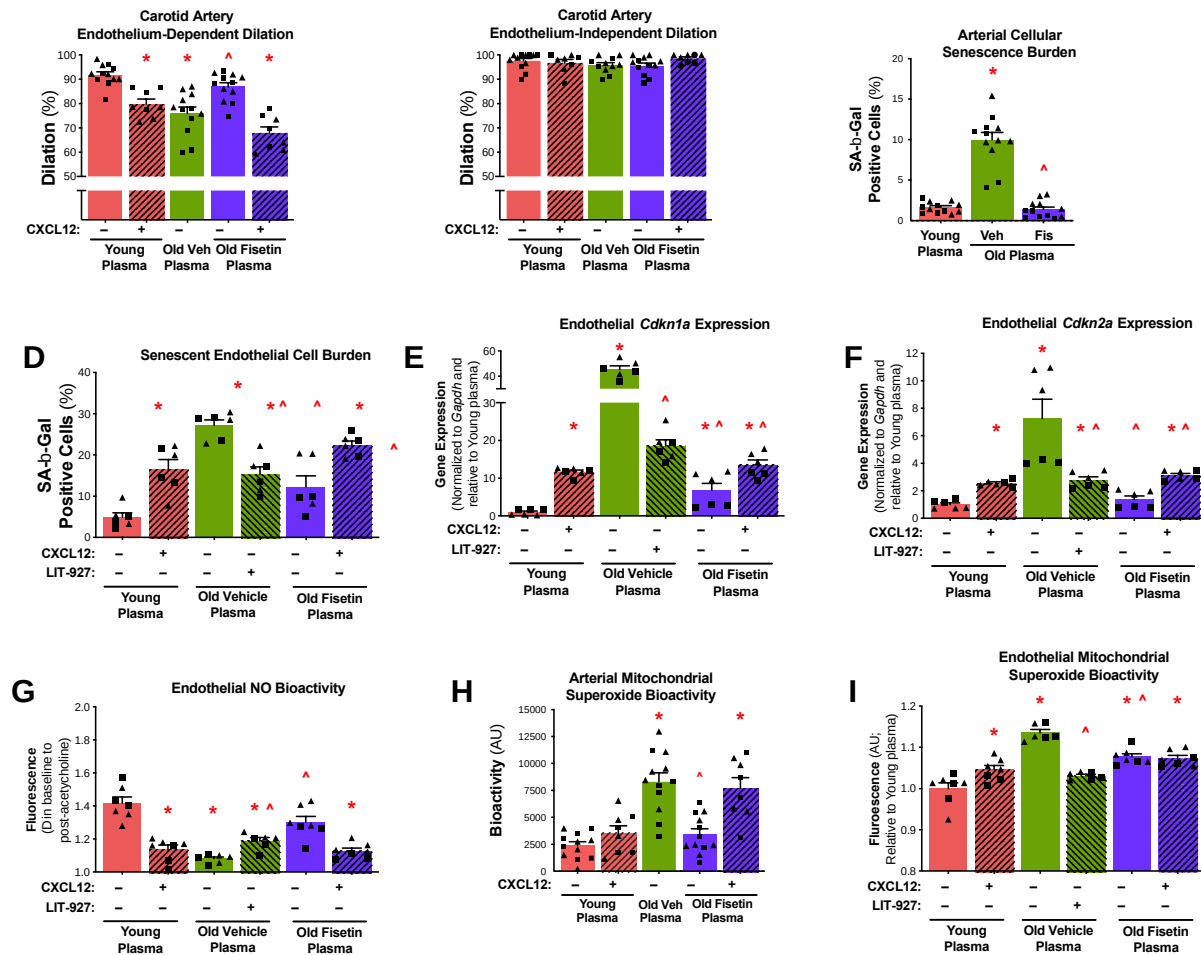


Figure 6. Fisetin supplementation prevents circulating SASP- and CXCL12-induced endothelial dysfunction with aging. Endothelium-dependent dilation (**A**) and endothelium-independent dilation (**B**) in isolated carotid arteries following plasma perfusion with and without the addition of recombinant mouse CXCL12, Nitric oxide bioactivity in human aortic endothelial cells (HAECs) following plasma exposure with and without the addition of recombinant CXCL12 and CXCL12 inhibitor, LIT-927 (**C**). Mitochondrial superoxide bioactivity following plasma exposure with and without the addition of recombinant CXCL12 and CXCL12 inhibitor, LIT-927 in isolated arteries (**D**) and HAECs (**E**). Senescence-associated β -galactosidase (SA- β -gal) signal following plasma exposure with and without the addition of recombinant CXCL12 and CXCL12 inhibitor, LIT-927 in isolated arteries (**F**) and HAECs (**G**). Cellular senescence biomarker gene expression of *Cdkn1a* (**H**) and *Cdkn2a* (**I**) in HAECs following plasma exposure with and without the addition of recombinant CXCL12 and CXCL12 inhibitor, LIT-927. Values represent mean $\pm$ SEM; * p <0.05 vs. young control, ^ p <0.05 vs. old vehicle; triangles represent females, squares represent males.

Fisetin supplementation mitigates endothelial-to-mesenchymal transition (EndoMT) with aging.

CXCL12 plays a crucial role in promoting the EndoMT, a process by which endothelial cells lose their characteristic markers, while gaining increased expression of mesenchymal markers. Importantly, the EndoMT is a putative mechanism of endothelial cell senescence that leads to a shift toward contractile properties and reduced NO production and is highly implicated in vascular aging. As such, we next sought to determine if the EndoMT is an underlying mechanism mediating SASP- and CXCL12-induced endothelial dysfunction in senescent cells. As such, we first sought to determine if the senescent endothelial cells demonstrated EndoMT attributes and then if fisetin supplementation prevented circulating SASP- and CXCL12-induced EndoMT. We first measured cell-specific markers for endothelial cells (*Cdh5*, *Pecam1*, *Vwf*, and *Nos3*) and mesenchymal cells (*Tgfb1*, *Acta2*, *Vim*, and *Colla1*) in the senescent endothelial subcluster 10 and found a general reduction in endothelial characteristics (**Fig. 7A**) and increase in mesenchymal markers (**Fig. 7B**) in the old vehicle group compared to the others.

To determine if the EndoMT is promoted by the circulating SASP milieu and CXCL12 we exposed HAECs to plasma from the treated mice with and without recombinant mouse CXCL12 and LIT-927. HAECs exposed to plasma from old vehicle mice had reduced *Cdh5* (0.7-fold, $p < 0.0001$) and *Pecam1* (0.6-fold, $p < 0.0001$) expression compared to young plasma and this age-related increase was mitigated by exposure to old fisetin plasma in *Cdh5* (3-fold, $p < 0.0001$) and *Pecam1* (2-fold, $p < 0.0001$) (**Fig. 7C-D**). The addition of CXCL12 to plasma reduced *Cdh5* (0.4-fold, $p < 0.0001$) and *Pecam1* (0.6-fold, $p < 0.0001$) expression in HAECs exposed to young plasma and reduced *Cdh5* (0.4-fold, $p < 0.0001$) and *Pecam1* (0.5-fold, $p < 0.0001$) expression in HAECs exposed to old fisetin plasma (**Fig. 7C-D**). The addition of LIT-927 to old vehicle

plasma increased *Cdh5* (4-fold, $p<0.0001$) and *Pecam1* (2-fold, $p<0.0001$) compared to old vehicle alone (**Fig. 7C-D**). Contrary to endothelial markers, HAECs exposed to plasma from old vehicle mice had increased *Tgfb1* (21-fold, $p<0.0001$) and *Acta2* (8-fold, $p<0.0001$) expression compared to young plasma and this age-related increase was mitigated by exposure to old fisetin plasma in *Tgfb1* (8-fold, $p<0.0001$) and *Acta2* (9-fold, $p<0.0001$) (**Fig. 7E-F**). The addition of CXCL12 to plasma increased *Tgfb1* (2-fold, $p=0.0023$) and *Acta2* (2-fold, $p=0.0001$) expression in HAECs exposed to young plasma and increased *Tgfb1* (5-fold, $p<0.0001$) and *Acta2* (6-fold, $p<0.0001$) expression in HAECs exposed to old fisetin plasma (**Fig. 7E-F**). The addition of LIT-927 to old vehicle plasma reduced *Tgfb1* (6-fold, $p<0.0001$) and *Acta2* (4-fold, $p<0.0001$) compared to old vehicle alone (**Fig. 7E-F**). In combination, these data indicate that the circulating SASP milieu and CXCL12 promote EndoMT in senescent endothelial cells and that these age-related inductions are prevented by fisetin supplementation.

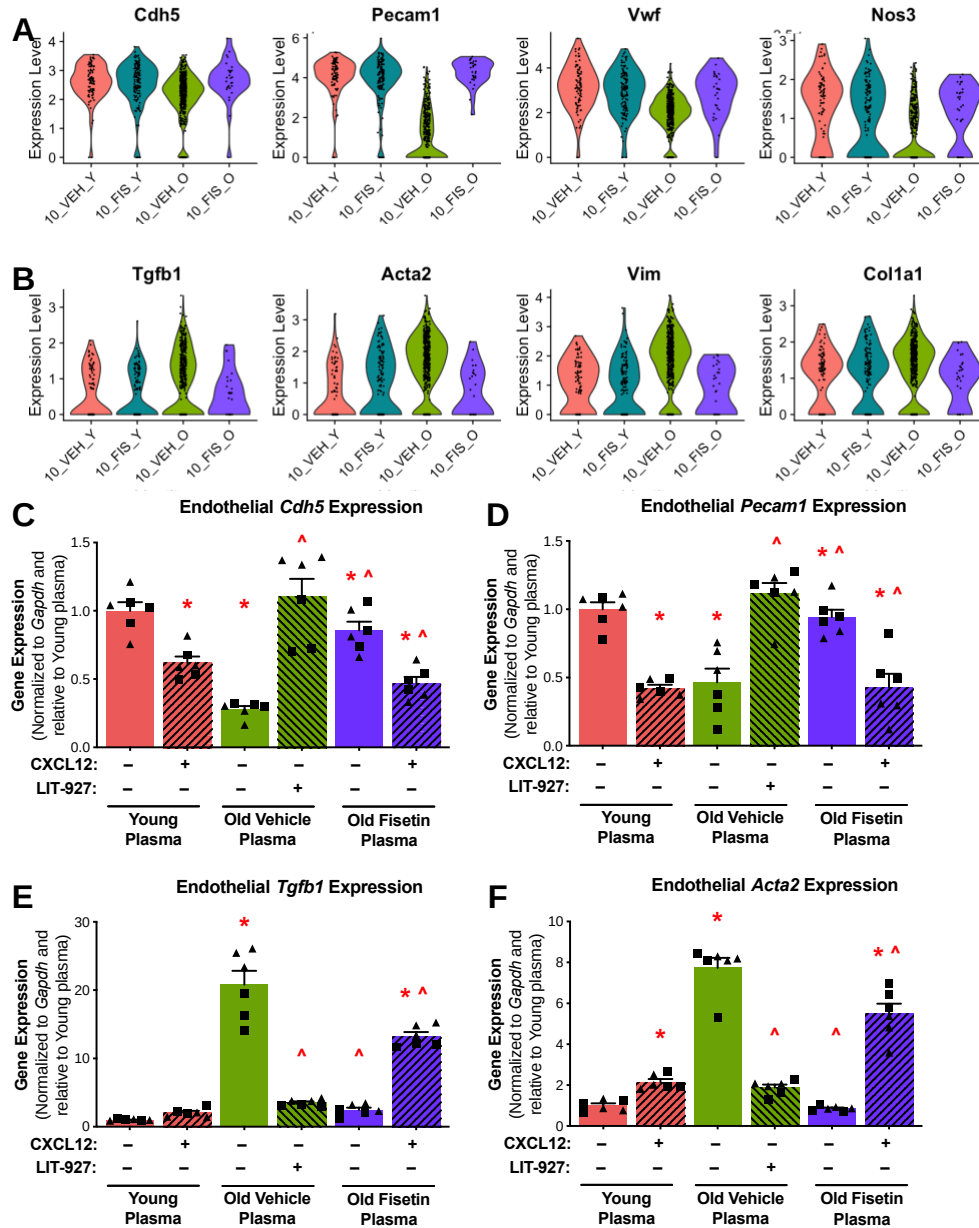


Figure 7. Fisetin supplementation prevents circulating SASP- and CXCL12-induced endothelial-to-mesenchymal transition (EndoMT). Violin plots of the endothelial (A) and mesenchymal (B) cell markers in senescent endothelial subcluster 10. EndoMT biomarker gene expression of *Cdh5* (C), *Pecam1* (D), *Tgfb1* (E), and *Acta2* (F) in human aortic endothelial cells (HAECs) following plasma exposure with and without the addition of recombinant CXCL12 and CXCL12 inhibitor, LIT-927. Values represent mean $\pm$ SEM; * $p < 0.05$ vs. young control, ^ $p < 0.05$ vs. old vehicle; triangles represent females, squares represent males.

DISCUSSION

In the present study, we characterized vascular senescent cells, the vascular SASP profile, and the circulating SASP milieu in the context of aging. Our findings demonstrate that endothelial cells are particularly susceptible to cellular senescence with aging and that fisetin, a naturally occurring senolytic compound, effectively eliminates senescent endothelial cells in old mice. Leveraging differences observed in senescent endothelial cell transcription, we establish the circulating SASP milieu as an underlying mechanism of age-related endothelial dysfunction and demonstrate the causal role of the SASP factor CXCL12 in mediating impaired endothelial function. These results contribute to a growing body of literature supporting the role of senescent cells and the SASP in vascular aging and the potential of senolytic therapy to mitigate age-related vascular dysfunction. However, these results go beyond descriptive analyses to reveal a causal role of vascular senescent cells, the circulating SASP milieu, and CXCL12 in mediating age-related endothelial dysfunction.

Vascular cell senescence and SASP.

Cellular senescence and the SASP increase with advancing age in the vasculature^{6,11} and have previously been characterized in aortic samples from mice²⁴, primates³⁷, and humans³⁸. Senescent cells accumulate in the vasculature among various mammals and promote chronic, low-grade inflammation and excess oxidative stress through the SASP^{39,40}. As such, senescent cells and the SASP significantly contribute to CVDs including atherosclerosis⁴¹, hypertension⁴², and occlusive stroke⁴³. Previous studies utilizing senolytic treatments have demonstrated improvements in vascular function^{6,14} leading to an amelioration of CVD progression^{15,44} following excess senescent cell clearance. Here, we extend these observations by demonstrating the effects of senolytic intervention on specific senescent cell populations in the vasculature.

In old mouse aortas, our senescent cell characterization identified endothelial, macrophage and neutrophil-like cells to have high cellular senescence signatures with aging. Further, changes in cell composition demonstrated higher endothelial and lower T cell populations with aging. Given the increase in endothelial cell senescence burden, age-related increase in endothelial population composition, and robust contributions to vascular function, we further characterized the effects of senolytic treatment on endothelial cell senescence. Previous studies have demonstrated that fisetin reduces vascular cell senescence *in vivo*¹⁴ and in senescent endothelial cells *in vitro*^{14,17}. We extend these findings by demonstrating that *in vivo* fisetin supplementation suppresses cellular senescence in endothelial cells that exhibit high senescent cell signatures. Importantly, the senescent endothelial cell population that we identified was involved in pathways related to cardiovascular pathologies and revealed a high expression of SASP-related transcripts. As such, our findings align with previous studies reporting the efficacy of senolytic agents in reducing senescent cell burden in aging tissues. However, this study extends prior work by identifying specific vascular cell populations targeted by senolytic intervention.

Circulating SASP-mediated endothelial dysfunction.

Vascular endothelial dysfunction is a key manifestation of age-related vascular dysfunction and a major antecedent to overt CVDs, including atherosclerosis and occlusive stroke^{2,45}. Excessive levels of cellular senescence impair endothelial function⁶ and contribute to other endothelium-related manifestations of vascular dysfunction manifestations including arterial stiffening⁶ and impaired angiogenesis⁴⁶. The aged circulating milieu is a key driver of age-related endothelial dysfunction¹⁰; however, the specific component(s) of the circulation responsible for transducing impairments in endothelial function remains to be elucidated. One of

the key discoveries in the present study is the establishment of the circulating SASP milieu as an underlying mechanism of age-related endothelial dysfunction. Our plasma exposure experiments in isolated arteries and cultured endothelial cells established a causal role for the aged circulating SASP milieu in impairing endothelial function. Age-associated endothelial dysfunction is largely mediated by the excess accumulation of senescent cells and the associated circulating SASP milieu^{6,15}. The circulating SASP can also induce cellular senescence in neighboring cells, further promoting tissue-level dysfunction⁸. Senescent endothelial cells also contain dysfunctional mitochondria which produce high levels of superoxide^{36,47} and are key regulators of the SASP⁴⁸. In turn, senescent endothelial cells demonstrate deficient NO bioactivity due in part to increased superoxide which sequesters NO, reducing its bioavailability, and limits NO production through the uncoupling of endothelial NO synthase³⁵. Here, we showed that the exposure of young arteries and proliferating endothelial cells to the aged circulating SASP milieu could induce paracrine senescence, reduce NO bioactivity, and stimulate excessive mitochondrial superoxide-related oxidative stress. Conversely, these age-related impairments induced by the aged circulating SASP milieu were at least in part prevented when exposed to the circulating milieu from old animals that received fisetin supplementation. Collectively, these findings establish the SASP as an underlying mechanism by which the circulating milieu drives endothelial dysfunction with aging.

CXCL12-mediated endothelial dysfunction.

We next sought to determine candidate factors in the circulating SASP milieu that may be responsible for transducing endothelial dysfunction with aging, as identification of these molecules would provide insight into novel mechanisms by which cellular senescence drives endothelial dysfunction with aging and potentially establish novel therapeutic targets. The

chemokine CXCL12 emerged from this study as a factor that was most highly upregulated in senescent endothelial cells with age, with its expression reduced following fisetin supplementation. Additionally, CXCL12 protein levels were elevated in the plasma of aged animals and restored to young levels by fisetin supplementation. Given that CXCL12 has been implicated in inflammatory^{29,49} and cellular senescence pathways^{21,22}, its role as a SASP factor in vascular aging provides mechanistic insight into how endothelial dysfunction develops with aging. CXCL12, through its interaction with the receptor CXCR4, controls the transduction of downstream signaling pathways involved in the endothelial-to-mesenchymal transition (EndoMT), a process where endothelial cells gain enhanced contractile properties^{28,29,49}. The manipulation of CXCL12 levels in our plasma exposure experiments supports a causal role for CXCL12 in age-related endothelial dysfunction, as inhibition of CXCL12 improved endothelial responses, whereas exogenous CXCL12 exposure recapitulated age-related dysfunction, suggesting fisetin supplementation mitigated circulating SASP milieu-related endothelial dysfunction, in part, by targeting CXCL12. Like the SASP milieu, circulating CXCL12 independently induced paracrine senescence, reduced NO bioactivity, and stimulated excess mitochondrial oxidative stress to confer dysfunction in isolated arteries and cultured endothelial cells. Although endothelial cell-derived CXCL12 has been identified to promote atherosclerosis progression, our findings also demonstrate that CXCL12 promotes endothelial dysfunction through the EndoMT, a manifestation that is prevented by fisetin supplementation. The ability of fisetin to ameliorate CXCL12-driven endothelial dysfunction highlights its therapeutic potential in mitigating vascular aging and establishes CXCL12 as a novel therapeutic target to mitigate vascular cell senescence.

Experimental considerations.

Despite these promising findings, particular experimental considerations should be noted. First, while our study establishes a link between CXCL12 and endothelial senescence, additional mechanistic studies are needed to determine the precise pathways through which CXCL12 contributes to vascular aging. Additionally, although fisetin effectively reduced CXCL12 levels and improved endothelial function in our mouse model, the translatability of these findings to humans remains to be fully explored. Ongoing clinical trials investigating fisetin supplementation in mid-life and older adults will provide further insights into its efficacy and safety in human populations. Although there were no distinct sex differences in the present study, sexual dimorphism in vascular aging and senescence warrants further exploration. Investigating sex-specific responses to fisetin treatment will be crucial in optimizing therapeutic strategies for both men and women.

CONCLUSIONS

In conclusion, this study identifies that senescent endothelial cells accumulate with advancing age and promote the systemic SASP program. Senolytic intervention with fisetin effectively eliminated senescent endothelial cells and highlighted CXCL12 as a critical endothelium-derived SASP factor that is modulated with age and senolytic treatment. Mechanistically, we uncovered that the circulating SASP milieu and CXCL12 induce cellular senescence, reduce NO bioactivity, and promote mitochondrial oxidative stress to confer endothelial dysfunction. By demonstrating that fisetin effectively reduces CXCL12 levels and improves endothelial function, our findings provide mechanistic insight into the therapeutic potential of senolytic compounds in vascular aging. These results may have important implications for ongoing and future clinical studies aimed at developing senolytic therapies to improve cardiovascular health in aging populations.

METHODS

Animals and experimental design.

Male and female C57Bl/6N (wildtype) mice were obtained from the National Institute of Aging colony (maintained by Charles River, Wilmington, MA). For the duration of the study, all mice were single housed at the National Institute of Aging animal facility with a 12hr:12hr light-dark cycle and allowed ad libitum access to an irradiated, fixed, and open rodent chow (Inotiv/Envigo 7917, stored at room temperature).

For the intervention period, young (6 months) and old (27 months) wildtype mice were randomly assigned to receive vehicle (10% ethanol, 30% PEG400 and 60% Phosal 50 PG) or fisetin (100 mg/kg/day in vehicle). For the young mice, 12 mice (6M/6F) received the vehicle and 12 mice (6M/6F) received fisetin, and for old mice, 8 mice (3M/5F) received vehicle and 10 mice (4M/6F) received fisetin. Treatment was administered via oral gavage using an intermittent dosing paradigm – one week on; two weeks off; one week on, as previously established senolytic dosing paradigm^{14,16}. Mice were sacrificed one to two weeks following the final dose to rule out any acute effects of the compound as the terminal half-life of fisetin is ~3.1 hours in plasma⁵⁰.

Aortic single-cell RNA sequencing.

Aortic single-cell transcriptomic analysis was performed, as previously described⁴⁴. Briefly, mice were anesthetized with ketamine and xylazine and blood was extracted using cardiac puncture. The whole aorta was collected after left ventricular perfusion with 10 ml of PBS and quickly transferred to cold PBS. To prepare a single-cell suspension, perivascular adipose tissue was removed and whole aortas from three mice (per sex and treatment group) were cut into ~ 1-mm pieces and digested with an enzyme solution consisting of 10 mg/ml Collagenase II (Gibco) and 1 mg/ml elastase (Worthington Biochemical Corp.) in Dulbecco's

Modified Eagle Medium (DMEM; 3 ml/sample) for 15 min at 37°C in a tissue culture incubator. The cell suspension was strained through a 40- μ m filter and centrifuged at 500 x g for 5 min. Cells were resuspended in DMEM supplemented with 15% fetal bovine serum (FBS) and 5 mM of EDTA and then stained with 2 μ g/ml of propidium iodide for 5 min to assess cell viability and single, viable cells were sorted using Fluorescence-Activated Cell Sorting (FACS) Aria FUSION (BD Bioscience) for 10x Genomics library preparation. The single-cell libraries were prepared with Chromium Next GEM Single Cell 3' Kit v3.11 (10x Genomics) with Chromium Next GEM Chip G Single Cell Kit (10x Genomics) according to manufacturer's protocol with Chromium Controller (10x Genomics). Briefly, ~10,000 single cells were used for GEM generation, and the cDNAs were then prepared checked on the Agilent Bioanalyzer with High Sensitivity DNA kit (Agilent). Prepared cDNAs were then used for library preparation that were checked on the Agilent Bioanalyzer with DNA 1000 kit (Agilent Technologies). The libraries were sequenced with Illumina NovaSeq 6000 sequencer at a depth of 70,000-120,000 reads per cell.

Single-cell RNA sequencing was processed following the Feature Barcoding tool pipeline in Cell Ranger (version 9.0.0; 10x Genomics) and the mouse mm10 reference genome. The obtained read count matrices were subsequently analyzed in R using the Seurat package, version 5.2.0 with default parameters in all functions, unless otherwise specified. Standard quality control filtering was applied to each sample to eliminate low-quality cells and potential doublets from downstream analysis. Filtering also removed cells containing more than 25% mitochondrial RNAs, expressing fewer than 200 or greater than 7,000. RNAs that were detected in less than 3 cells were excluded from the analysis. For each sample, over 90% of filtered cells remained and RNA data was analyzed with "LogNormalize" method followed by running the FindVariableFeatures function to select 2,000 most variable RNAs for dimensionality reduction.

The FindIntegrationAnchors function was applied to choose anchors for data integration. After performing Principal Component Analysis (PCA), the top 40 PCA dimensions were determined by the ElbowPlot method and used to create the Uniform Manifold Approximation and Projection (UMAP) with the resolution parameter set to 0.3 resulting in 29 clusters.

Differentially expressed marker RNAs for each cluster were identified with the FindAllMarkers function and the FindMarkers function was used to find differentially expressed genes between experimental conditions. The main cell types were identified using marker genes of each cluster in combination with data from literature. To identify subclusters within the cell types, the analysis was rerun separately on the cells of each cluster and the UMAPs were created using the resolution 0.3. Senescent cells assessment scoring was performed with Seurat function AddModuleScore. Pathway analysis was performed using KEGG.

Plasma protein quantification.

For multiplex analysis, plasma was thawed and centrifuged at 16,000 x g for 4 min. Custom murine Luminex Assay kits were designed by R&D Biosystems to include the following analytes: CCL4, CCL7, CCL8, CXCL12, FGF Basic, GDF15, IGFBP3, PAI1, S100A9, TIMP4. Plasma was diluted 1:10 using the Calibrator Diluent RD6-52 provided in the kit. Standards (provided with the kit), blanks, and plasma were incubated with the microparticle cocktail for 2 h at 25°C, followed by incubation with Biotin-Antibody cocktail for 1 h. The final incubation lasted 30 min with Streptavidin-PE and shaking at 25°C prior to running the plate on the Bio-Rad Bioplex-200 Instrument. Each incubation was followed by washing 3 times with Wash Buffer (provided in the kit). Instrument settings were adjusted to the following: 50 µl sample volume, Bio-Plex

MagPlex Beads (Magnetic), Double Discriminator Gates set at 8,000 and 23,000, low RP1 target value for the CAL2 setting, 50 count/region. The results were analyzed with the Bio-Plex Manager software.

Plasma-mediated vascular endothelial function.

To assess the role of the circulating SASP milieu (plasma) and CXCL12 on endothelial function, an *ex vivo* isolated artery model was leveraged, as previously described¹⁰. In brief, carotid arteries were excised from young (3-6 month), intervention-naïve wildtype mice and cannulated onto pressure myographs. Plasma collected from young vehicle, old vehicle and old-fisetin treated mice was diluted in a solution containing 5% sex-matched plasma, 1% penicillin-streptomycin antibiotic cocktail, and 94% modified Krebs buffer corrected to pH 7.3. The diluted plasma samples were perfused luminally through the pressurized arteries for 24h prior to assessing endothelial function. Recombinant mouse CXCL12 protein was added back to plasma with low CXCL12 levels (e.g., plasma from young and old fisetin-treated mice), such that concentration of CXCL12 matched the average levels in old vehicle-treated mice (3210 pg/ml). Following plasma perfusion, endothelial function was measured endothelium-dependent dilation (EDD) and endothelium-independent dilation (EID) in response to increasing doses of acetylcholine (ACh) and sodium nitroprusside (SNP), respectively, as described previously. In brief, after vessels were pre-constricted with phenylephrine (PE; 2 mM; Sigma-Aldrich), EDD was assessed by measuring increases in luminal diameter in response to increasing concentrations of ACh (1×10^{-9} to 1×10^{-4} M; Sigma-Aldrich). Following EDD, EID was assessed by measuring the increase in luminal diameter in response to increasing concentrations of SNP, an exogenous NO donor (1×10^{-10} to 1×10^{-4} M; Sigma-Aldrich). All dose response

data are presented as percent dilation relative to maximum diameter to account for differences in baseline vessel diameter.

Arterial mitochondrial superoxide bioactivity.

Arterial mitochondrial superoxide bioactivity was assessed using the mitochondrial-specific superoxide spin probe 1-hydroxy-4-[2-triphenylphosphonio-acetamido]-2,2,6,6-tetramethylpiperidine (mitoTEMPO-H; Enzo Life Sciences,) by electron paramagnetic resonance (EPR) spectrometry, as previously described^{6,14}. In short, 1 mm aortic rings were incubated in 5% sex-matched plasma from young, old vehicle- or old fisetin-treated mice in DMEM for 24h. Recombinant mouse CXCL12 protein was added back to plasma from young and old fisetin-treated mice, such that concentration of CXCL12 matched the average levels in old vehicle-treated mice (3210 pg/ml) and 1 µg/ml of LIT-927 was added to the old vehicle plasma³¹. Following plasma exposure, aortic rings incubated in Krebs/HEPES buffer containing 0.5 mM mitoTEMPO-H at 37°C for 1 h. Samples were analyzed using aMS300 Xband EPR spectrometer (Magnettech).

Cultured endothelial cell nitric oxide (NO) and mitochondrial superoxide bioactivity.

Human aortic endothelial cells (HAECs) (PromoCell; used at passag 3-4; female, age: 80 years, non-smoker, free from known CVD) were cultured in a 96-well glass bottom plate (CellVis) under standard culture conditions (37.5°C, 100% relative humidity, 5% CO₂). HAECs were grown in basal media (Endothelial Cell Growth Medium-2 [EGM-2] BulletKit; PromoCell) supplemented with 5% plasma collected young, old vehicle, and old fisetin-treated mice for 2 h. Recombinant mouse CXCL12 protein was added back to plasma from young and old fisetin-treated mice, such that concentration of CXCL12 matched the average levels in old vehicle-treated mice (3210 pg/ml) and 1 µg/ml of LIT-927 was added to the old vehicle plasma³¹.

HAECs grown in basal media and FBS were used as a control condition. HAECs were stained with the fluorescent probe Hoechst (nuclei stain; Thermo Fisher) and either 10 μ M diaminorhodamine-4M AM (DAR-4M AM; Sigma-Aldrich) (to quantify NO production) for 45 min or 5 μ M MitoSOX (Thermo Fisher) (to quantify mitochondrial superoxide bioactivity) for 30 min. Live HAECs were imaged at 20x using wide-field fluorescence microscopy (EVOS M7000 Imaging System; Thermo Fisher) under standard incubation conditions. HAECs stained with DAR-4M AM were imaged before and 6 min after addition of 100 μ M acetylcholine (Sigma) to stimulate NO production⁵¹ and mitochondrial superoxide bioactivity was assessed under basal (unstimulated) conditions⁵². Images were quantified using Celleste 5.0 Image Analysis Software (Thermo Fisher) as previously described^{51,52} and normalized to the FBS condition on each plate to control for potential variation between plates.

Senescence-associated β -galactosidase (SA- β -gal) staining.

Arterial and cultured endothelial cell SA- β -gal staining was performed using the Senescence Detection Kit (Abcam) according to the manufacturer's instructions. Briefly, 1 mm aortic rings or HAECs (described above) were incubated in 5% sex-matched plasma from young, old vehicle- or old fisetin-treated mice in DMEM or EGM-2, respectively, for 24 h. Recombinant mouse CXCL12 protein was added back to plasma from young and old fisetin-treated mice, such that concentration of CXCL12 matched the average levels in old vehicle-treated mice (3210 pg/ml) and 1 μ g/ml of LIT-927 was added to the old vehicle plasma³¹. Following plasma exposure, aortas and HAECs were washed with PBS, fixed (fixing solution provided in kit), and incubated with the X-gal solution overnight. Aortic rings were then washed, frozen in OCT compound, and stored at -80°C until the time of sectioning. Aortic samples were later sectioned (7 μ m; Leica CM300, Leica Biosystems) and plated in poly-L-lysine coated slides. Aortas and

HAECs were stained with DAPI overnight and images were captured using bright-field and fluorescent microscopy (EVOS M7000 Imaging System; Thermo Fisher) at 10X magnification and quantified using ImageJ, as described ³³.

Cultured endothelial cell gene expression.

mRNA gene expression was measured in HAECs following incubation with 5% plasma from young, old vehicle or old fisetin-treated mice cultured in EGM-2 for 24 h. Recombinant mouse CXCL12 protein was added back to plasma from young and old fisetin-treated mice, such that concentration of CXCL12 matched the average levels in old vehicle-treated mice (3210 pg/ml) and 1 µg/ml of LIT-927 was added to the old vehicle plasma³¹. Briefly, RNA was extracted using the RNeasy mini kit (Qiagen). cDNA was synthesized using the iScript cDNA synthesis kit (Bio-Rad Laboratories). Transcripts of cellular senescence markers, *Cdkn2a* and *Cdkn1a* were analyzed using a StepOnePlus Real-Time PCR System (Applied Biosystems) in 96-well plates and the Taqman OpenArray (Applied Biosystems) was used as a master mix, as described. SimpleSeq DNA sequencing (Quintara Biosciences) was used to validate PCR products.

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CHAPTER V

CONCLUSIONS

The purpose of this dissertation was to investigate the natural senolytic compound fisetin on its ability to suppress excess vascular cell senescence and the senescence-associated secretory phenotype (SASP), mechanisms-of-action, and improve age-related vascular dysfunction.

Despite advancements in senotherapeutics drug development, research investigating age-related changes in cellular senescence and the SASP has not conclusively determined whether these differences are causally implicated in vascular dysfunction or feasible for clinical translation. In Chapter II, I analyzed scientific literature pertaining to mechanisms by senescent cells and the SASP elicit their detrimental effects on vascular function. I reviewed evidence of senotherapeutic intervention on vascular function and sought to explore clinical groups that exhibit high senescent cell burden and early vascular dysfunction and may stand to benefit the most from senotherapeutic treatment. Despite the promise of senotherapeutics, I also highlighted the potential disadvantageous effects of senescent cell clearance especially in vascular diseases emphasizing the need to further understand the role of senescent cells in vascular dysfunction and disease.

In Chapter III, I sought to test the senolytic effects of fisetin supplementation on the vasculature. I first identified optimal concentrations of fisetin that would eliminate senescent, but not proliferating endothelial cells in culture. Translating to old mice, I found that late-life fisetin supplementation reduced vascular cell senescence and improved age-related arterial stiffness and endothelial function. These improvements in vascular function were mediated by a suppression of excess cellular senescence, improvement in nitric oxide (NO) bioavailability, a reduction in whole-cell and mitochondrial-specific oxidative stress, and favorable arterial wall remodeling.

These findings provided initial evidence that fisetin may be an effective senolytic therapy in the vasculature and holds promise for clinical translation to healthy mid-life and old adults.

In alignment with this observation, in chapter IV, I sought to characterize senescent cells and the SASP in the vasculature to better understand the cellular-molecular mechanisms by which vascular senescent cells change with aging and senolytic intervention. Using aortic single-cell transcriptomic techniques, I identified that endothelial cells composition increased and were highly susceptible to becoming senescent in the context of aging. Fisetin supplementation was used to investigate the cellular-molecular changes that occur with senolytic intervention in the aorta and demonstrated robust effects on senescent endothelial cell populations. Senescent endothelial cells exhibited pathways related to cardiovascular pathology and highly expressed SASP factors. Of the expressed SASP factors, the *Cxcl12* gene expression was the most highly upregulated with age and downregulated with senolytic treatment and this pattern persisted in circulating CXCL12 protein levels in plasma. As such, we next sought to determine the role of the circulating SASP milieu and CXCL12 in regulating endothelial function. *Ex vivo* isolated artery and endothelial cell culture exposure to plasma from treated mice revealed that the aged circulating SASP milieu directly induced endothelial dysfunction through cellular senescence induction, reduced NO bioactivity, and increased mitochondrial oxidative stress. These circulating SASP-mediated impairments were in part transduced by CXCL12 and prevented by senolytic intervention. These findings establish the circulating SASP milieu as an underlying mechanism of endothelial dysfunction and identify CXCL12 as a mediator and therapeutic target of cellular senescence-induced endothelial dysfunction.

Collectively, these dissertation studies identify cellular senescence and the SASP as mechanisms of age-related vascular dysfunction, offering insight into potential preventative and

therapeutic targets. Senotherapeutic, particularly fisetin, emerged as promising strategies to target senescent cells and the SASP and mitigate cardiovascular disease risk associated with aging.

FUTURE DIRECTIONS

With the growing number of older adults there is a pressing need for effective therapies to reduce the risk and prevalence of cardiovascular diseases. These dissertation studies have identified the accumulation of cellular senescence and the SASP as mechanisms by which aging induces vascular dysfunction. As such, targeting excess cellular senescence and the SASP may be an effective approach for improving vascular function and reducing risk of vascular-related clinical disorders of aging. Notably, based on the findings presented in this dissertation in mice, a randomized, placebo-controlled trial is currently being conducted by my co-mentor and dissertation committee member Dr. Matthew Rossman to determine the efficacy of fisetin supplementation for reducing vascular cell senescence and improving vascular function in healthy mid-life and older adults (NCT06133634).

To propel cellular senescence, SASP, and senolytic research forward, there is a need to move beyond descriptive analyses and develop mechanistic understandings of microbial influences in humans. Excitingly, new approaches are being pioneered to do just that. Novel biomarkers of cellular senescence, the SASP, and senescent cell clearance are emerging to better quantify burden and clearance. Preclinical transgenic models are being advanced to have abilities to perform cell type- or tissue-specific senescent cell clearance which will allow for the investigation of the role of cellular senescence from certain regions and influence on other tissues and systems. The recent development of our laboratory's circulating milieu bioassays allows for the causal investigation of components and specific factors of the bloodstream,

including the circulating SASP milieu and SASP factors. Finally, new senolytic compounds are being developed with greater potency, specificity, and safety profiles that may have high implications in clinical medicine.

Overall, as the field progresses towards deeper mechanistic insights via the continued understanding of molecular underpinnings of senescent cells, tools to measure and test the role of cellular senescence and the SASP and development of cellular senescence- and SASP-targeted therapeutic strategies that will impact vascular function and mitigate vascular-related disorders during aging. This collective pursuit promises to advance our understanding and pave the way for transformative therapeutic strategies in combating cardiovascular diseases among mid-life and older adults.

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